

Parallelism

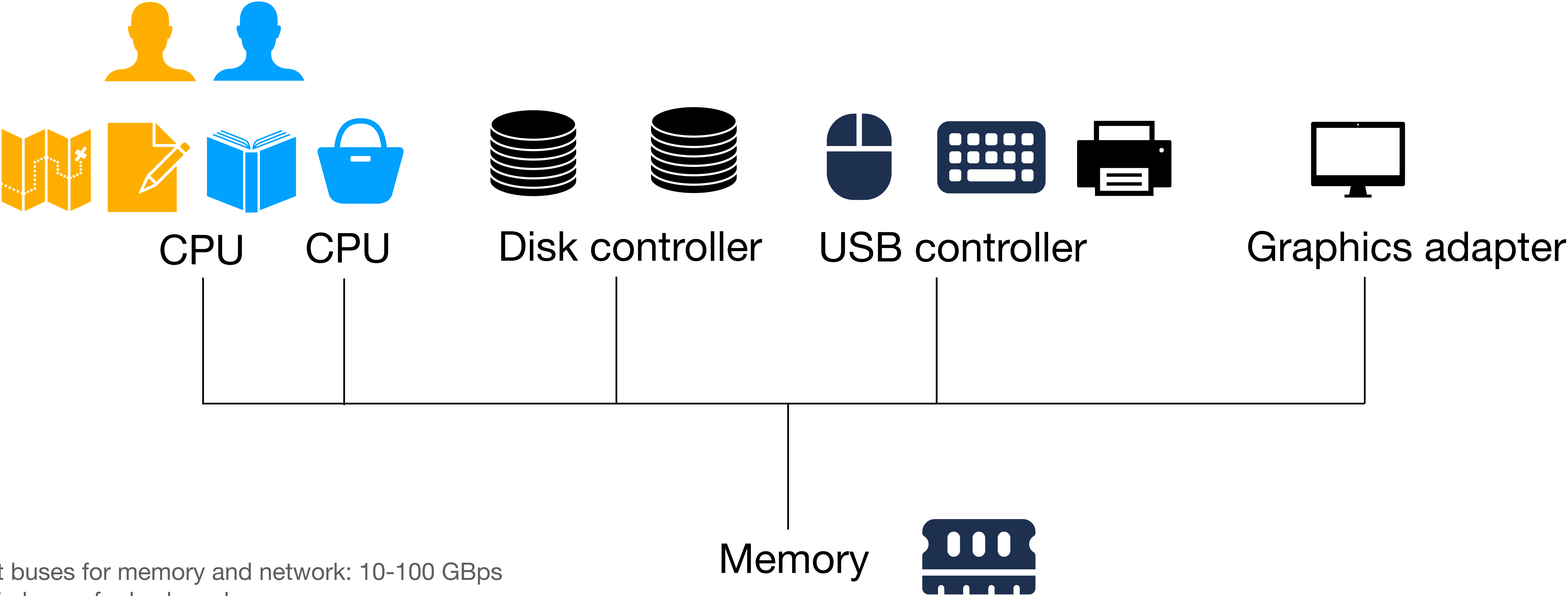
Abhilash Jindal

Agenda

xv6 book Ch4, OSTEP Ch 25-32

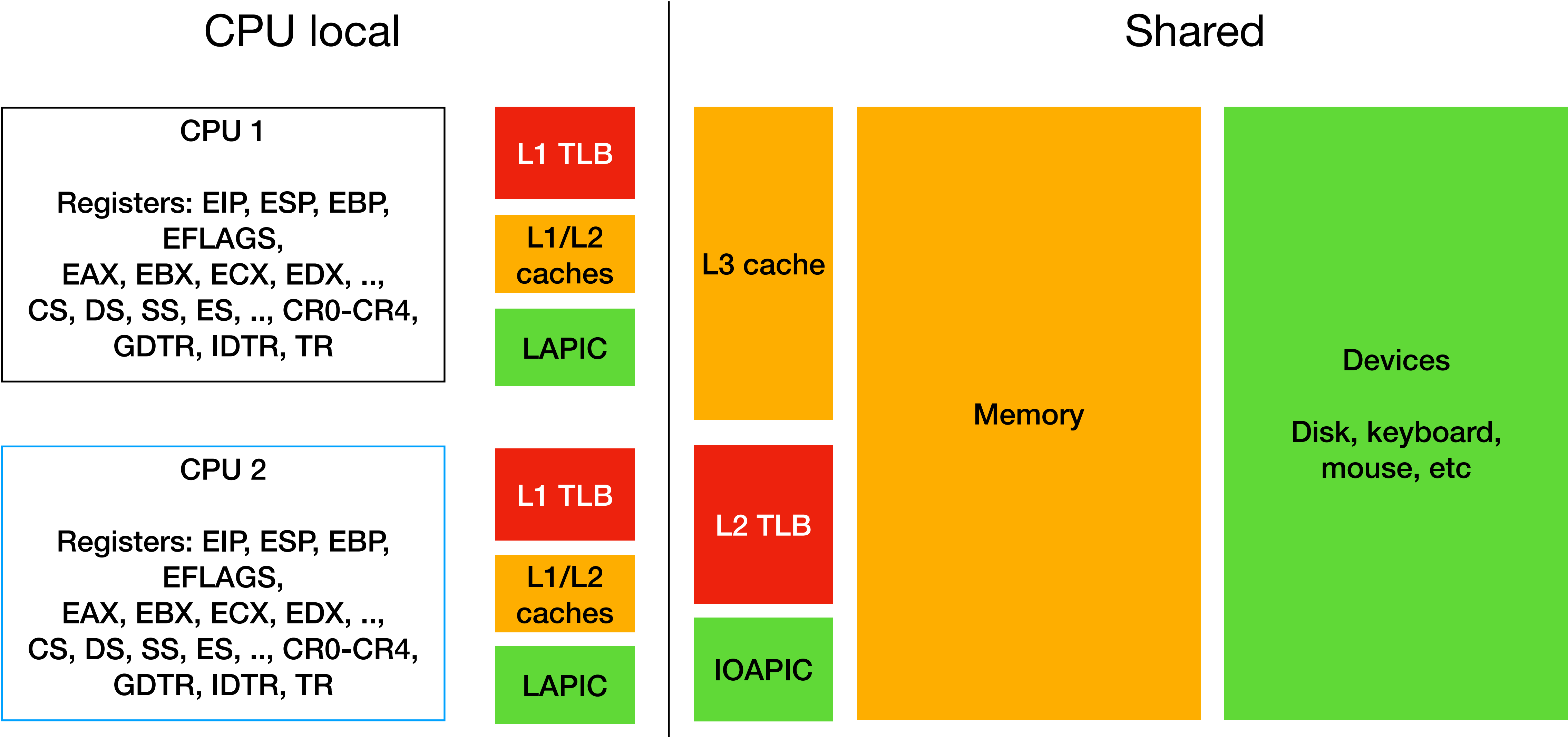
- Multi-processing hardware
 - xv6 setting up other processors
- Threads
 - Race conditions
- Design of locks
 - Spin locks, conditional variables, semaphores, read-write locks
 - Atomic instructions and memory consistency models
- Difficulties with using locks: lost wakeup, deadlocks, locking discipline

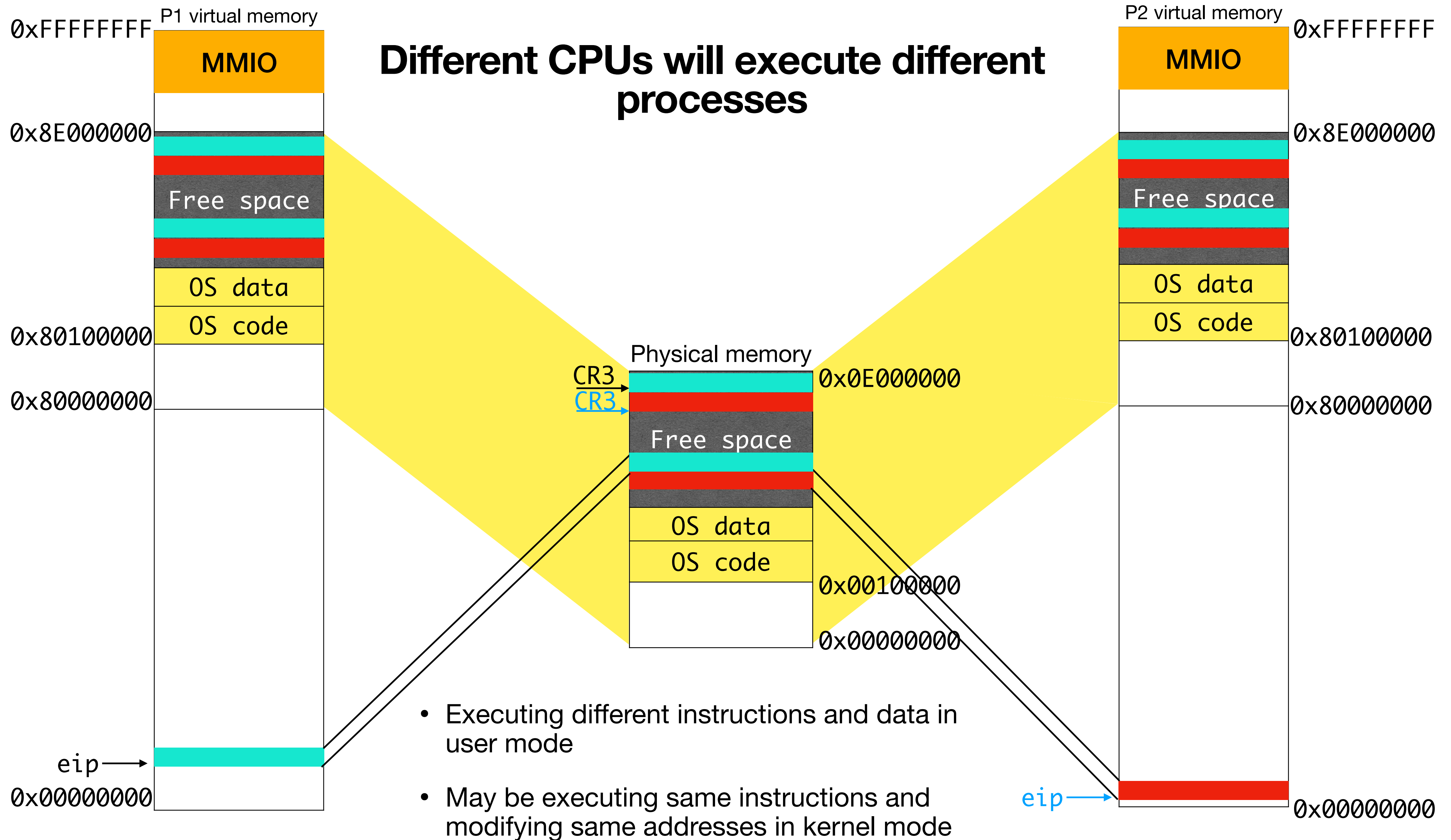
Computer organization

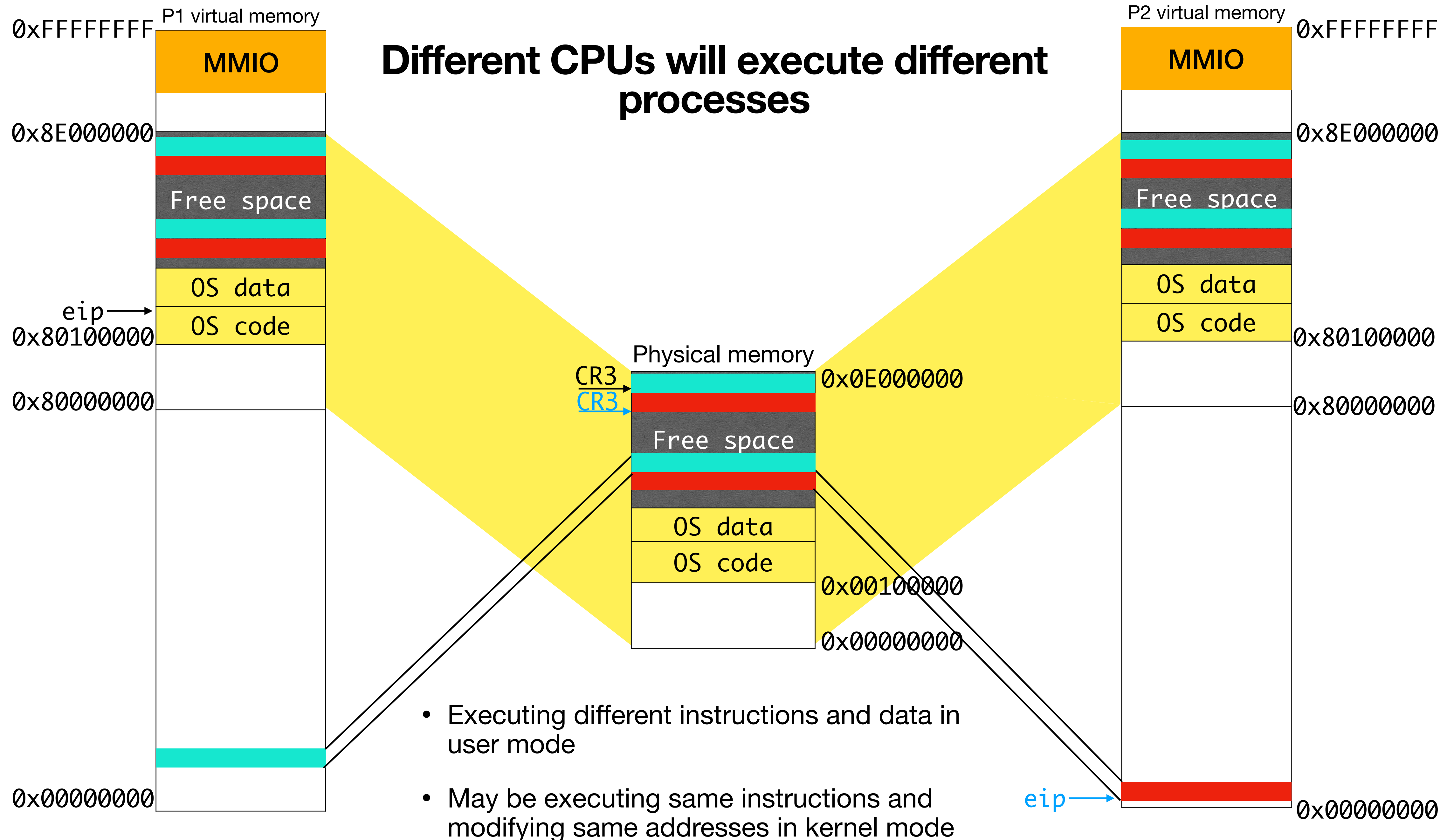


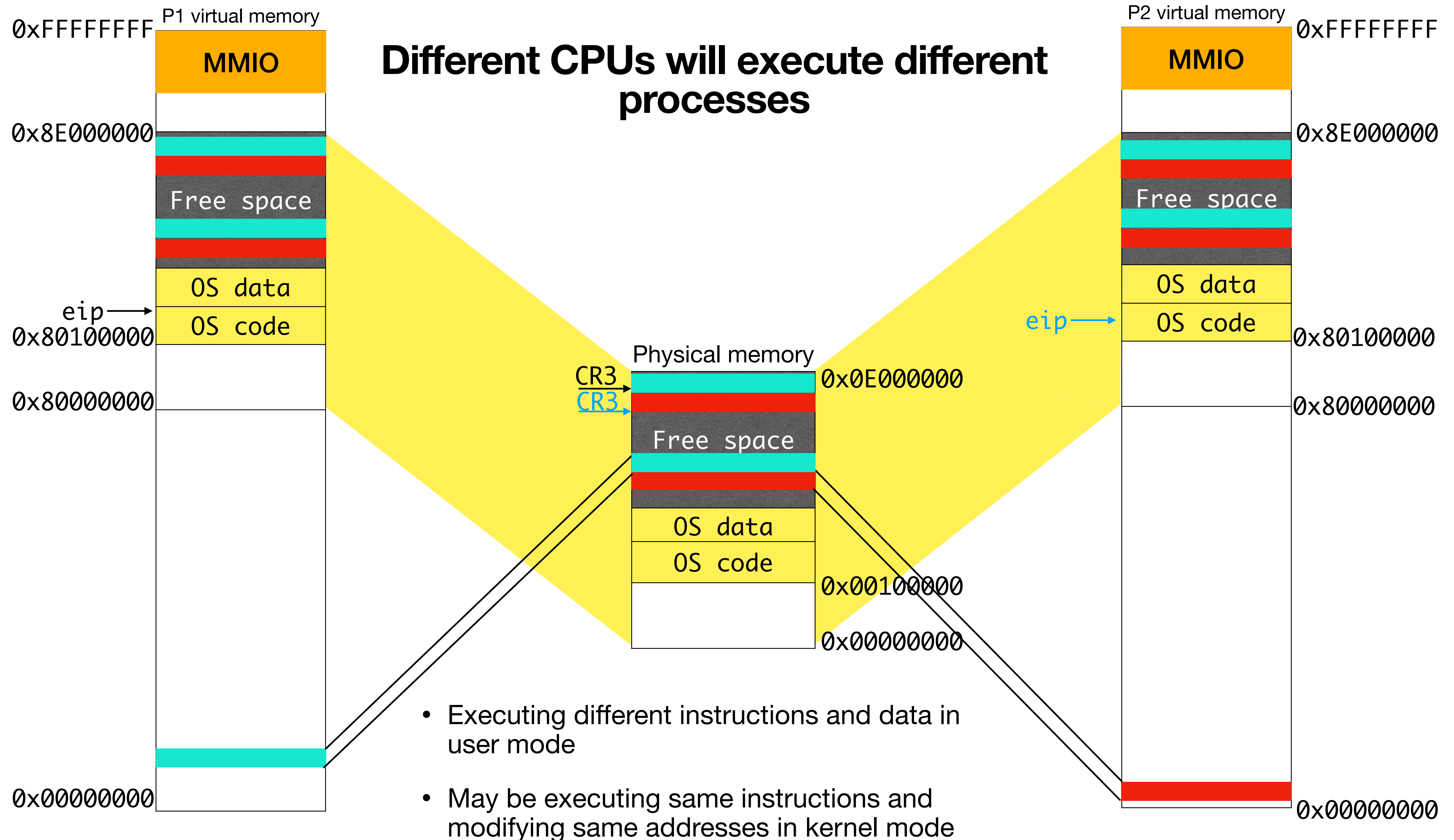
Fat buses for memory and network: 10-100 GBps
Thin buses for keyboard, mouse

Computer organization



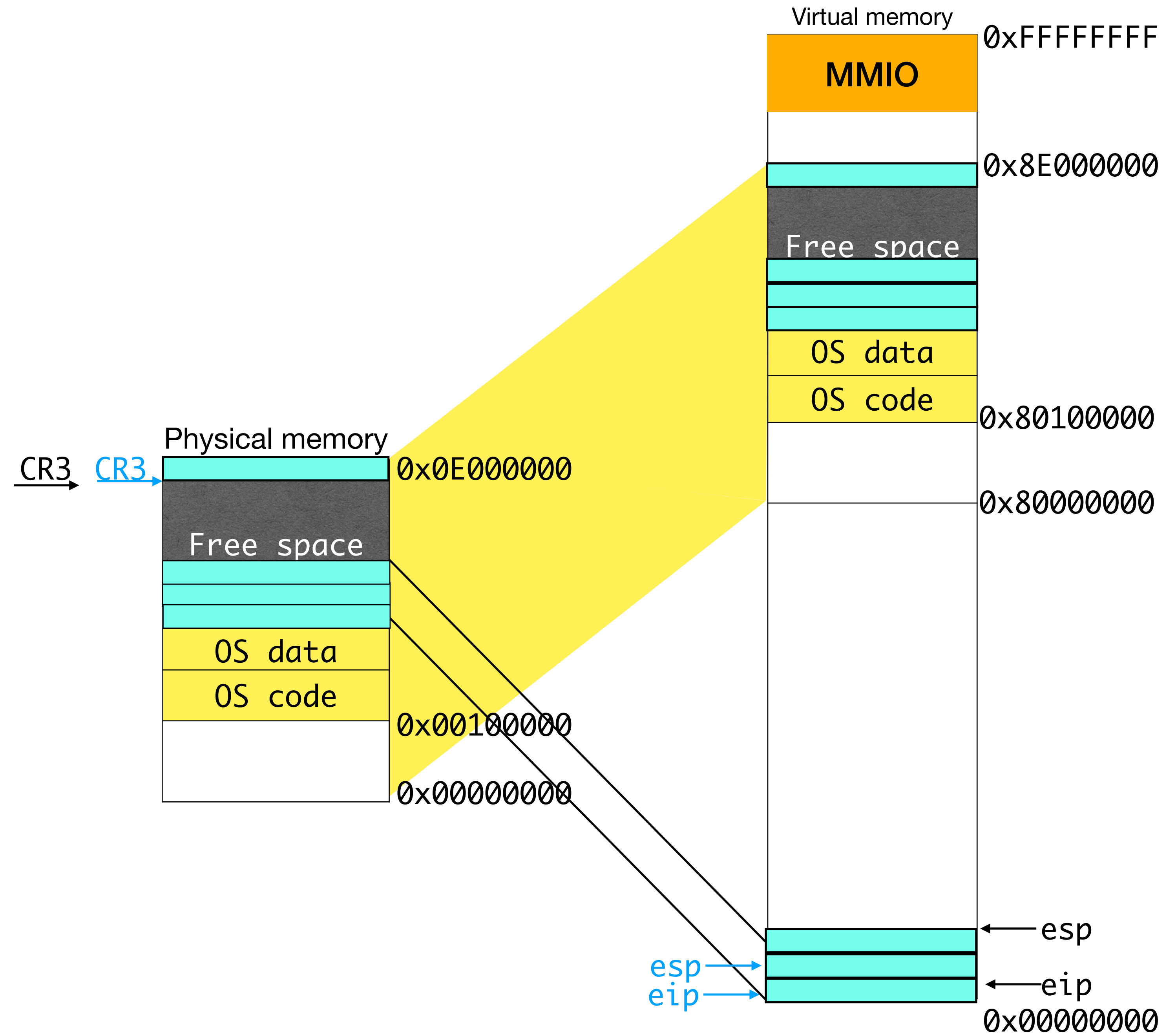




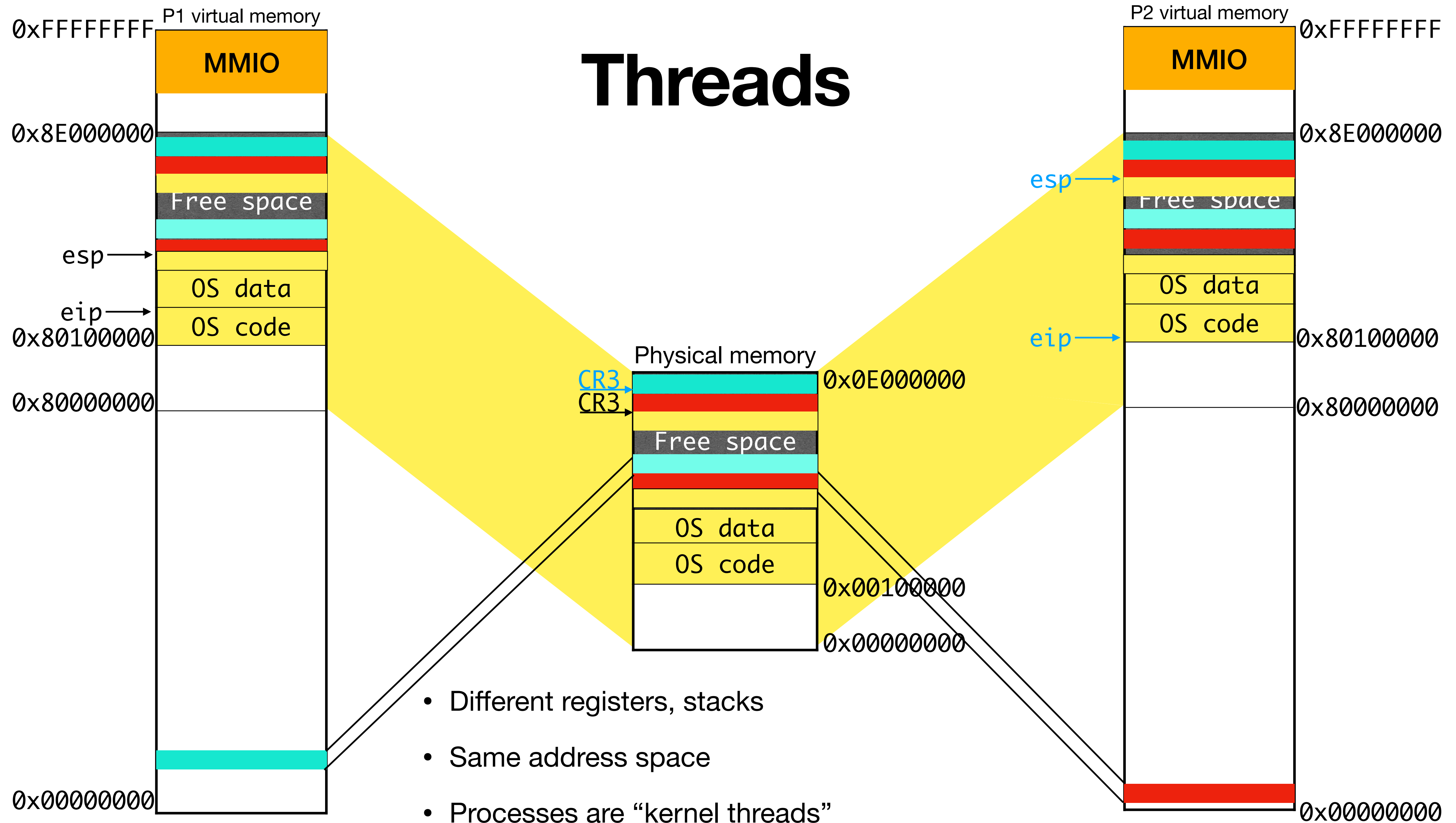


User threads

- Different registers, stacks
- Same address space



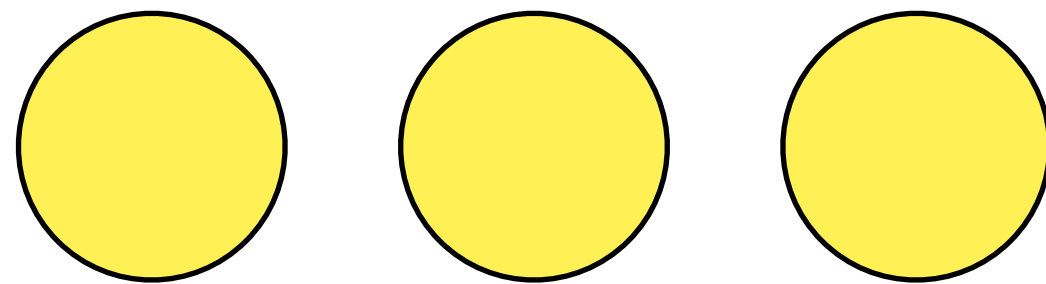
Threads



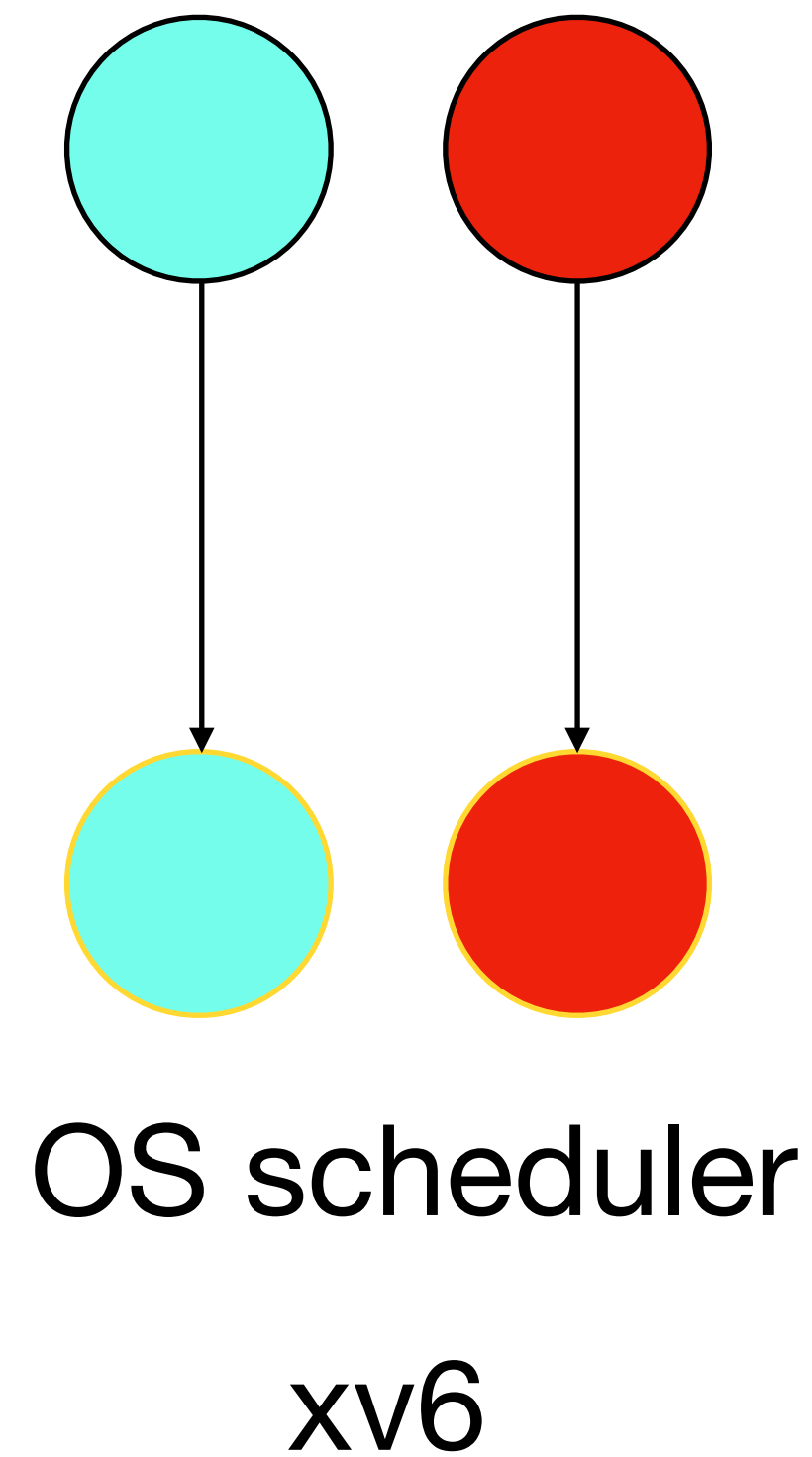
Processes are kernel threads

- When executing in OS mode:
 - Different kernel stacks, different registers
 - Same address space

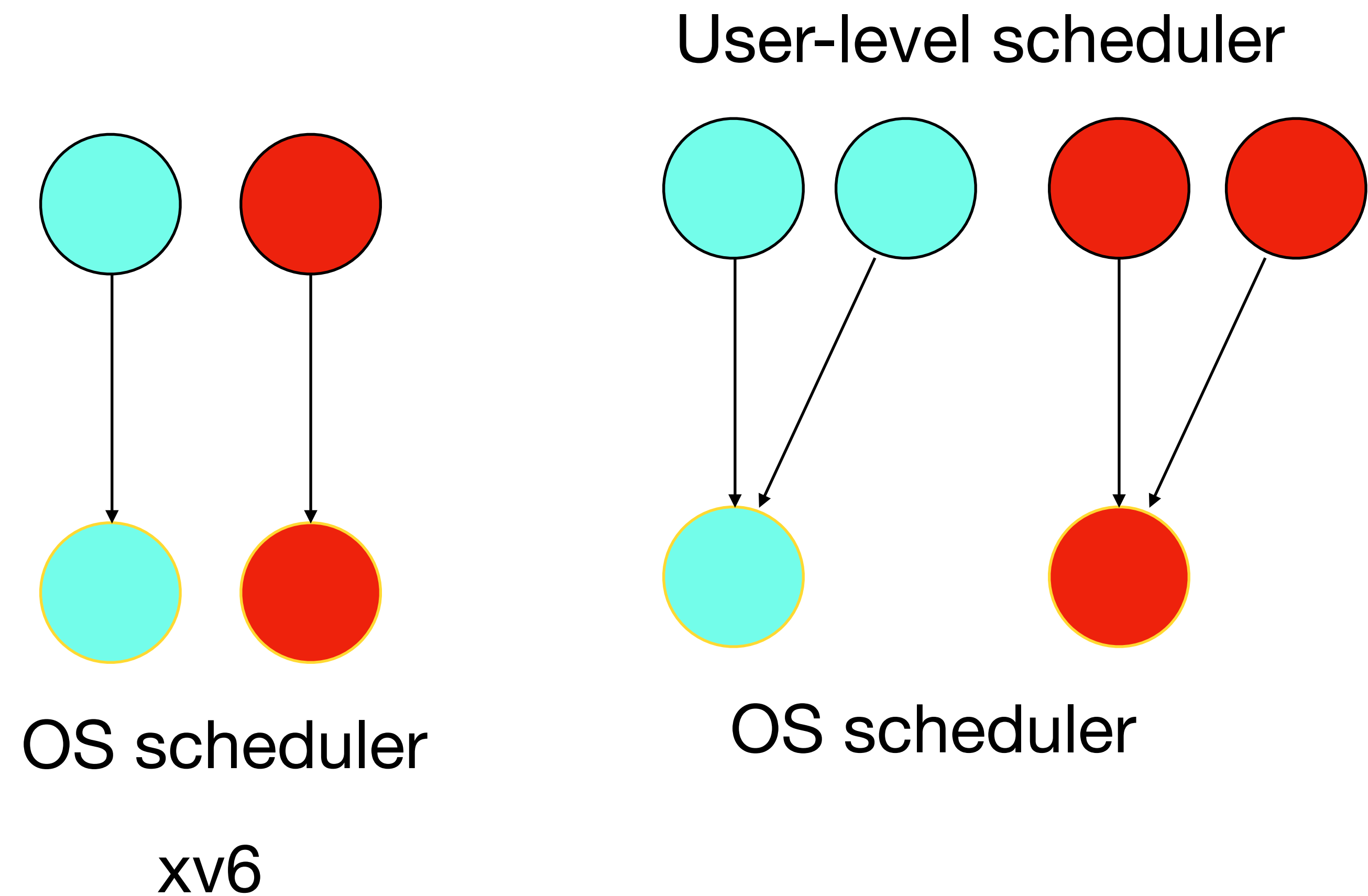
OS scheduler



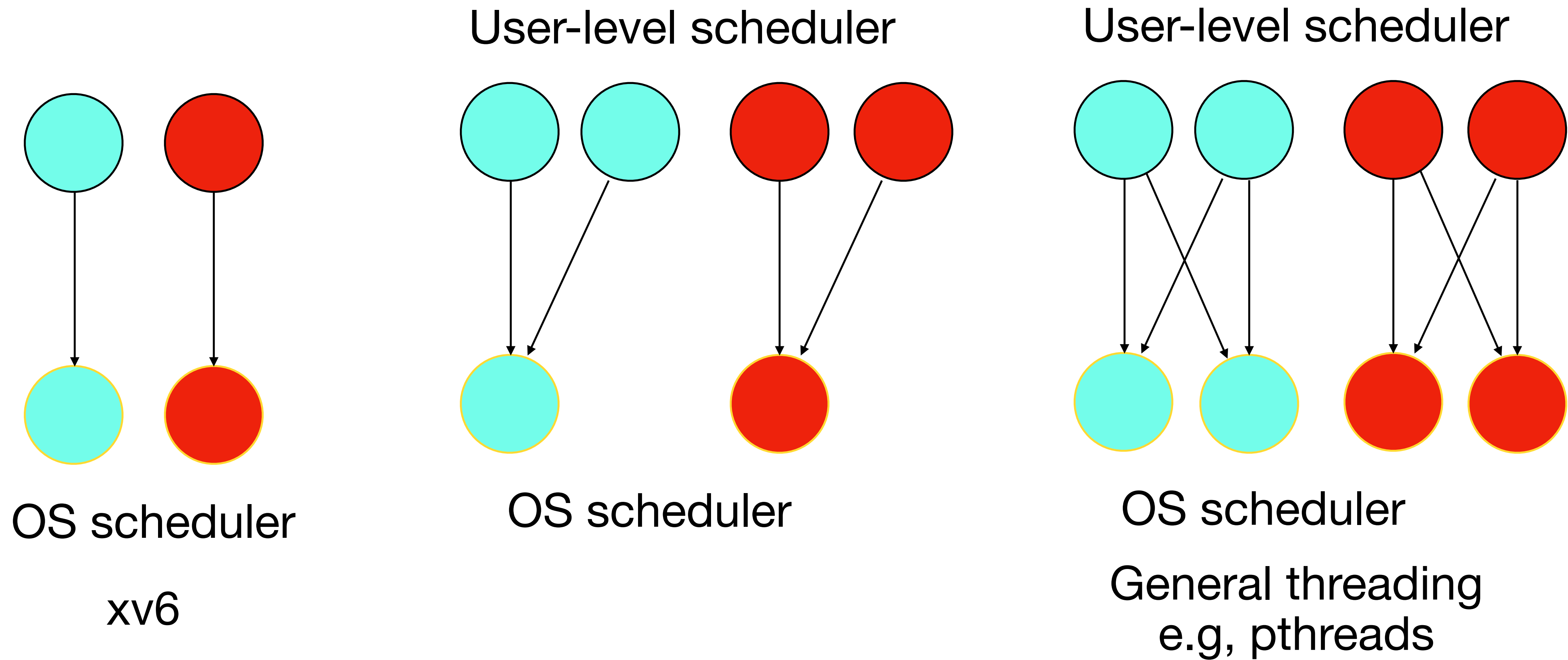
User threads and kernel threads



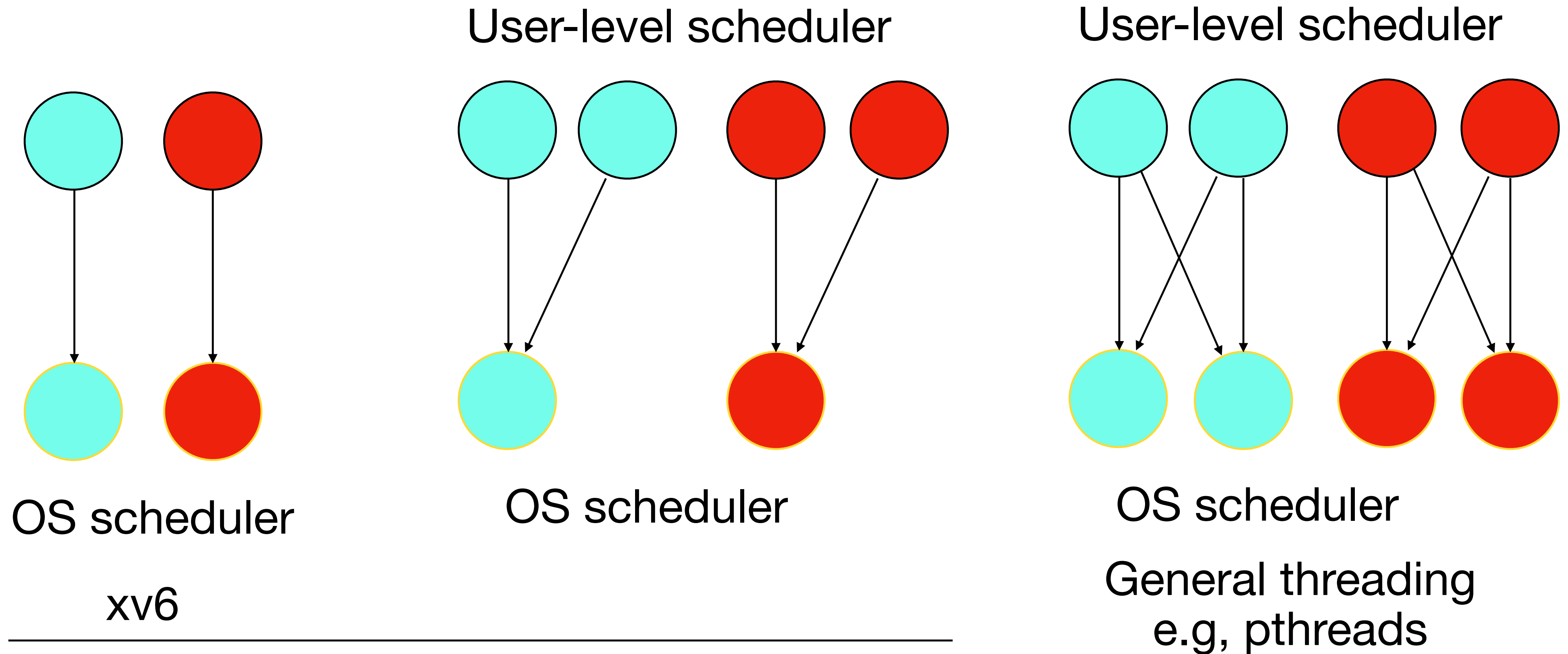
User threads and kernel threads



User threads and kernel threads



User threads and kernel threads



Cannot exploit parallelism within a process: two CPUs
will never run with same address space in user code

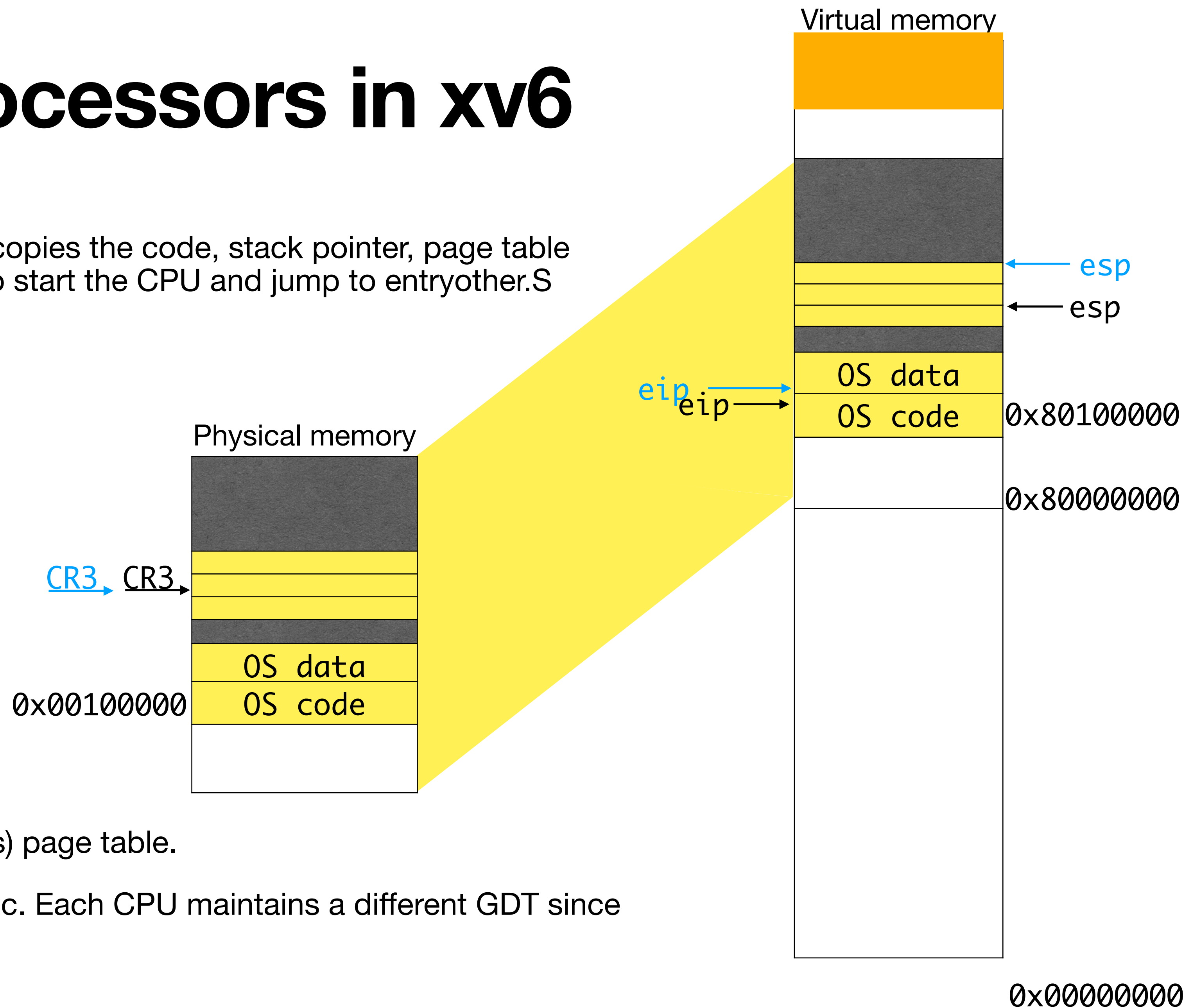
Starting other processors in xv6

startothers:

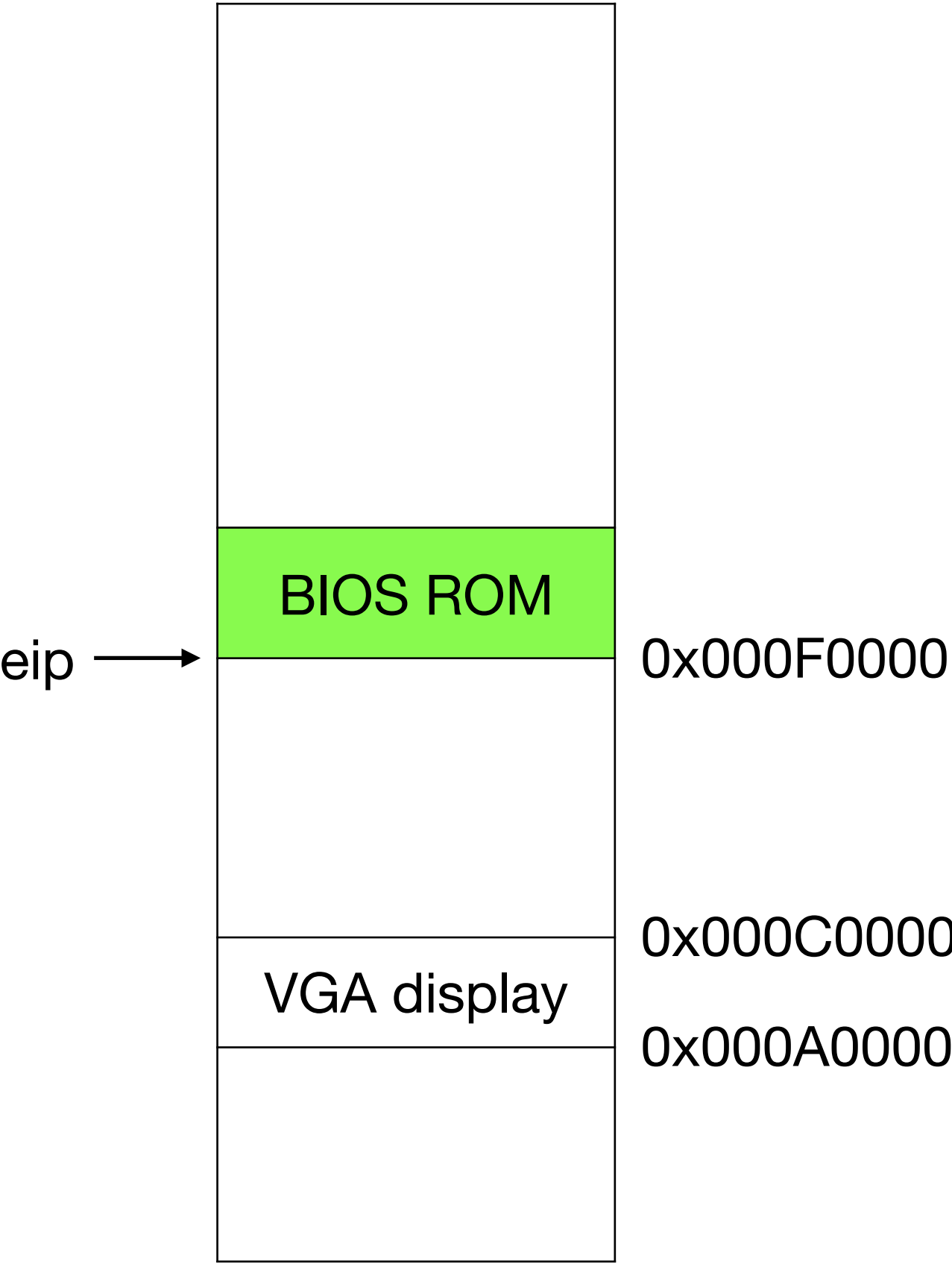
- Allocates a separate stack for the other CPU, copies the code, stack pointer, page table pointer in low 1MB, asks other CPU's LAPIC to start the CPU and jump to entryother.S

entryother.S does what bootloader+entry.S did

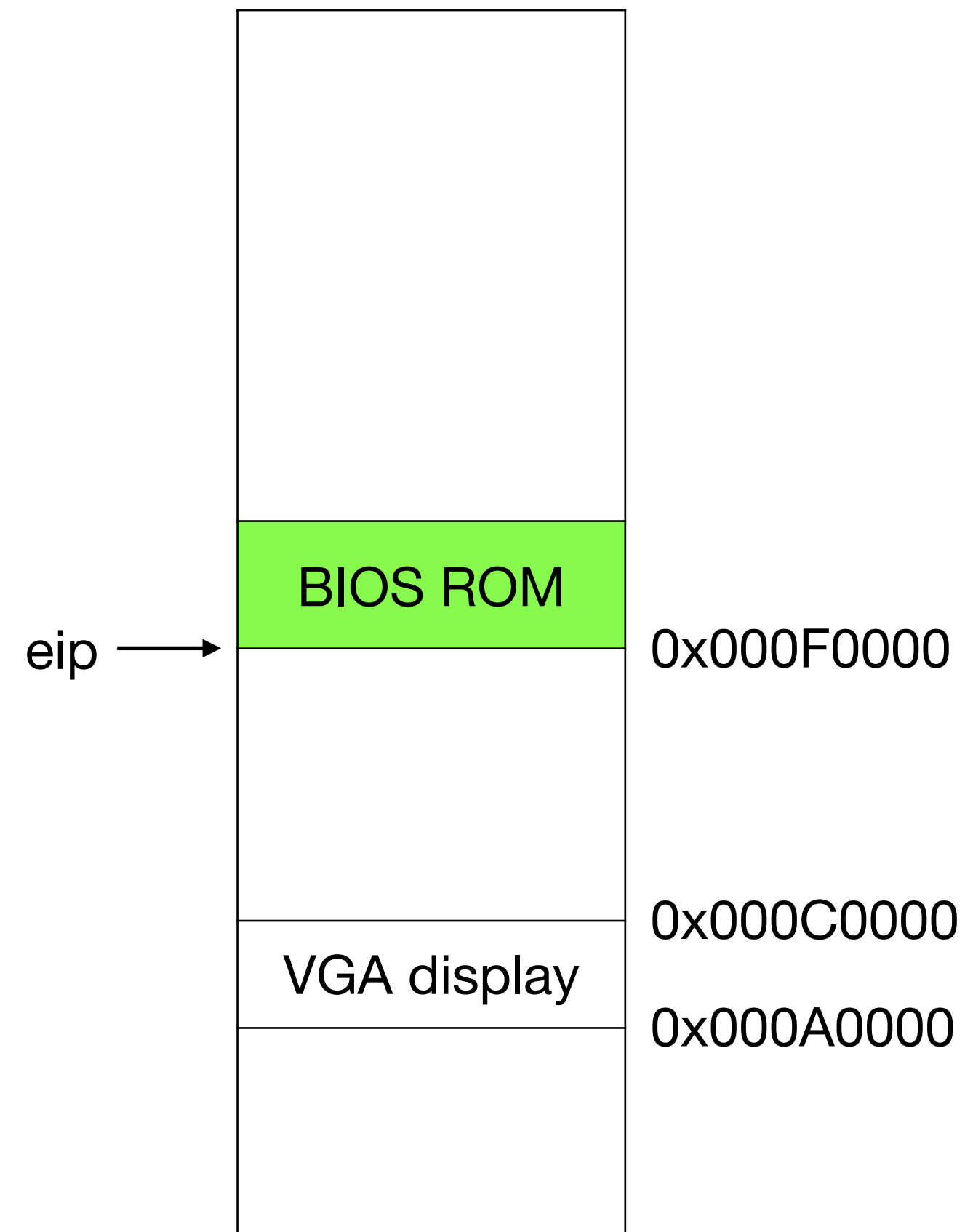
- clears interrupts
- sets up temporary GDT, GDTR
- Switch to 32 bit mode, set segment selectors
- enable paging with temporary page table, sets up stack pointer
- jumps to mpenter
- mpenter does what main did
 - Switch to (OS only in high virtual addresses) page table.
 - Sets up new GDT with KCODE, UCODE, etc. Each CPU maintains a different GDT since each CPU will write TSS in its own GDT
 - Sets up IDTR and jump into scheduler



Boot up sequence (1): BIOS

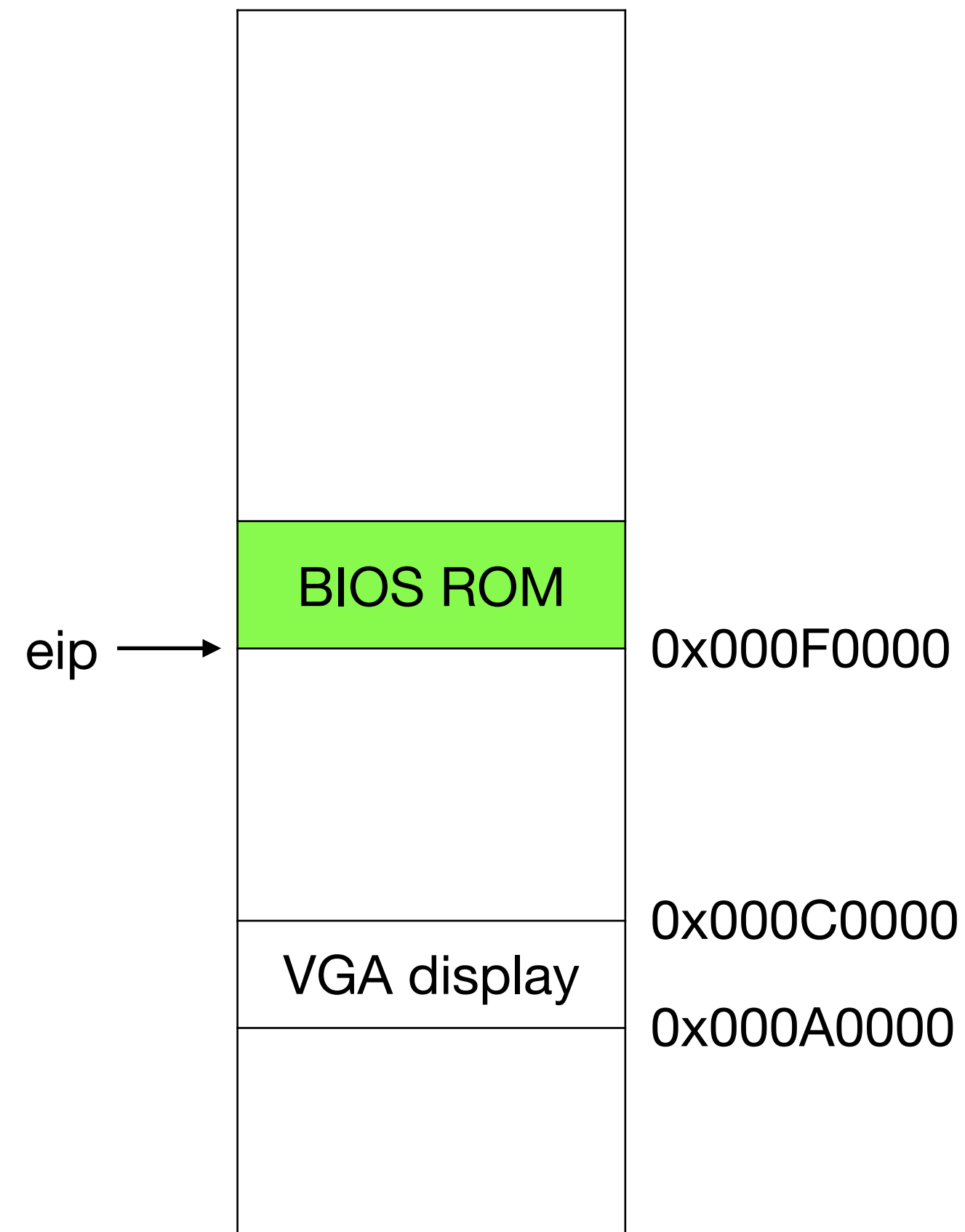


Boot up sequence (1): BIOS



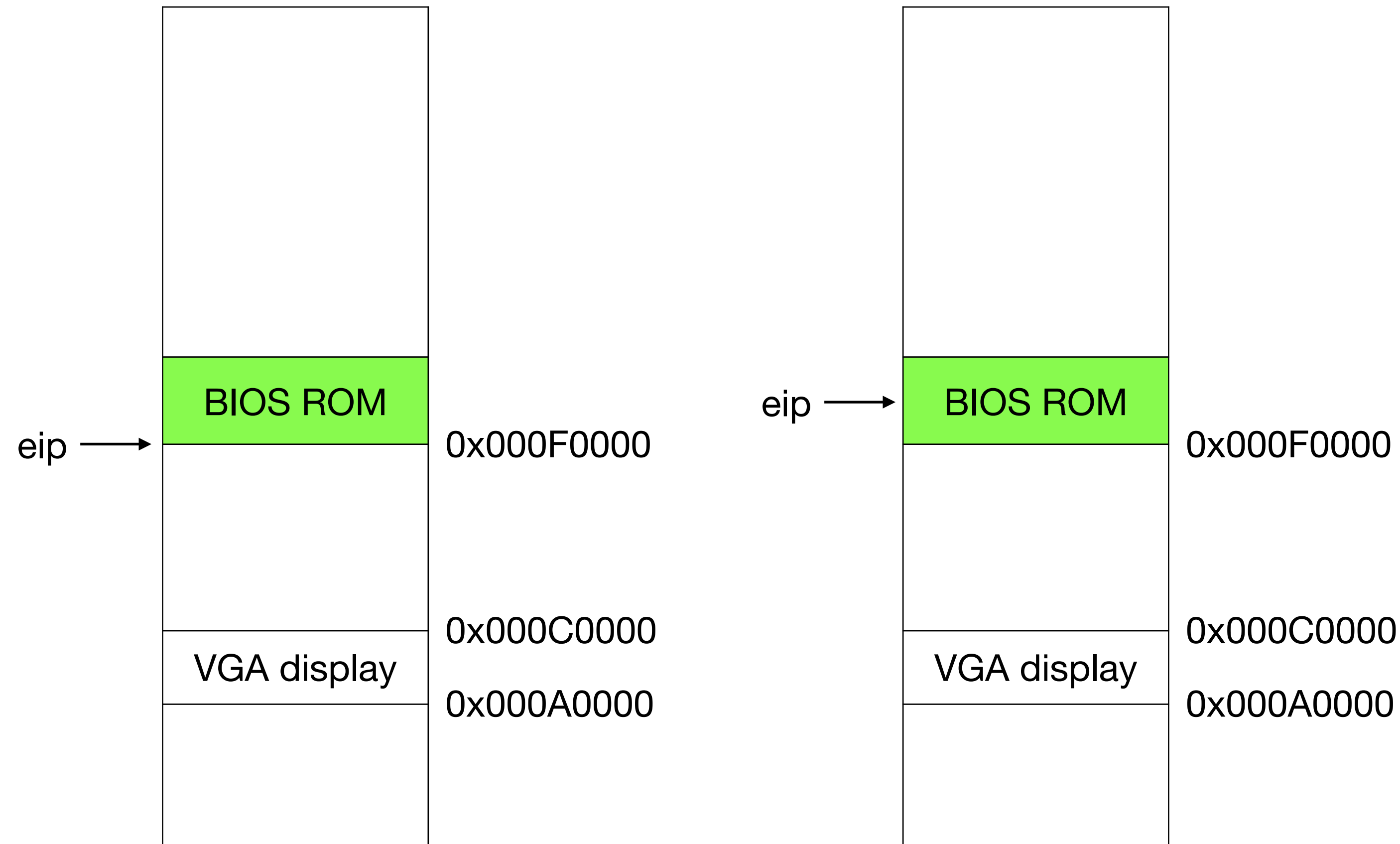
- For backward compatibility, PC boots in 16-bit mode

Boot up sequence (1): BIOS



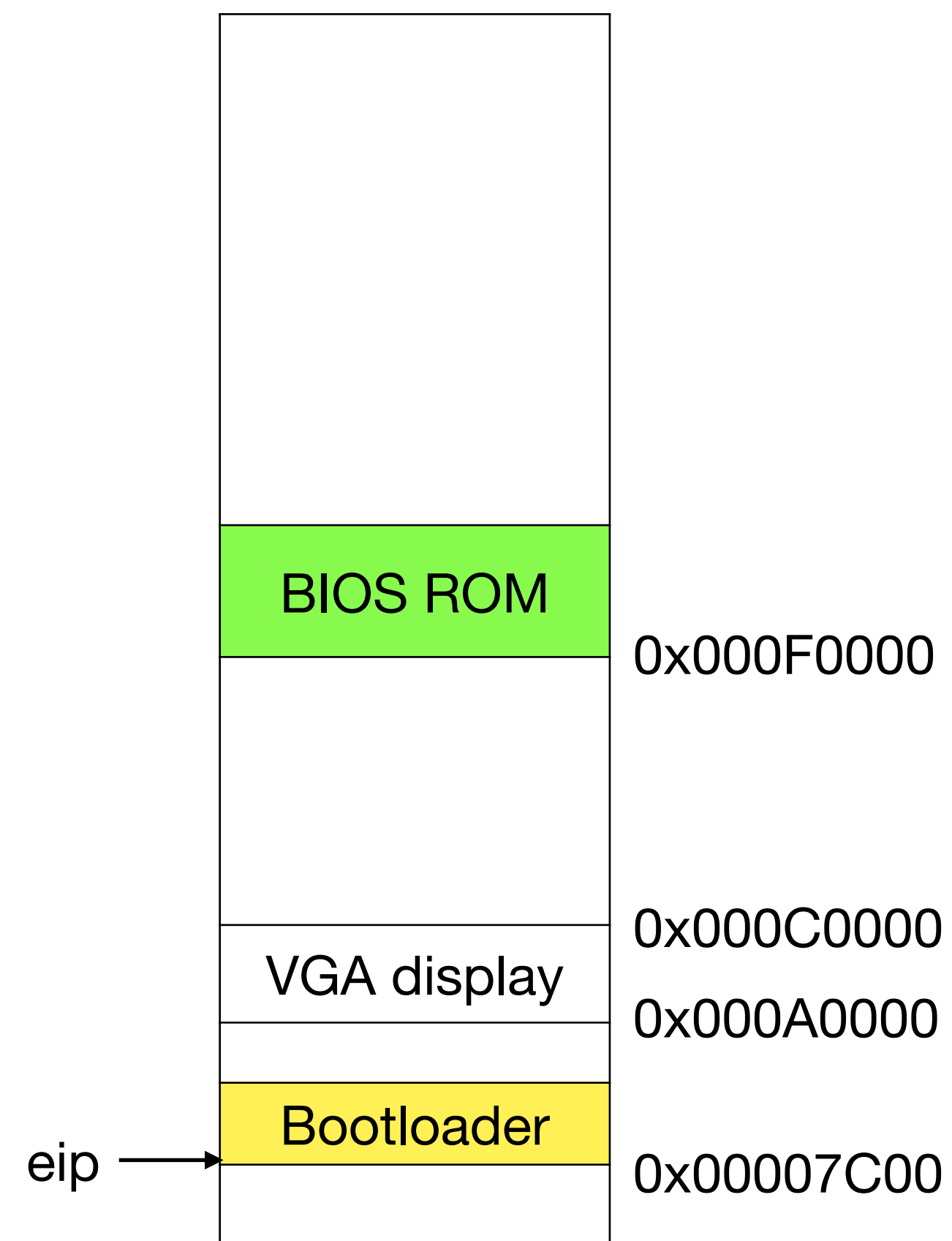
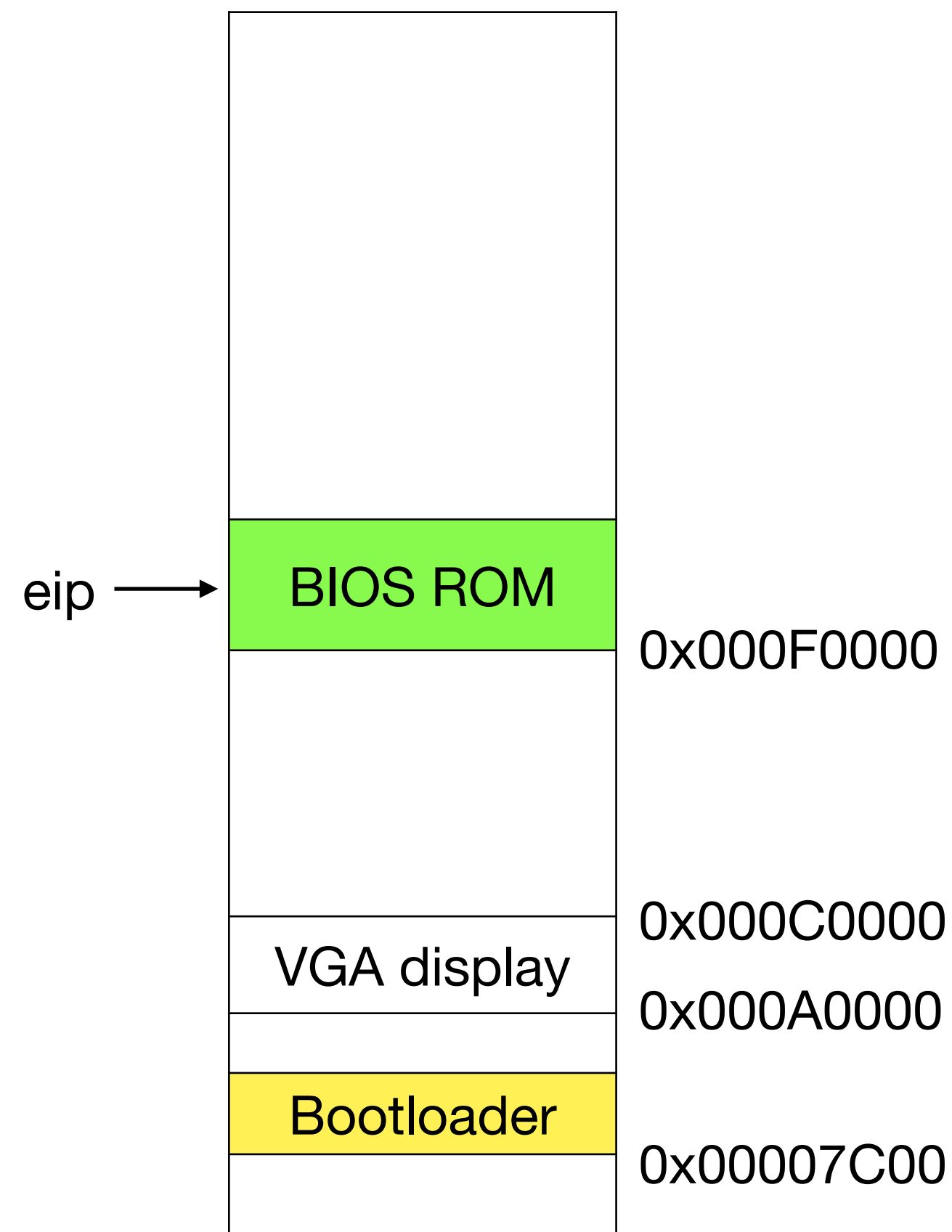
- For backward compatibility, PC boots in 16-bit mode
- BIOS does initial hardware check: are CPUs, memory, disk functional?

Boot up sequence (1): BIOS



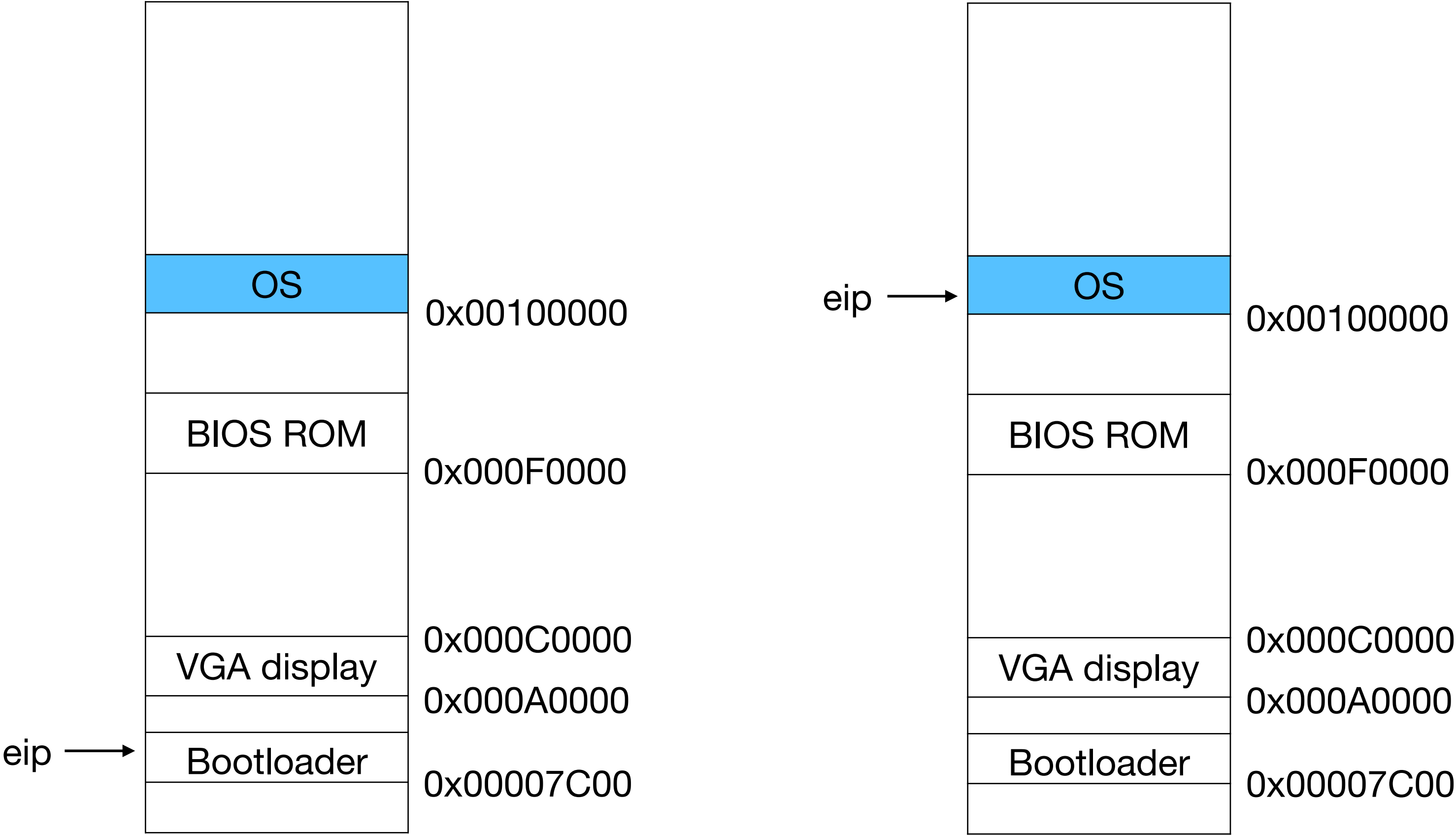
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Boot up sequence (2): Bootloader

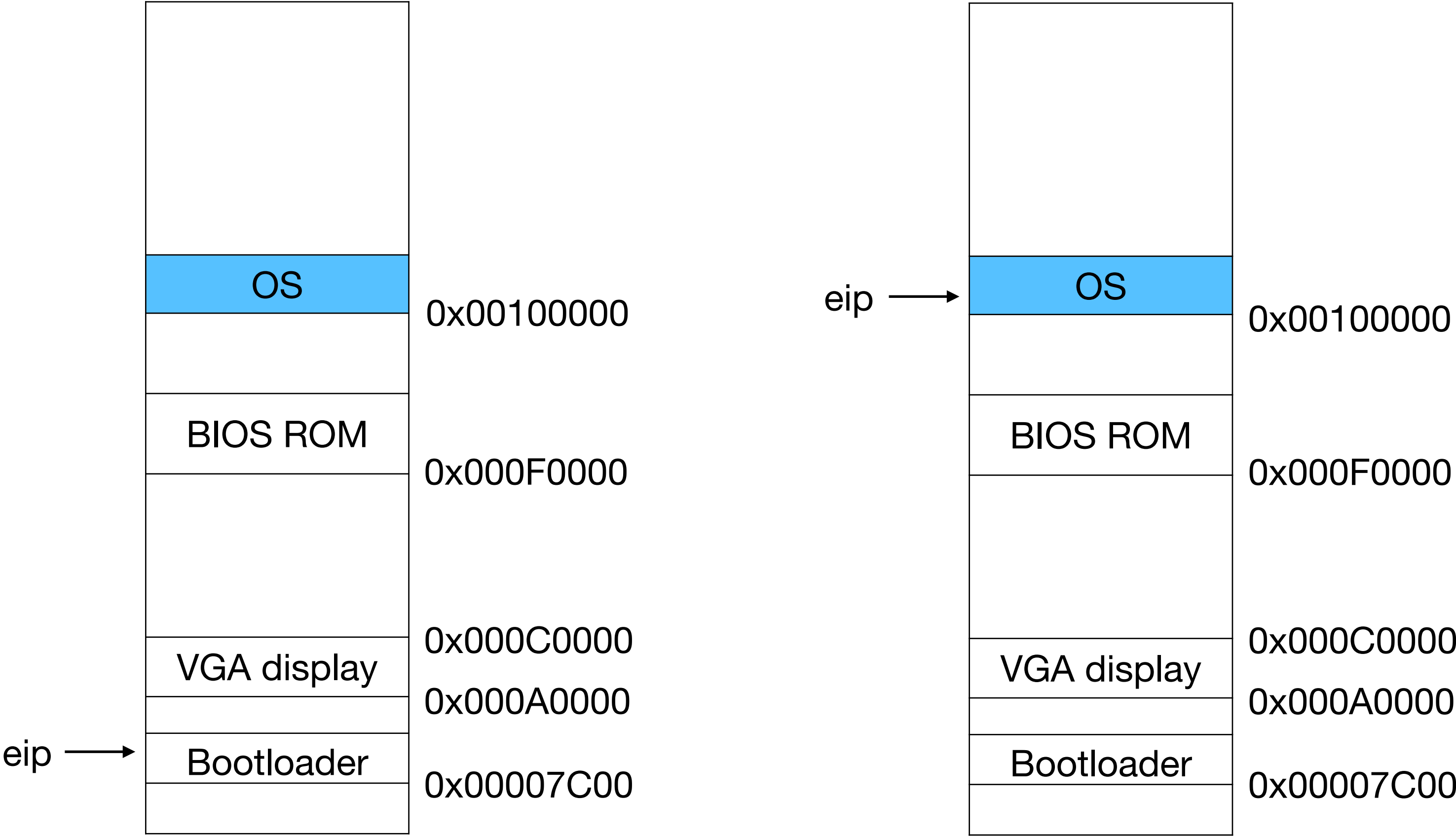


- BIOS loads first disk sector (512 bytes) at 0x7c00 and gives control
 - We need to write a boot loader that fits in the first sector of disk
 - We need to tell gcc that boot loader will load at 0x7c00 (virtual memory to physical memory translation is not setup yet!)
- Boot loader changes to 32 bit mode

Boot up sequence (3): OS

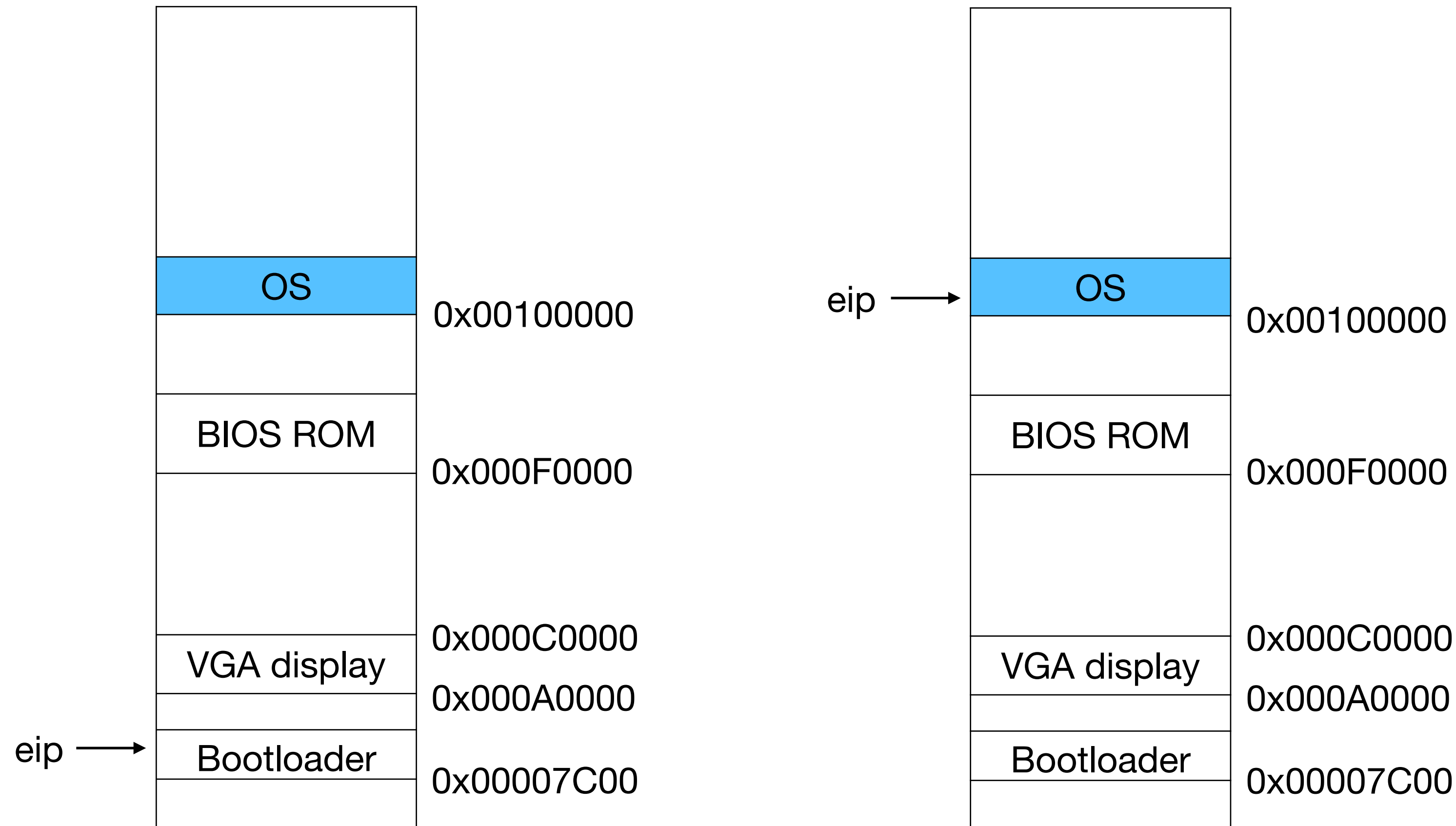


Boot up sequence (3): OS



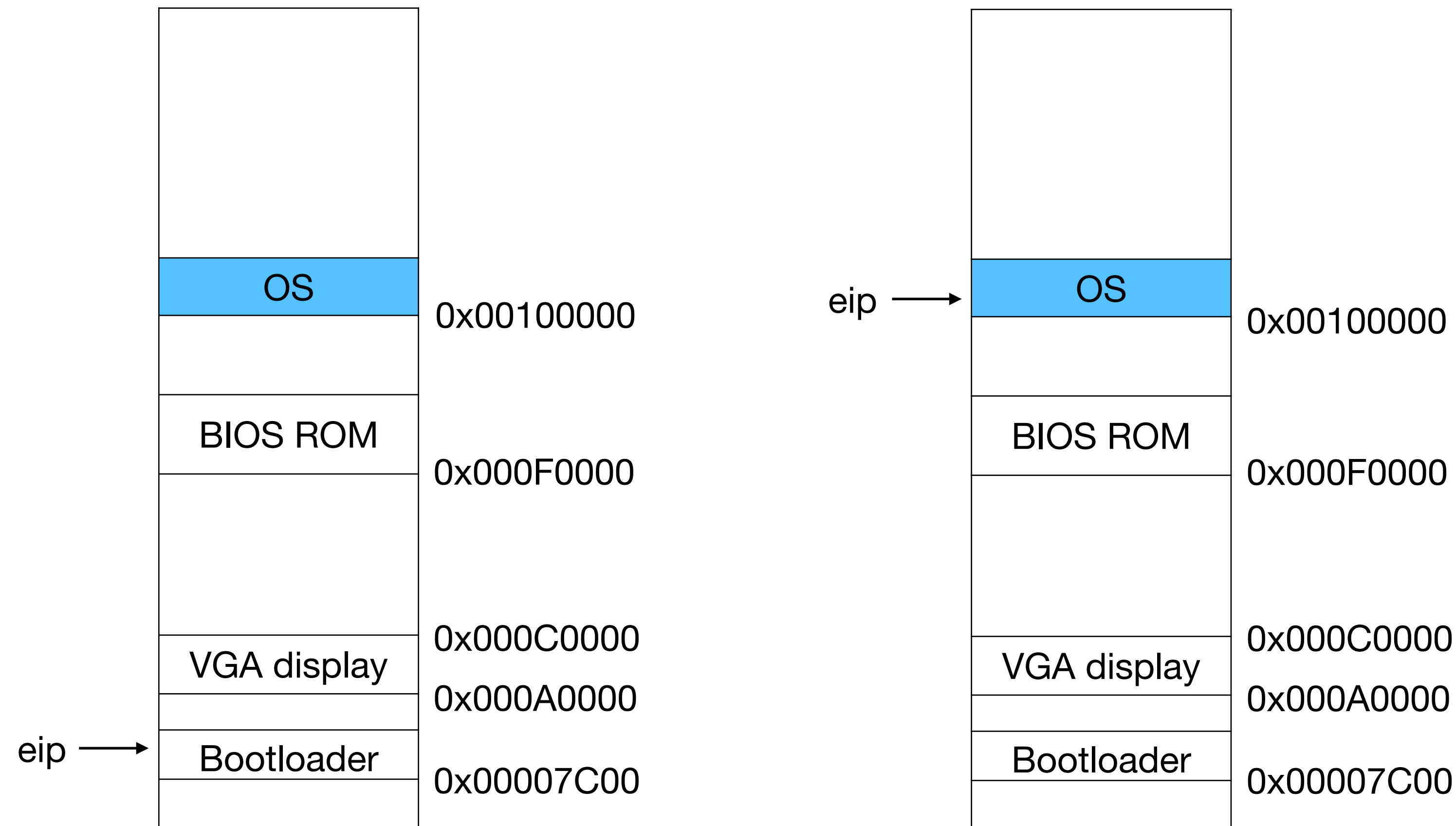
- gcc prepares OS image in Executable and Linkable Format (ELF)

Boot up sequence (3): OS



- gcc prepares OS image in Executable and Linkable Format (ELF)
- Bootloader copies OS image starting from disk sector 1 to 0x00100000 and transfers control to it

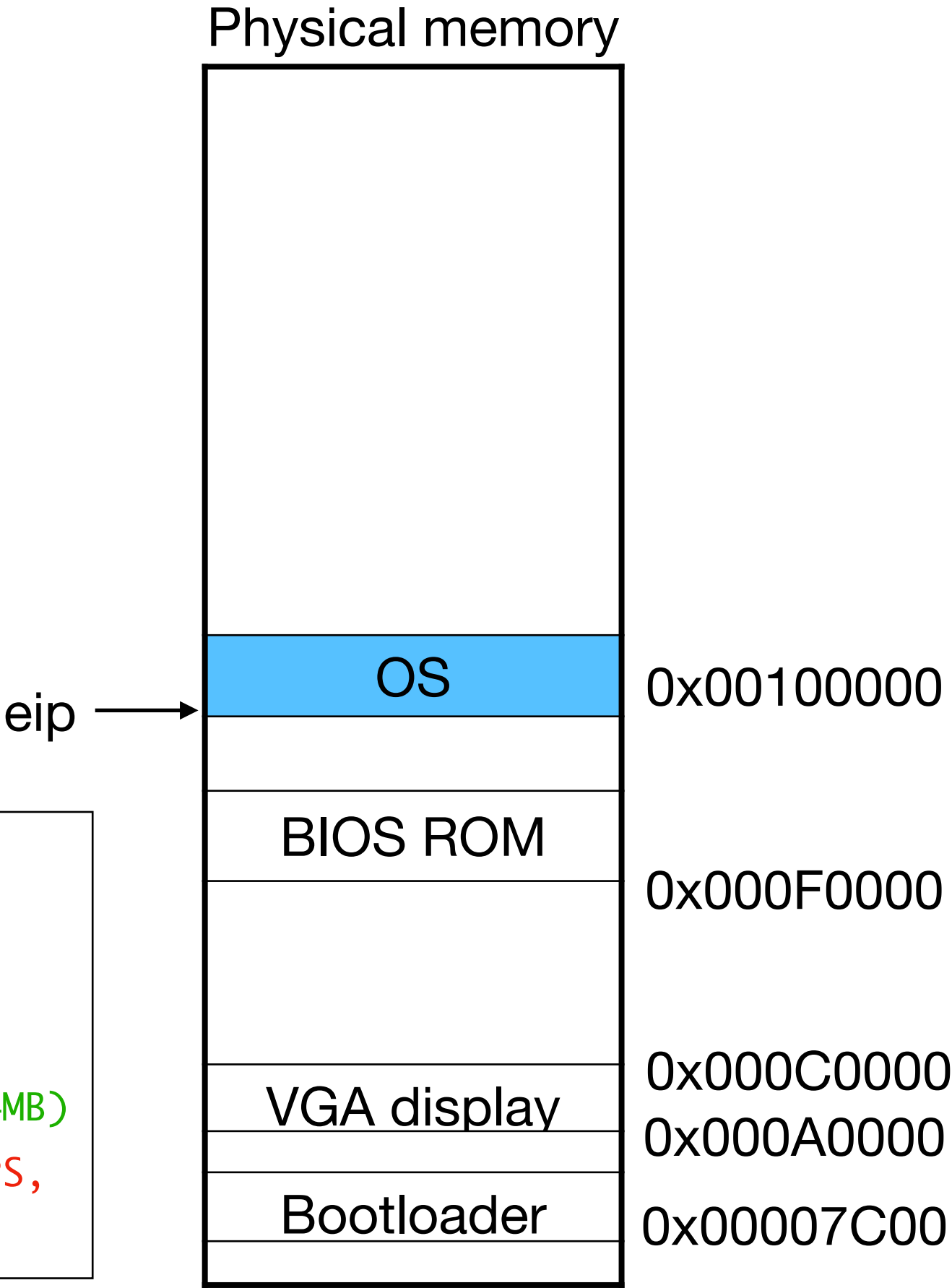
Boot up sequence (3): OS



- gcc prepares OS image in Executable and Linkable Format (ELF)
- Bootloader copies OS image starting from disk sector 1 to 0x00100000 and transfers control to it
- We need to tell gcc that OS image will be loaded at physical address 0x00100000

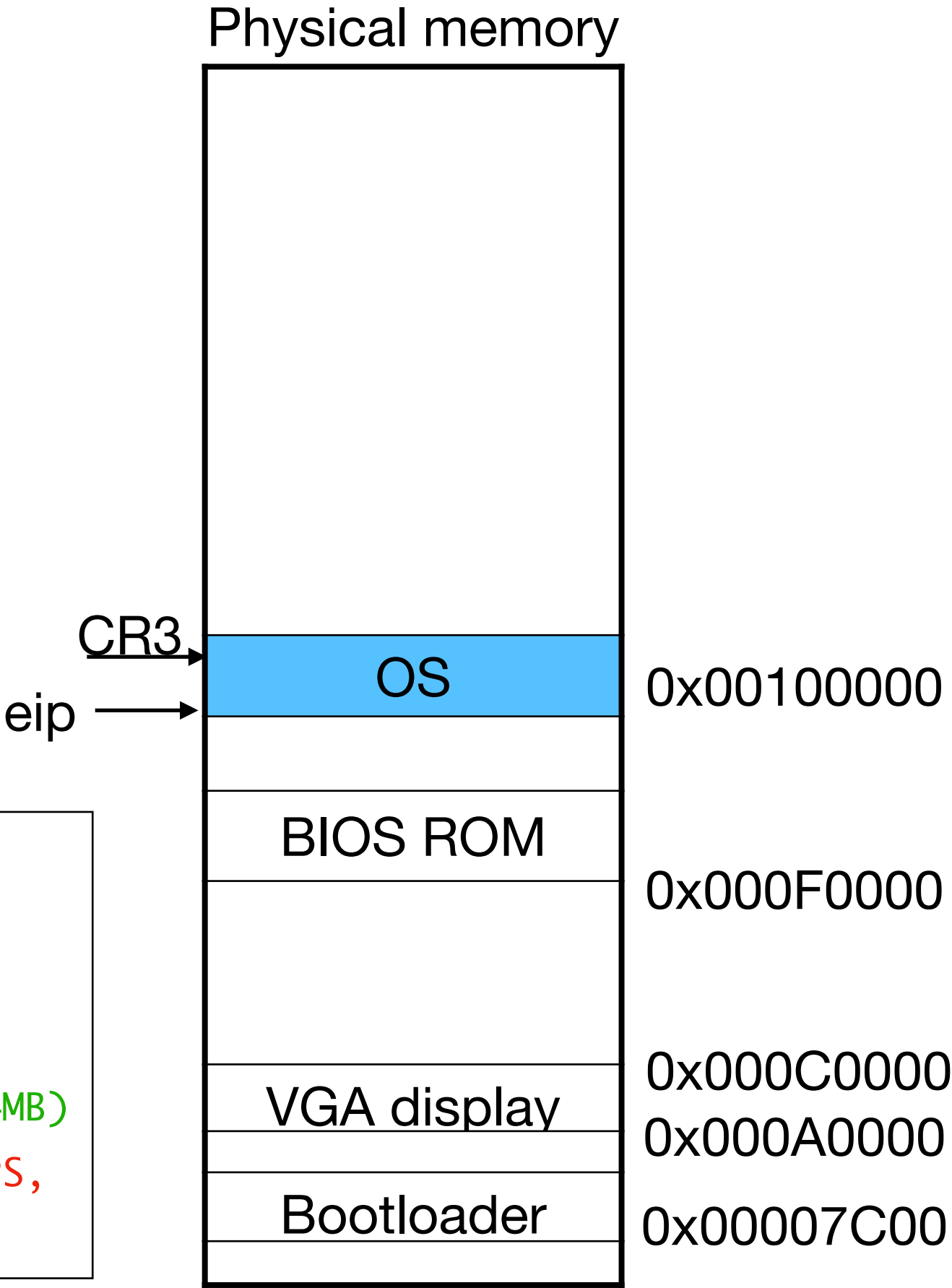
Boot up sequence (4): OS sets up initial page table

```
__attribute__((__aligned__(PGSIZE)))
pde_t entrypgdir[NPDENTRIES] = {
    // Map VA's [0, 4MB) to PA's [0, 4MB)
    [0] = (0) | PTE_P | PTE_W | PTE_PS,
    // Map VA's [KERNBASE, KERNBASE+4MB) to PA's [0, 4MB)
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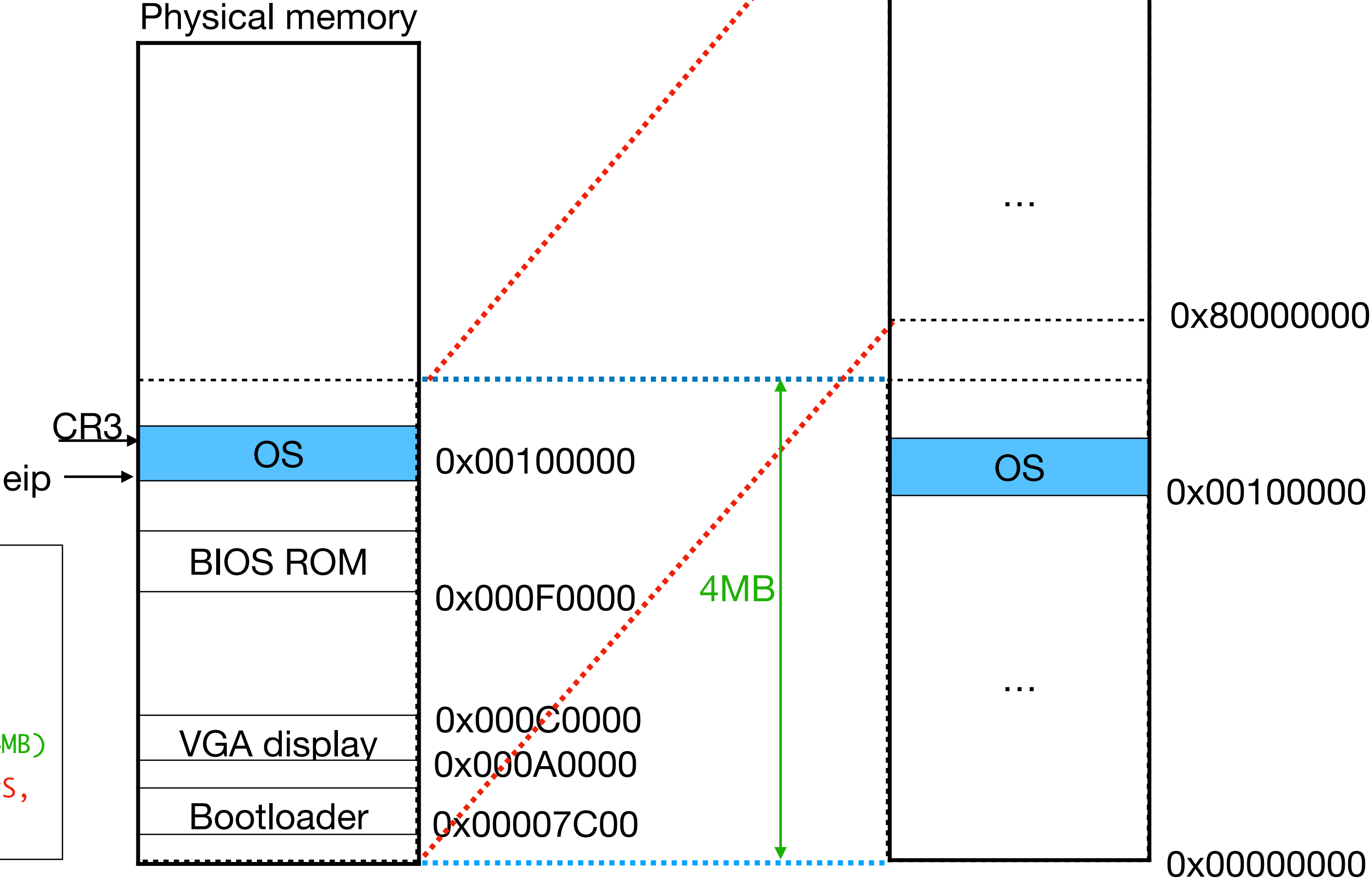
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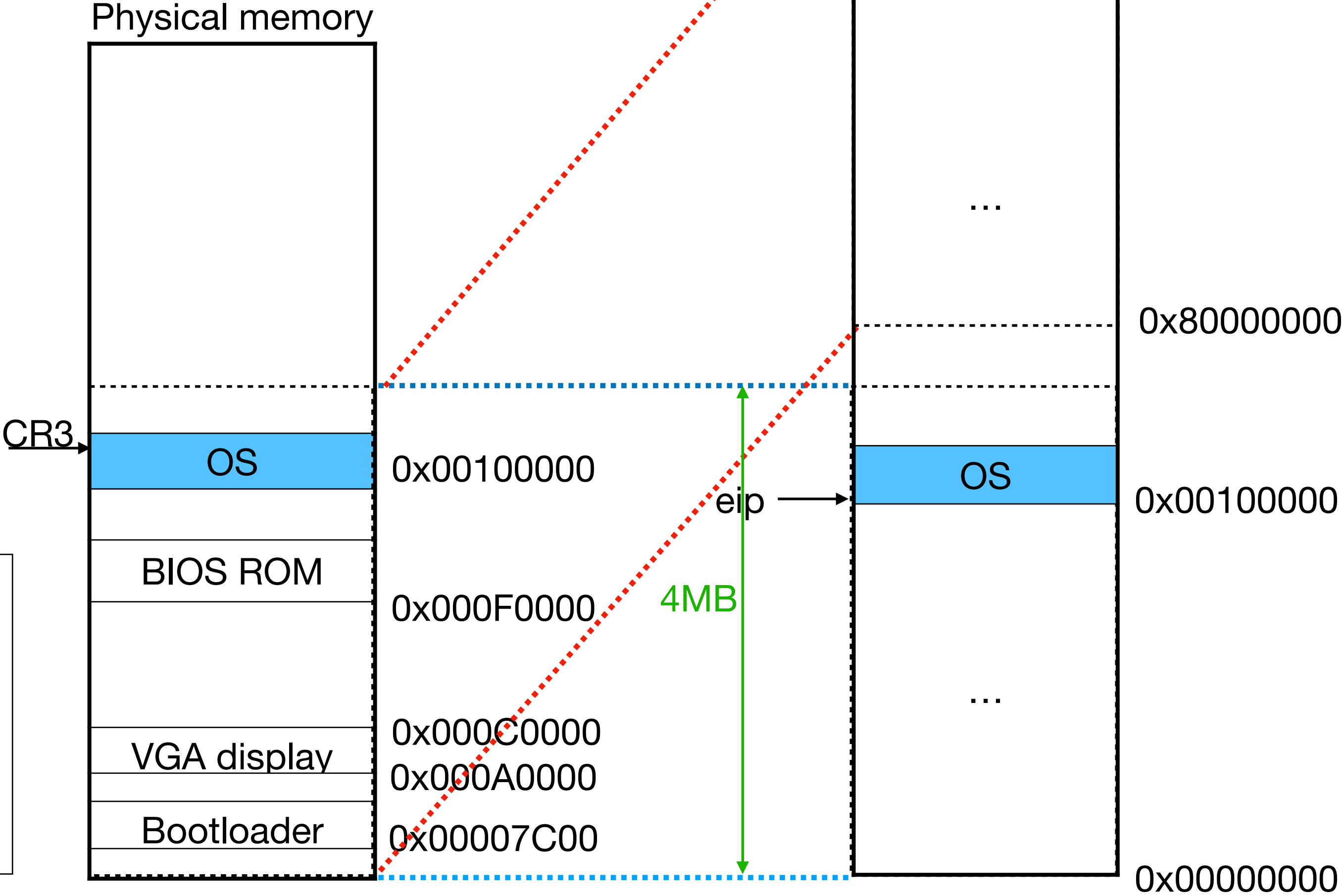
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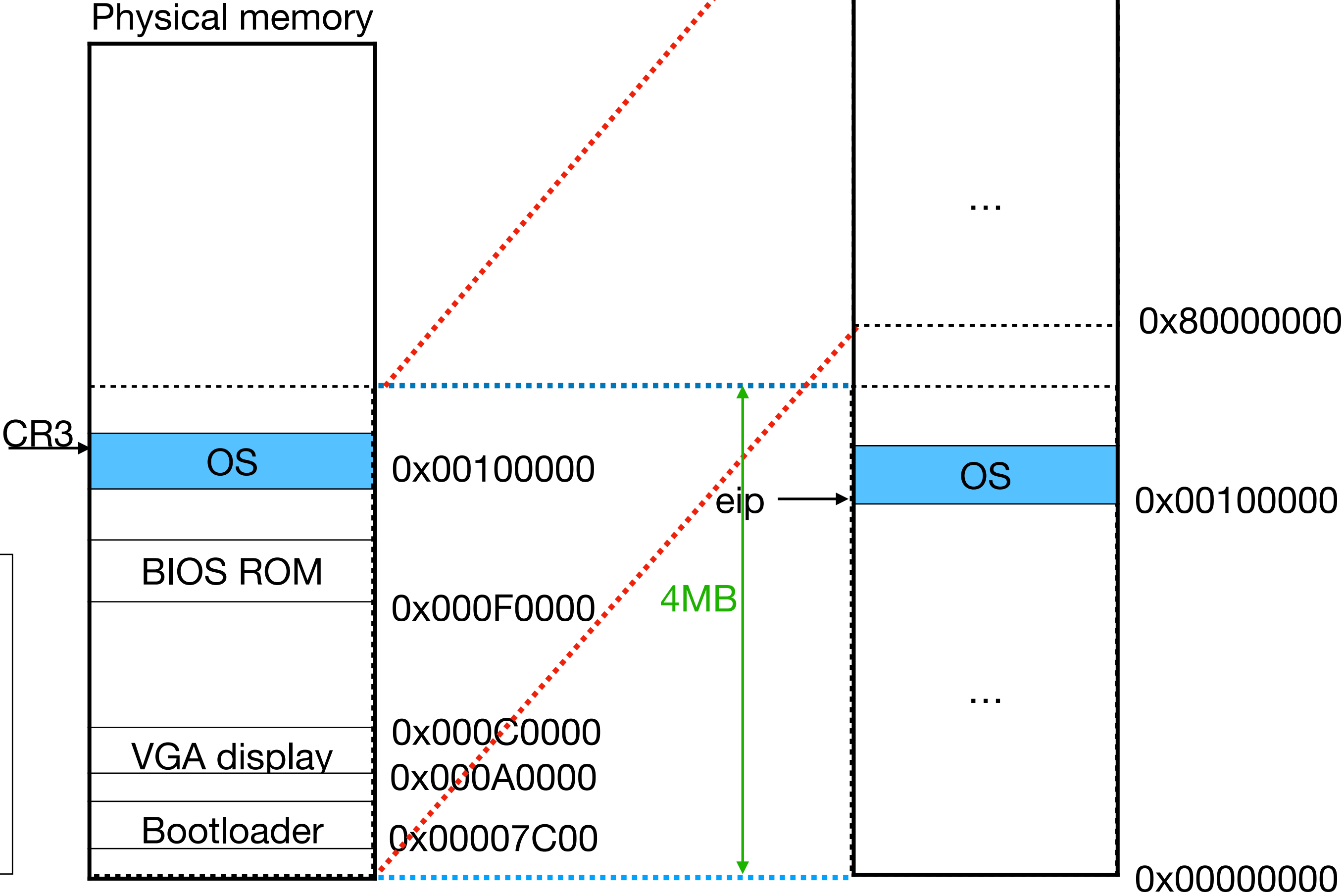
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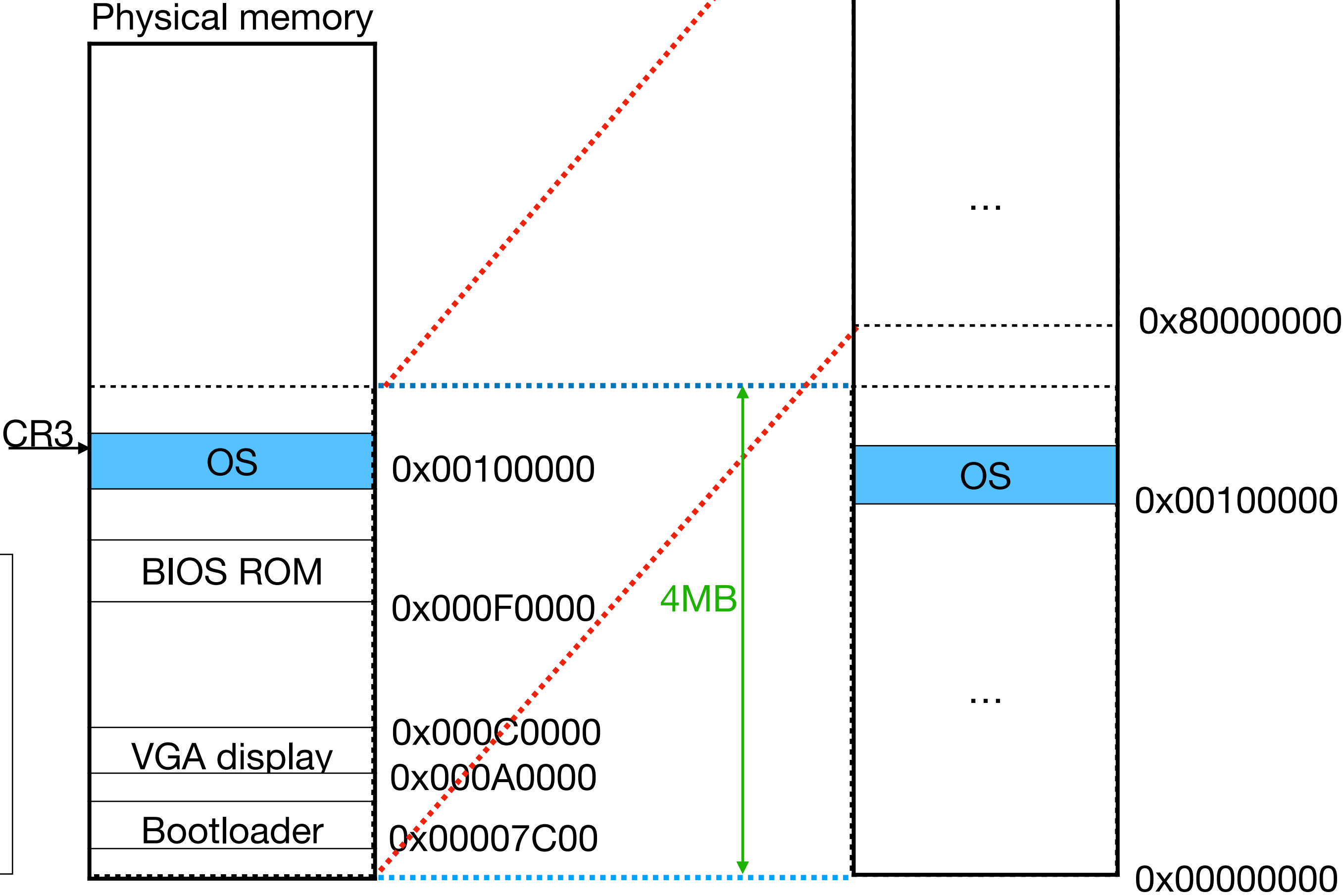
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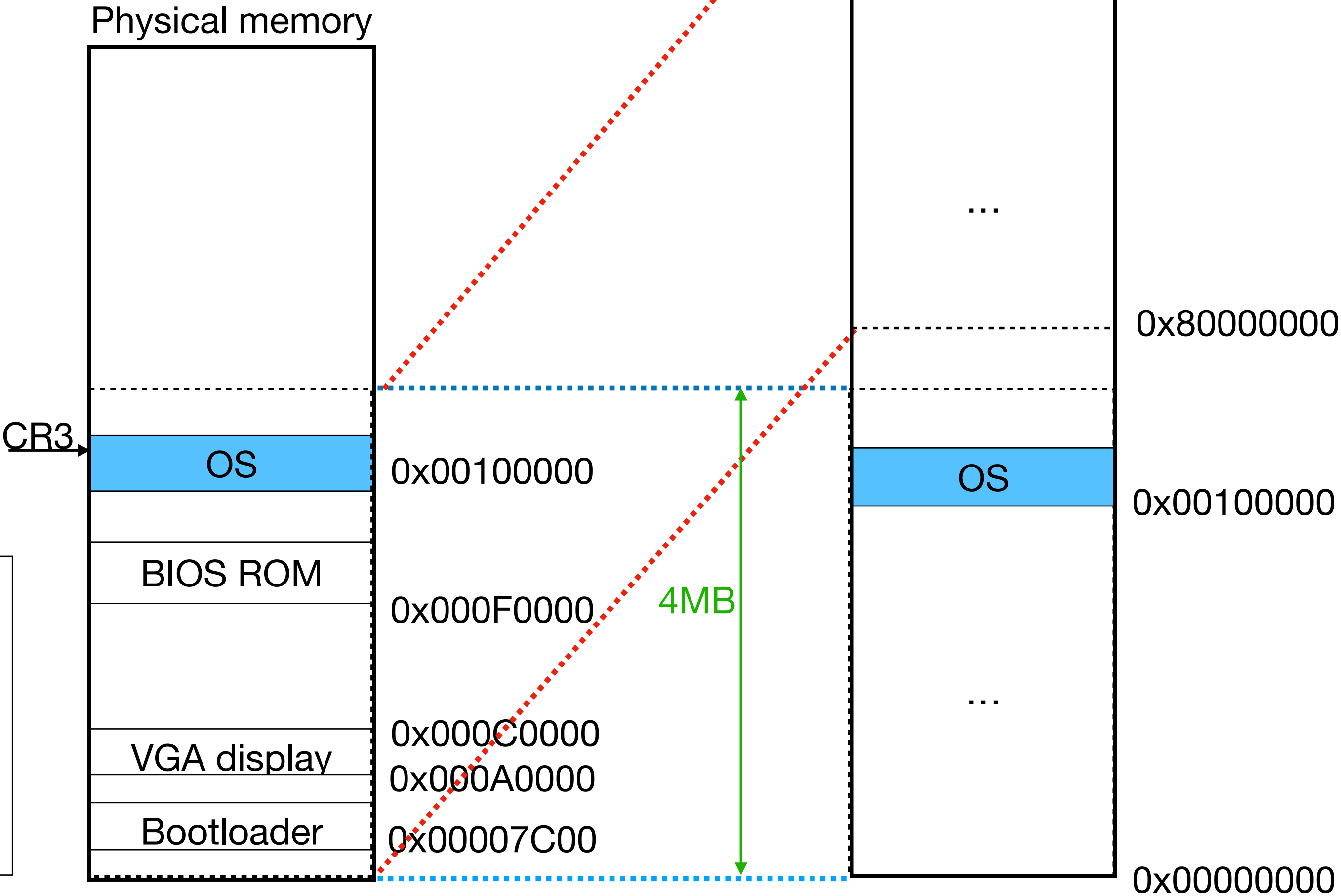


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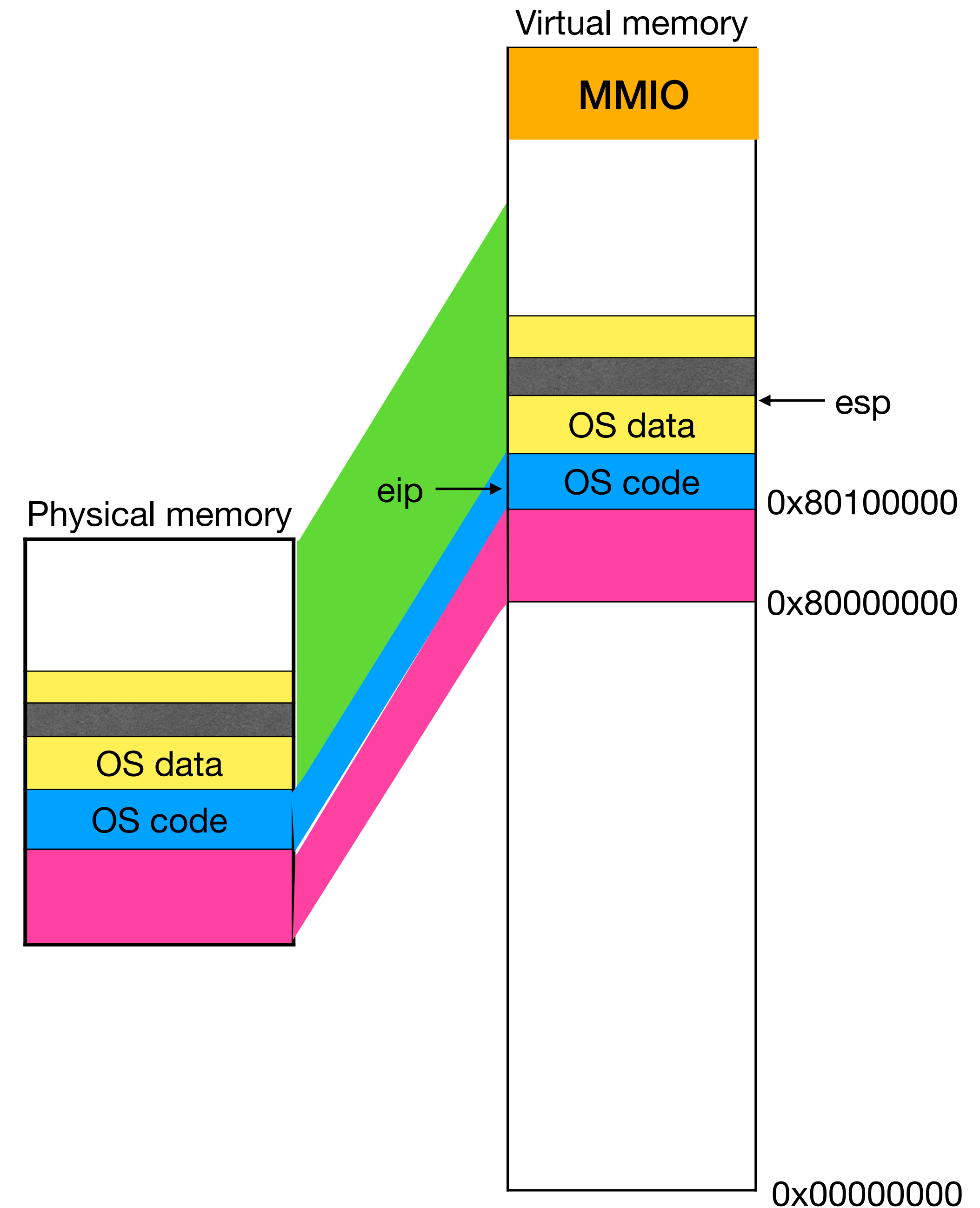
- Jump from physical addresses to virtual addresses

CR3→

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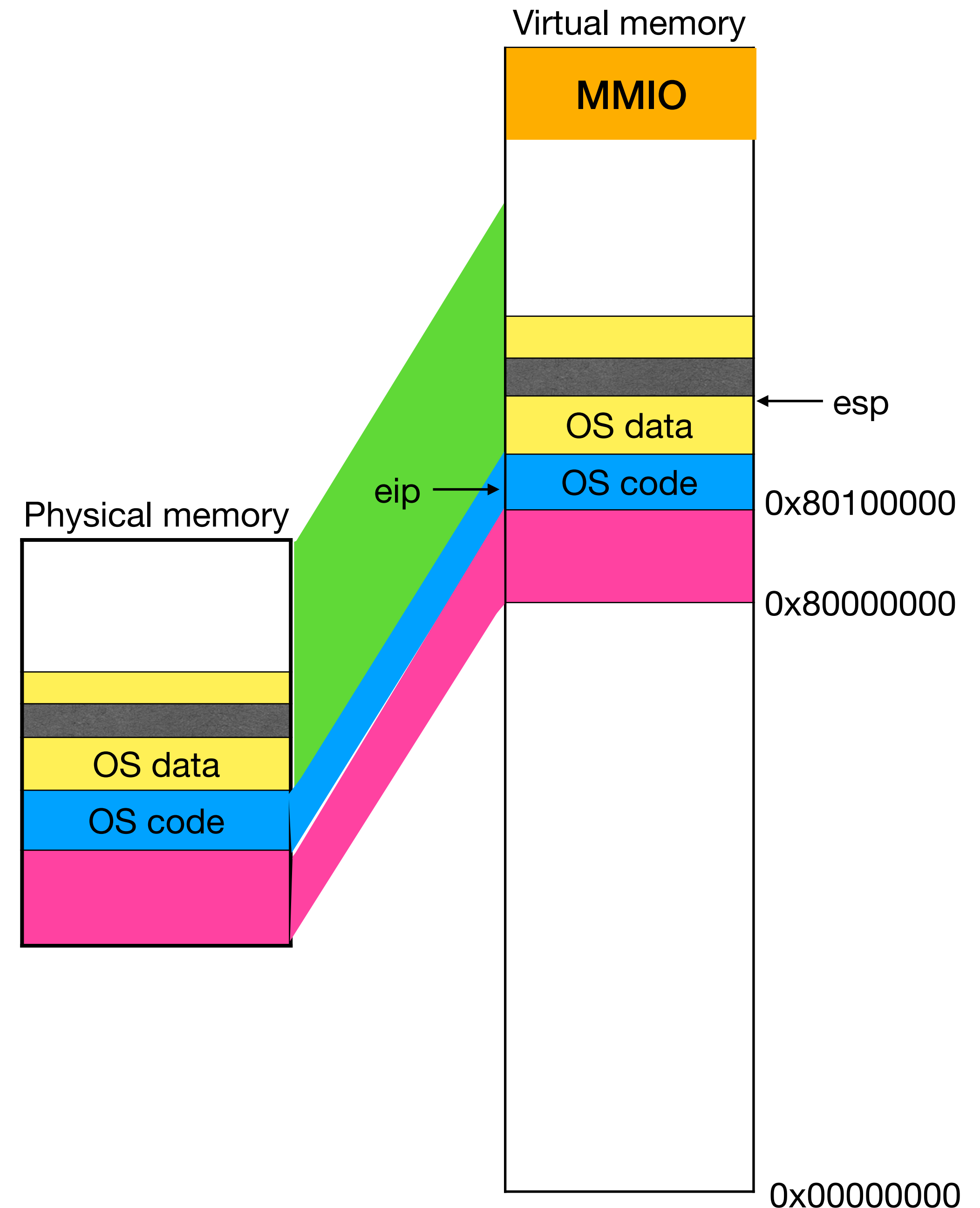


Boot up sequence (5): OS sets up a new page table



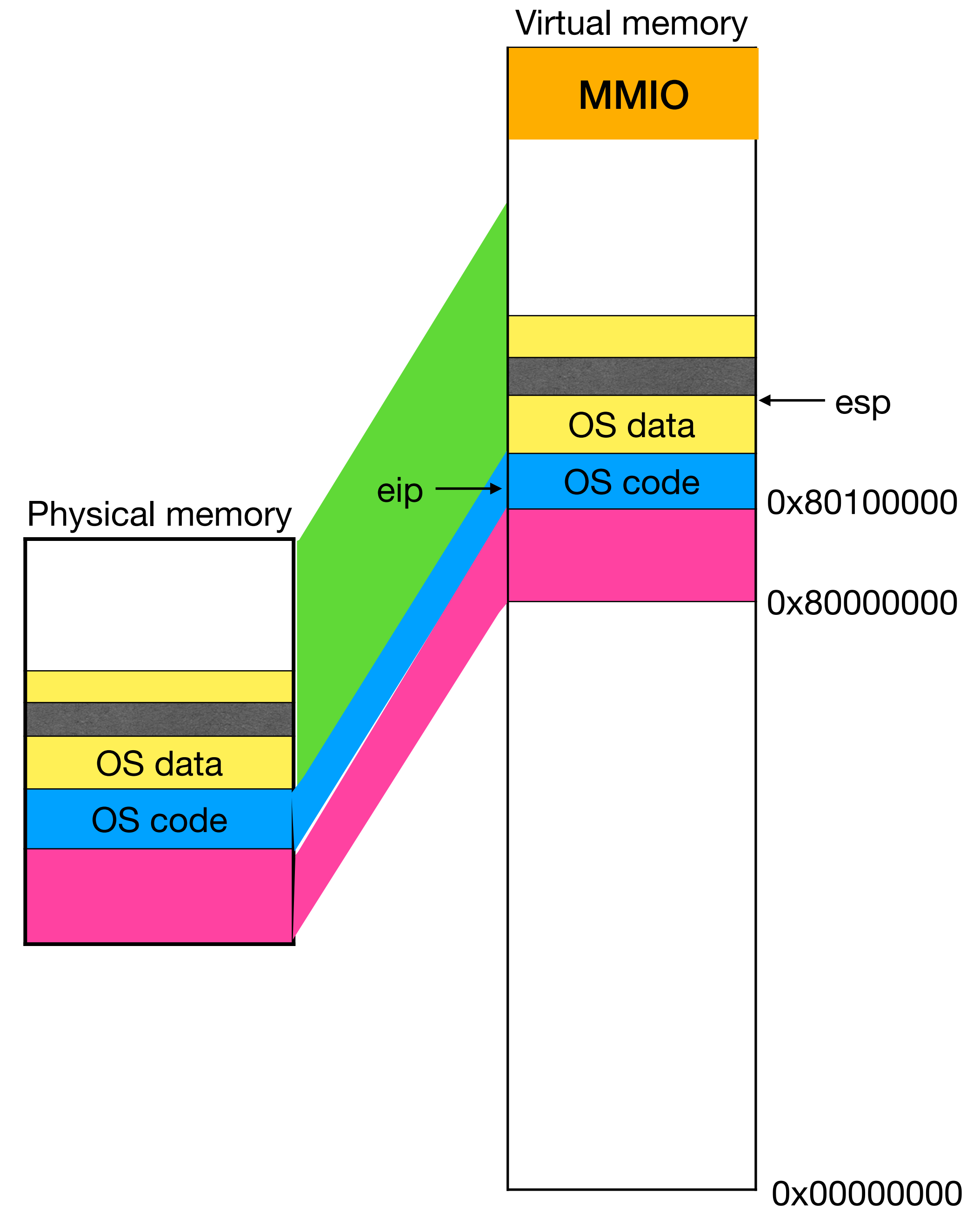
Boot up sequence (5): OS sets up a new page table

- Mark code as read-only



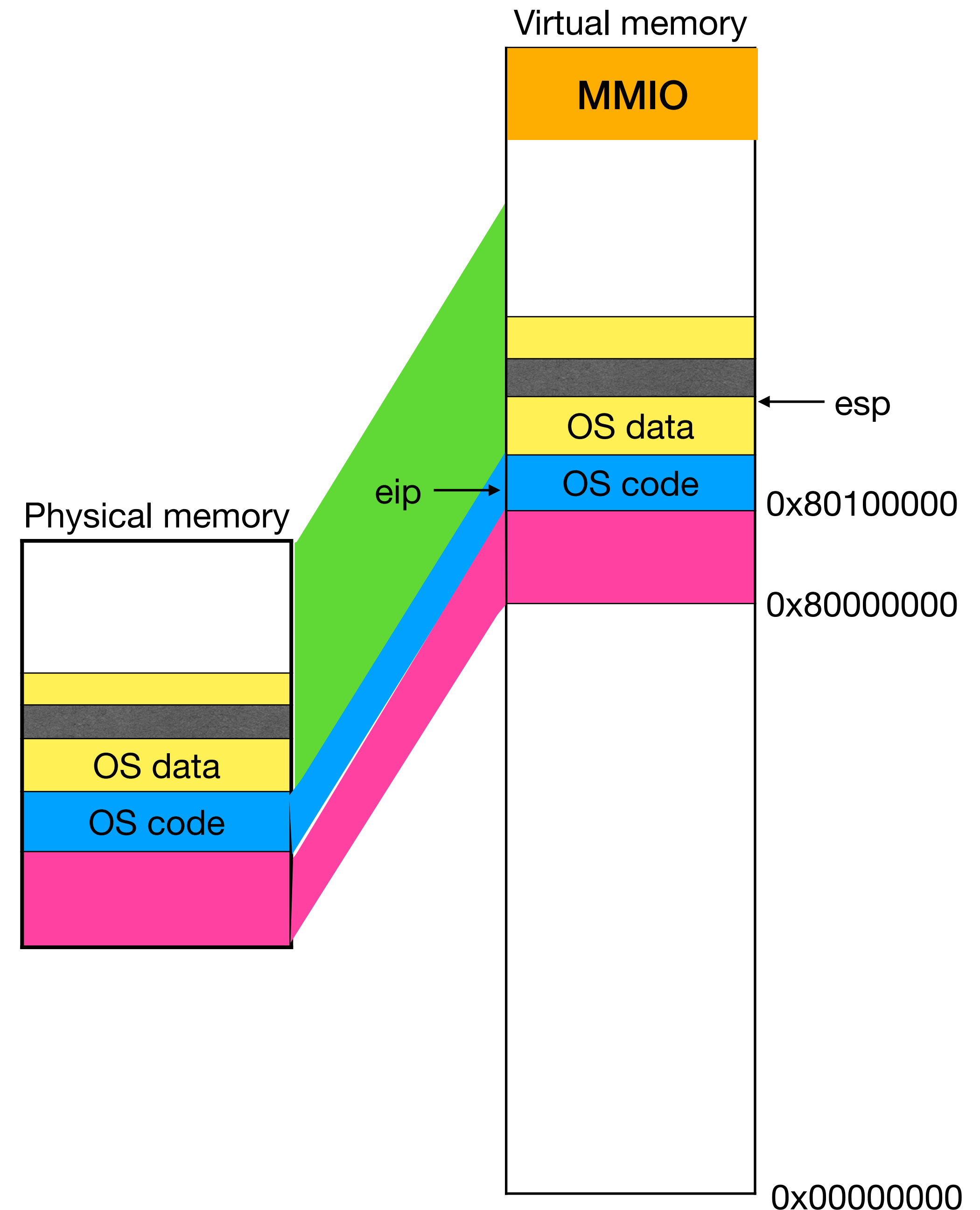
Boot up sequence (5): OS sets up a new page table

- Mark code as read-only
- Map entire physical memory



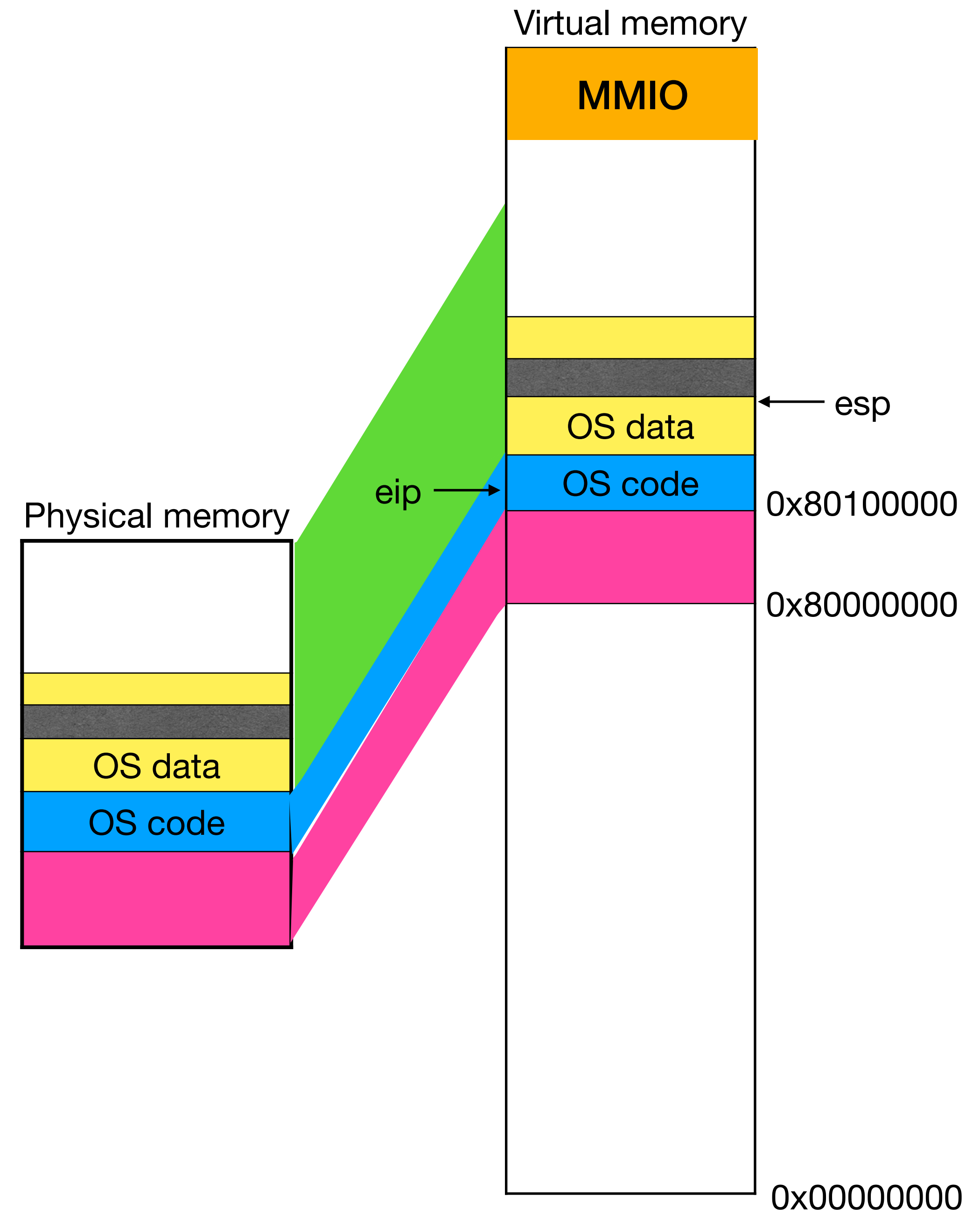
Boot up sequence (5): OS sets up a new page table

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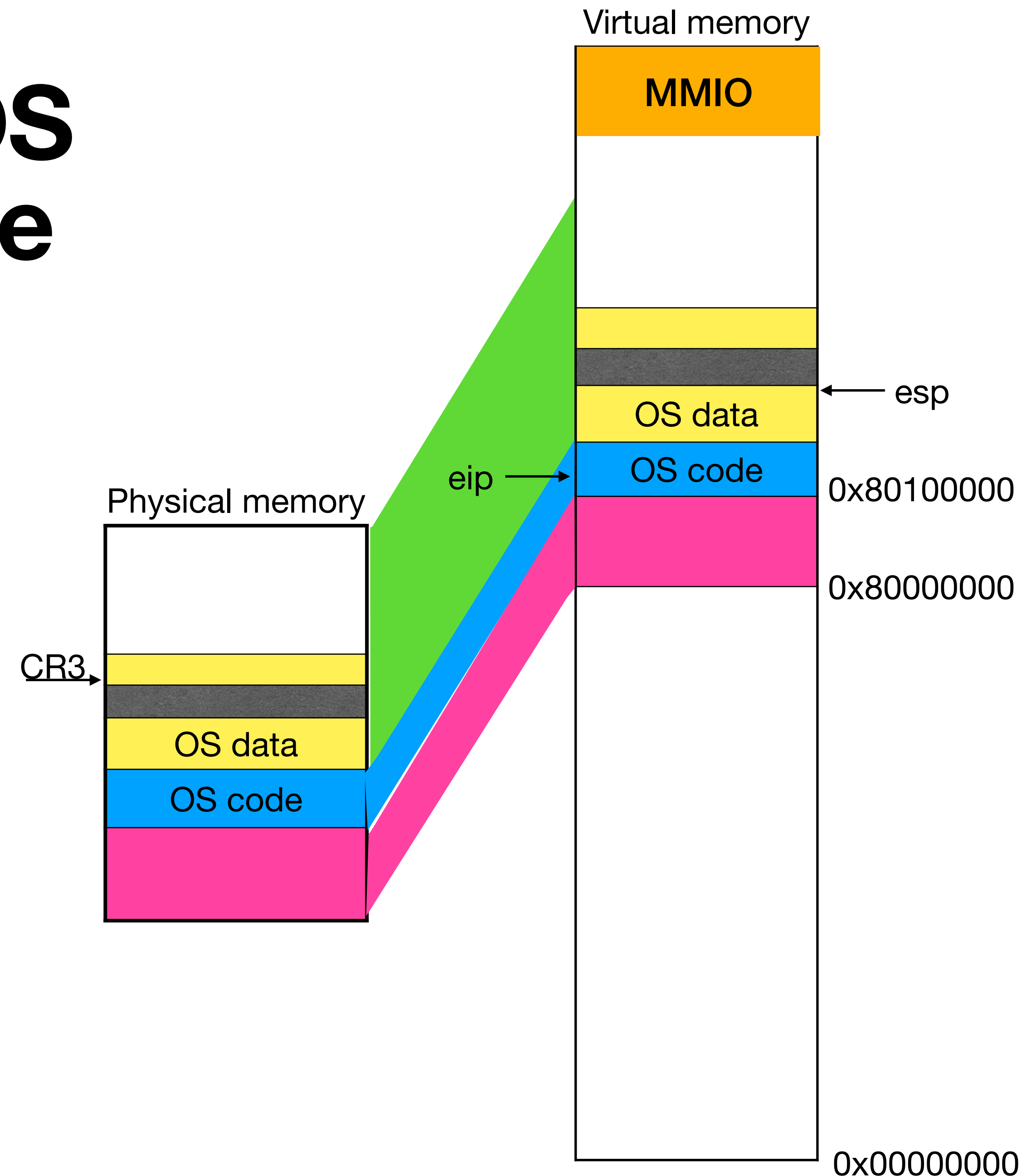
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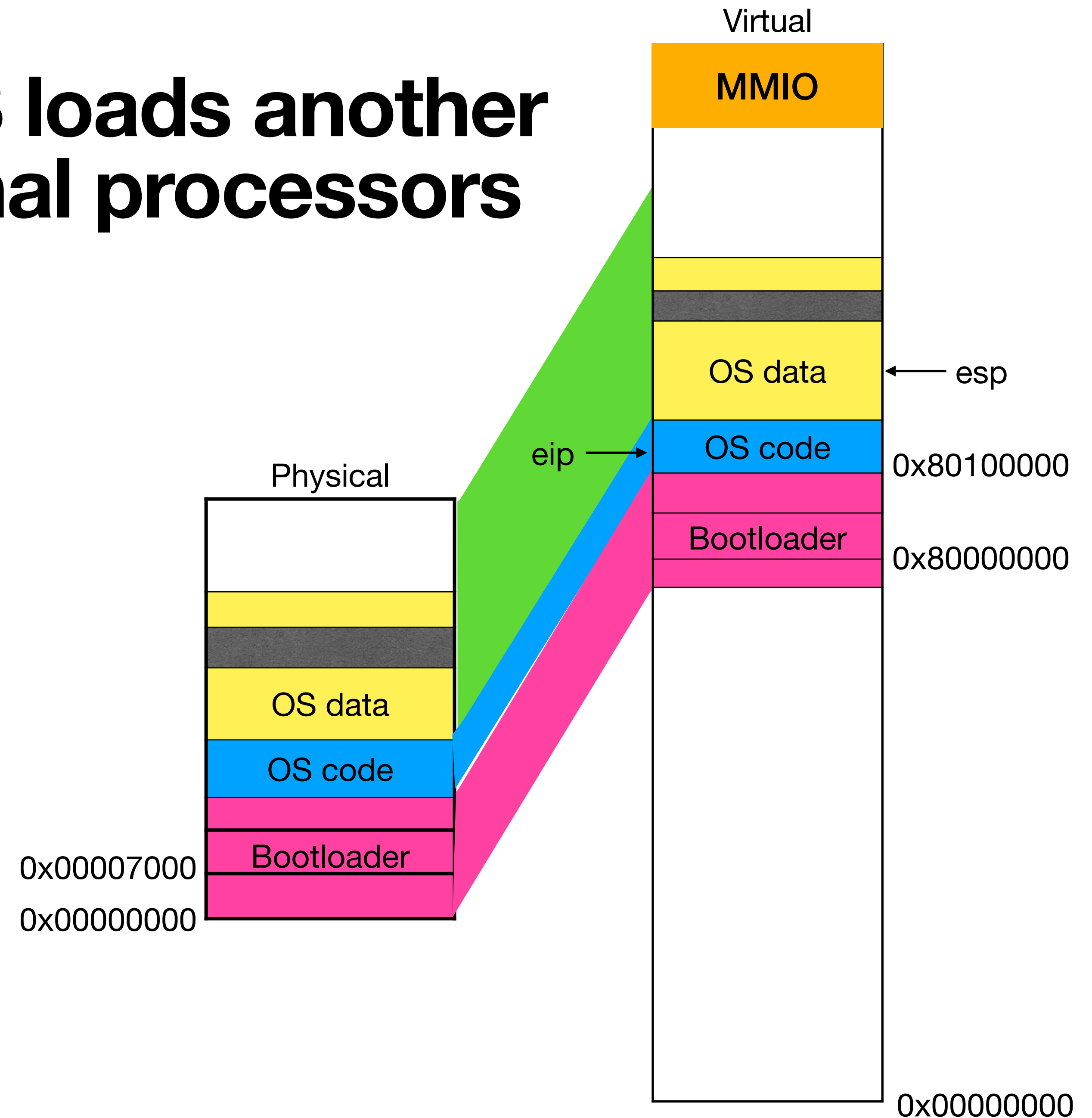
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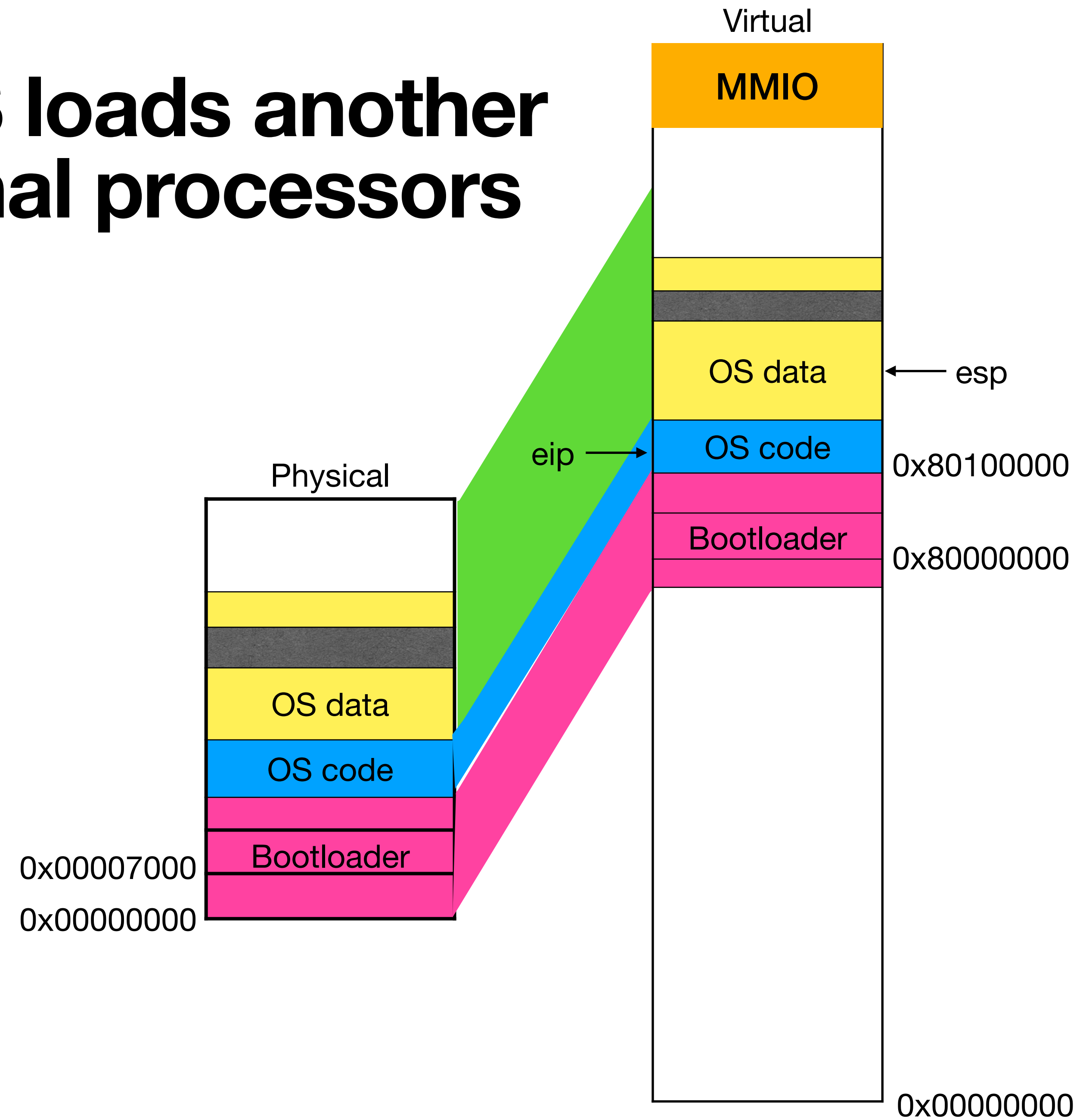
Boot up sequence (6): OS loads another “boot loader” for additional processors

```
int main(void) {  
    ..  
    startothers();    // start other processors  
  
static void startothers(void) {  
    code = P2V(0x7000);  
    memmove(code, _binary_entryother_start,  
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}
```



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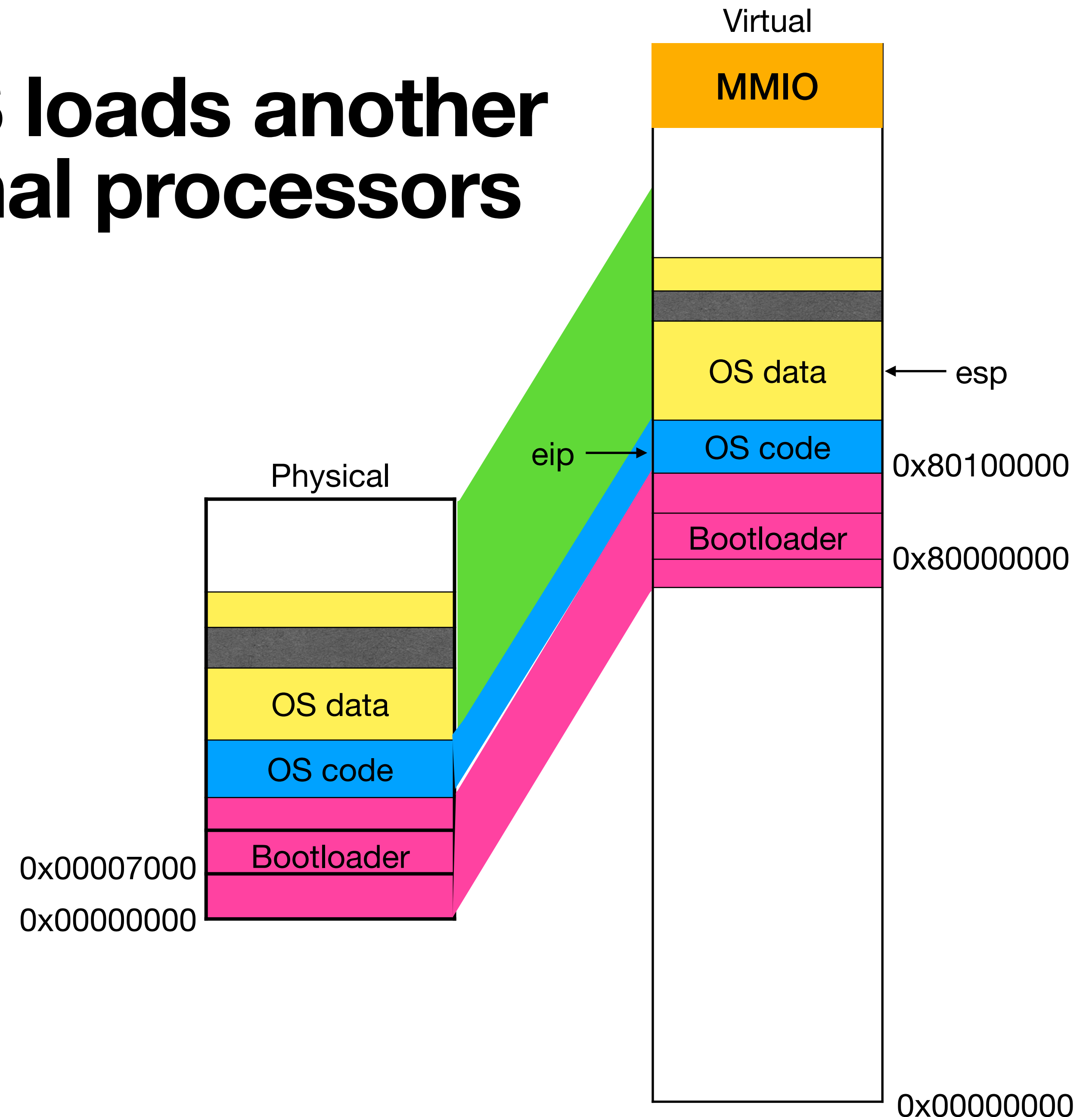
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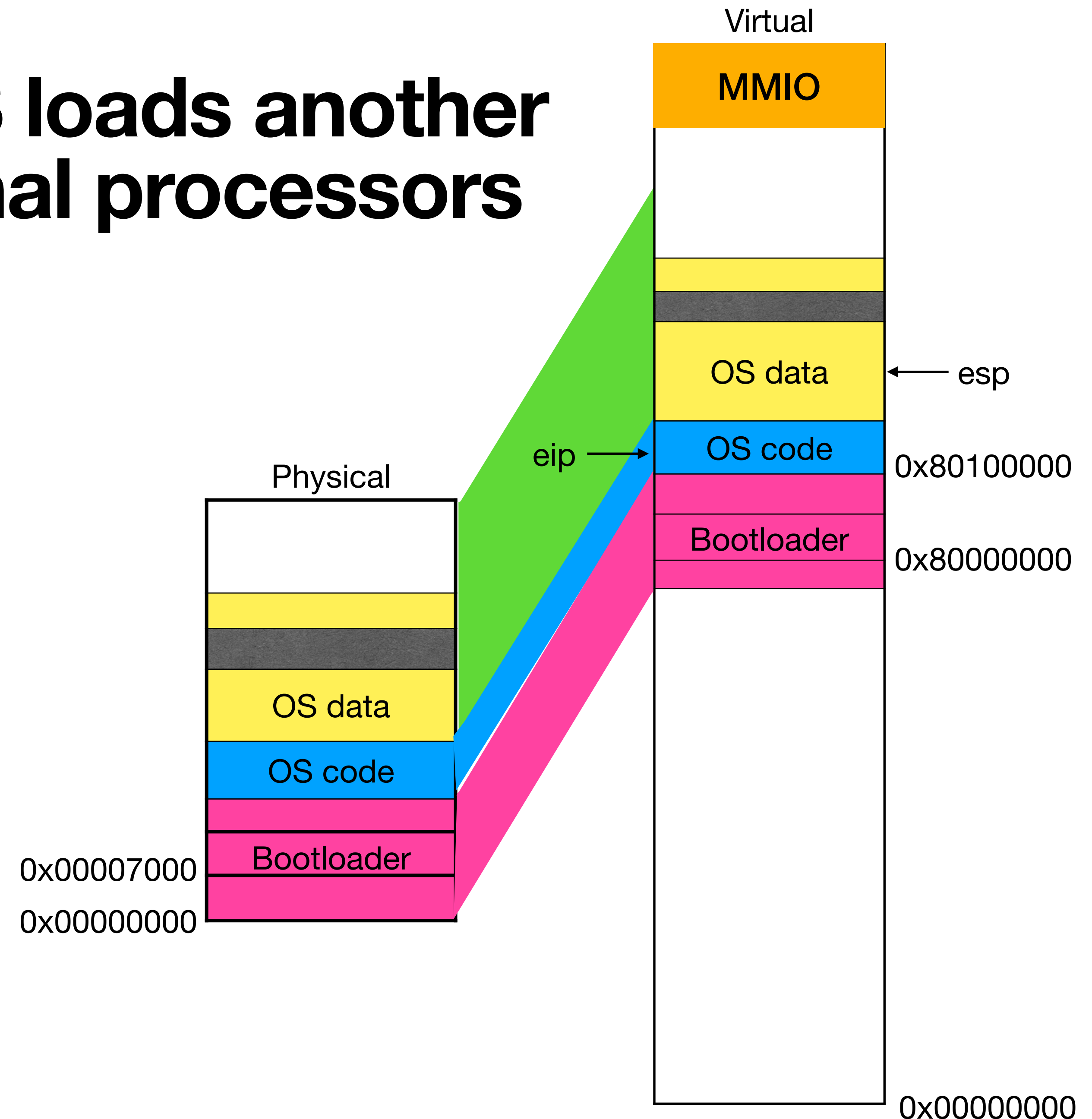
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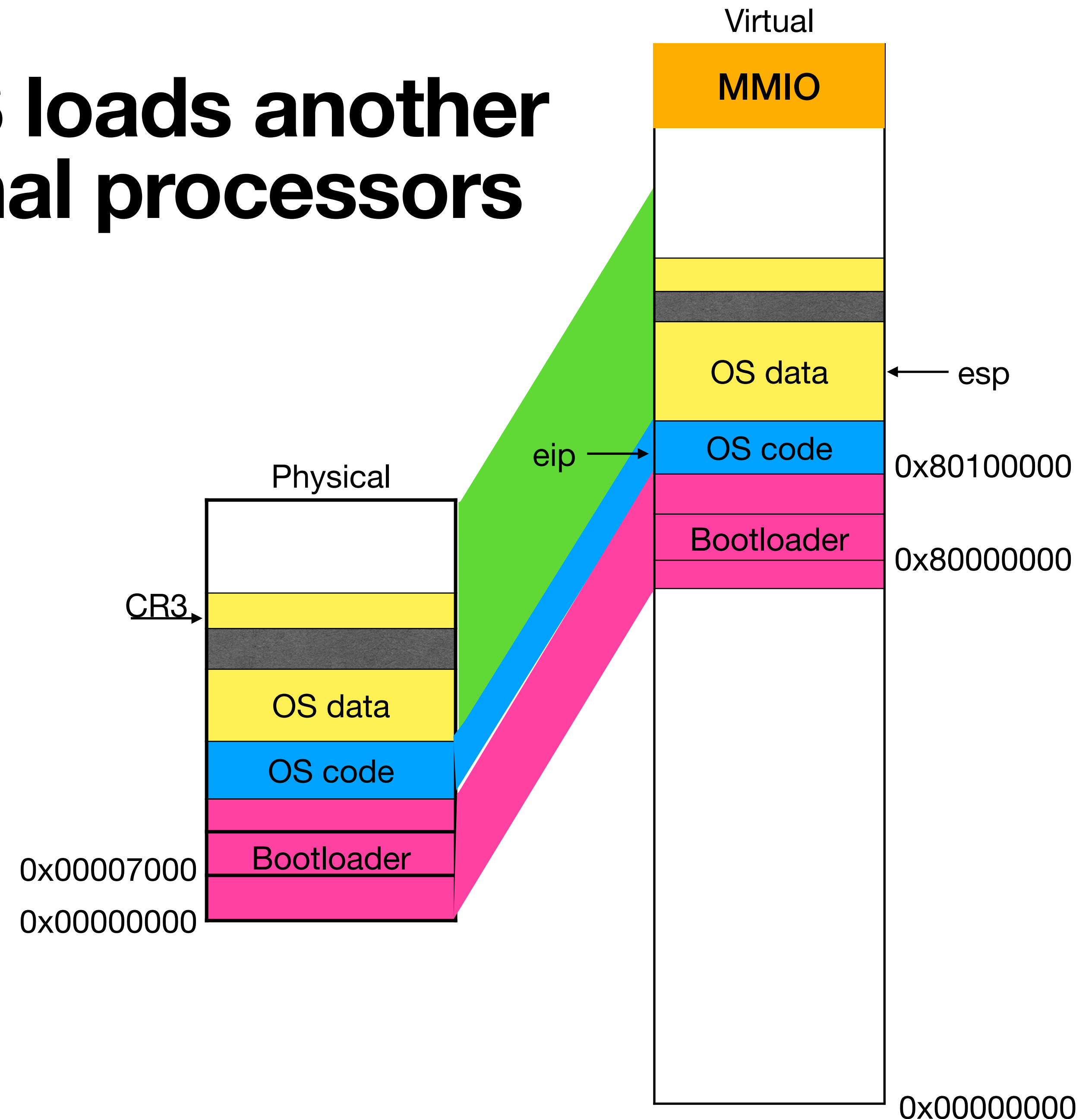
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Preparing “boot loader block” for APs

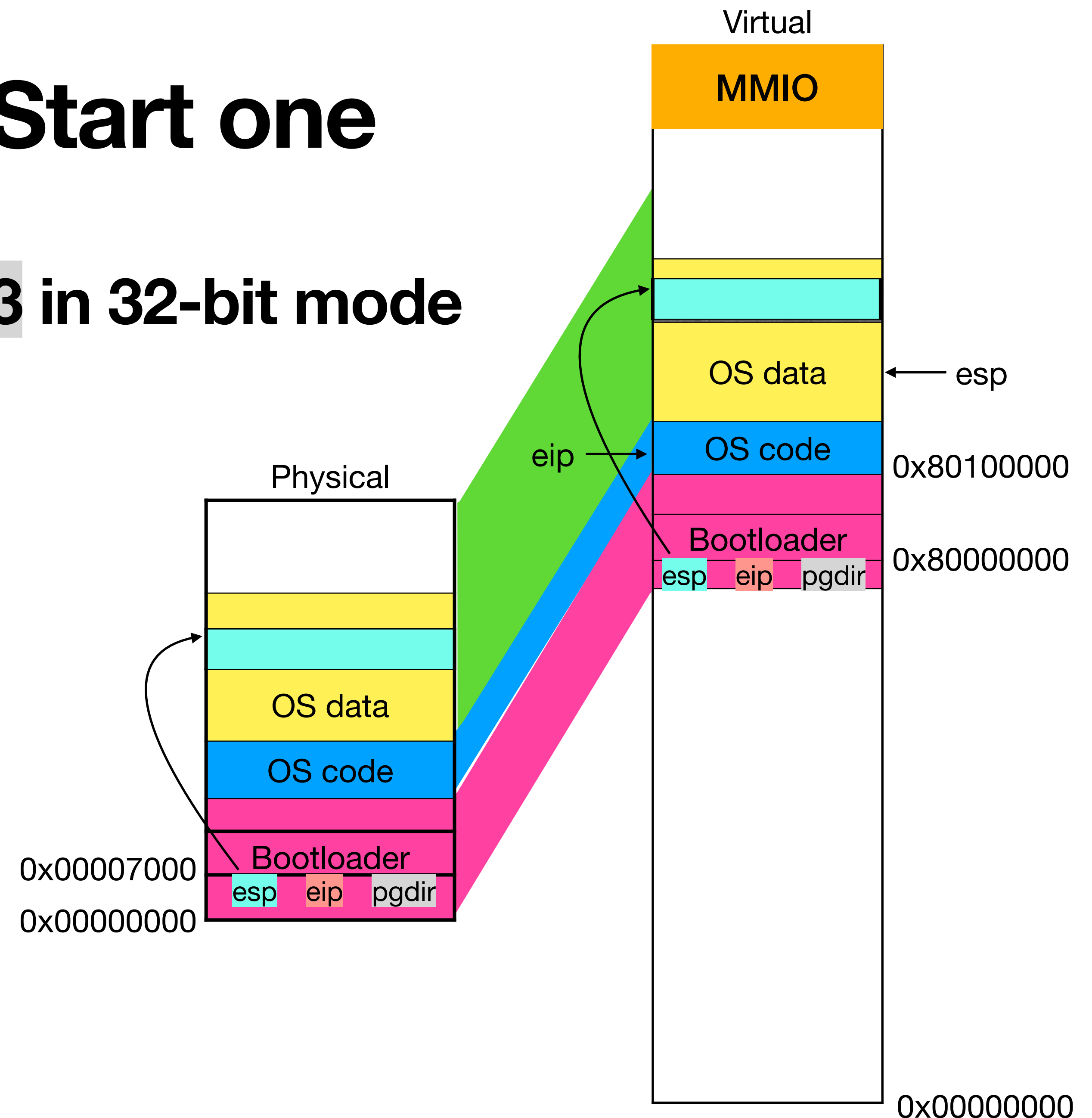
```
entryother: entryother.S
$(CC) $(CFLAGS) -fno-pic -nostdinc -I. -c entryother.S
$(LD) $(LDFLAGS) -N -e start -Ttext 0x7000 -o
                                bootblockother.o entryother.o

$(OBJCOPY) -S -O binary -j .text bootblockother.o entryother
$(OBJDUMP) -S bootblockother.o > entryother.asm
```

Boot up sequence (7): Start one CPU at a time

Instruct where to set **esp**, **eip**, **cr3** in 32-bit mode

```
static void startothers(void) {  
    code = P2V(0x7000);  
    ..  
    for(c = cpus; c < cpus+ncpu; c++){  
        stack = kalloc();  
        *(void**) (code-4) = stack + KSTACKSIZE;  
        *(void**) (void) (code-8) = mpenter;  
        *(int**) (code-12) = (void *) V2P(entrypgdir);  
  
        ..  
        // wait for cpu to finish mpmain()  
        while(c->started == 0);  
    }  
}
```

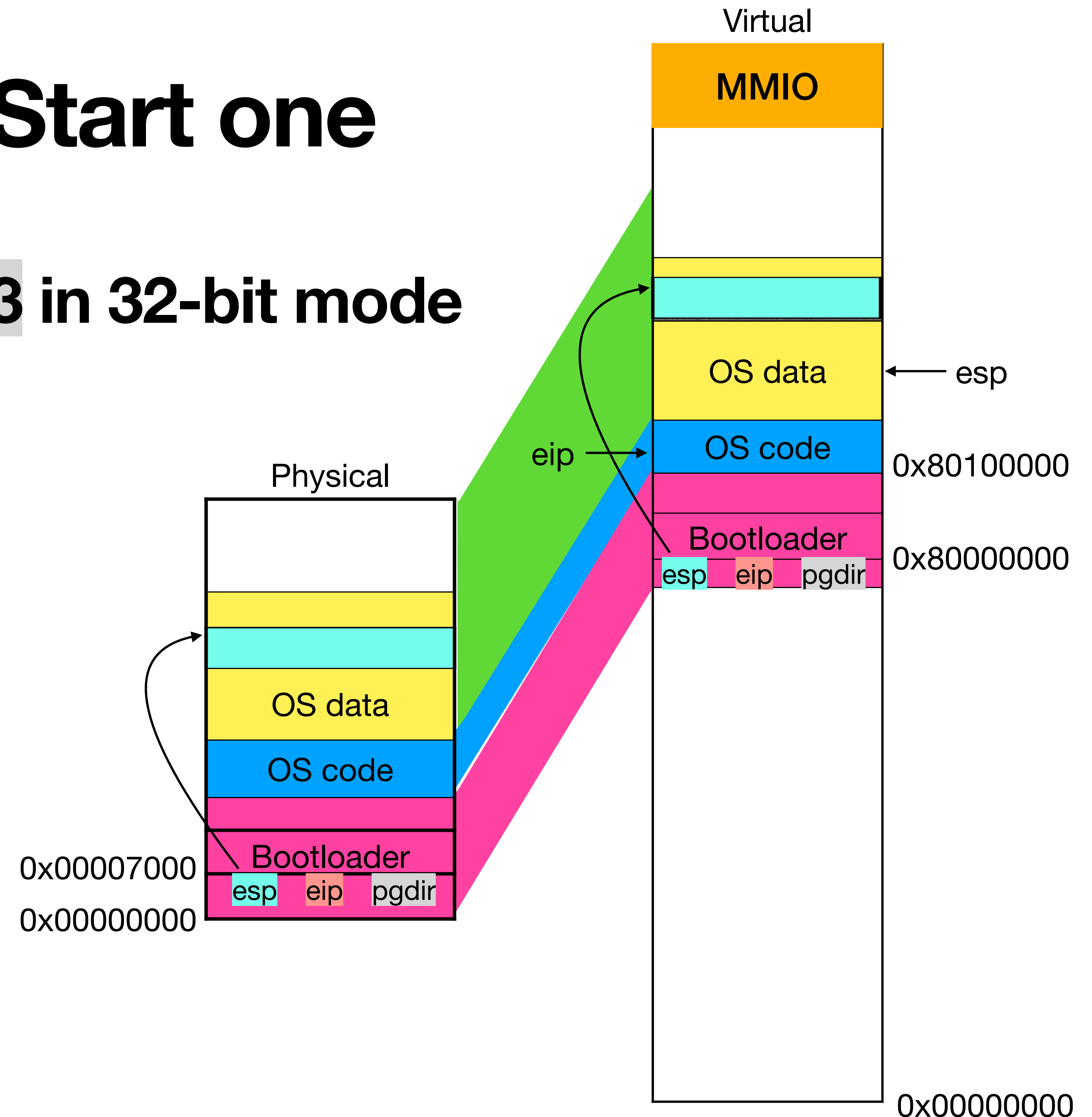


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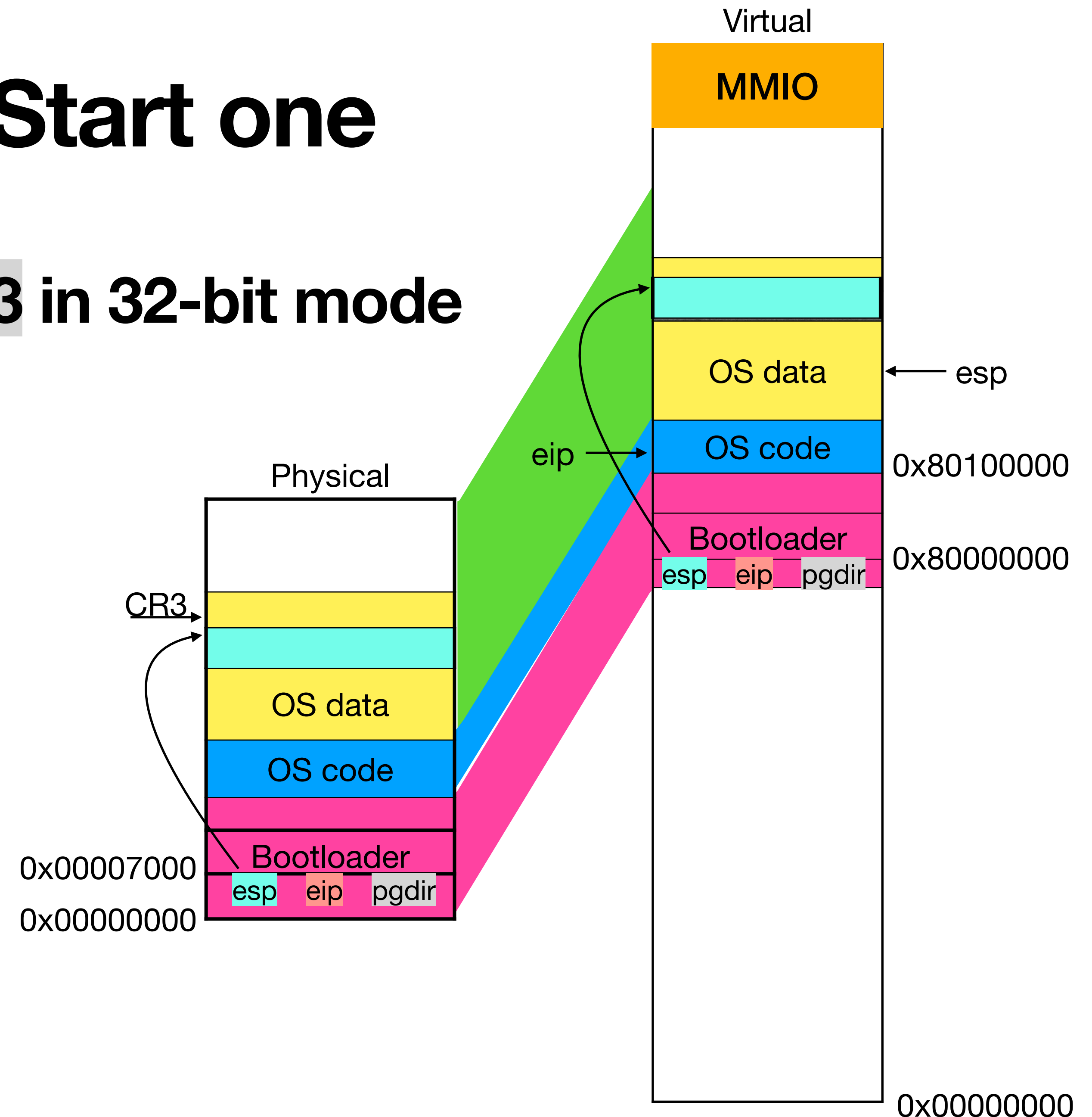
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    }
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```



Boot up sequence (7): Start one CPU at a time

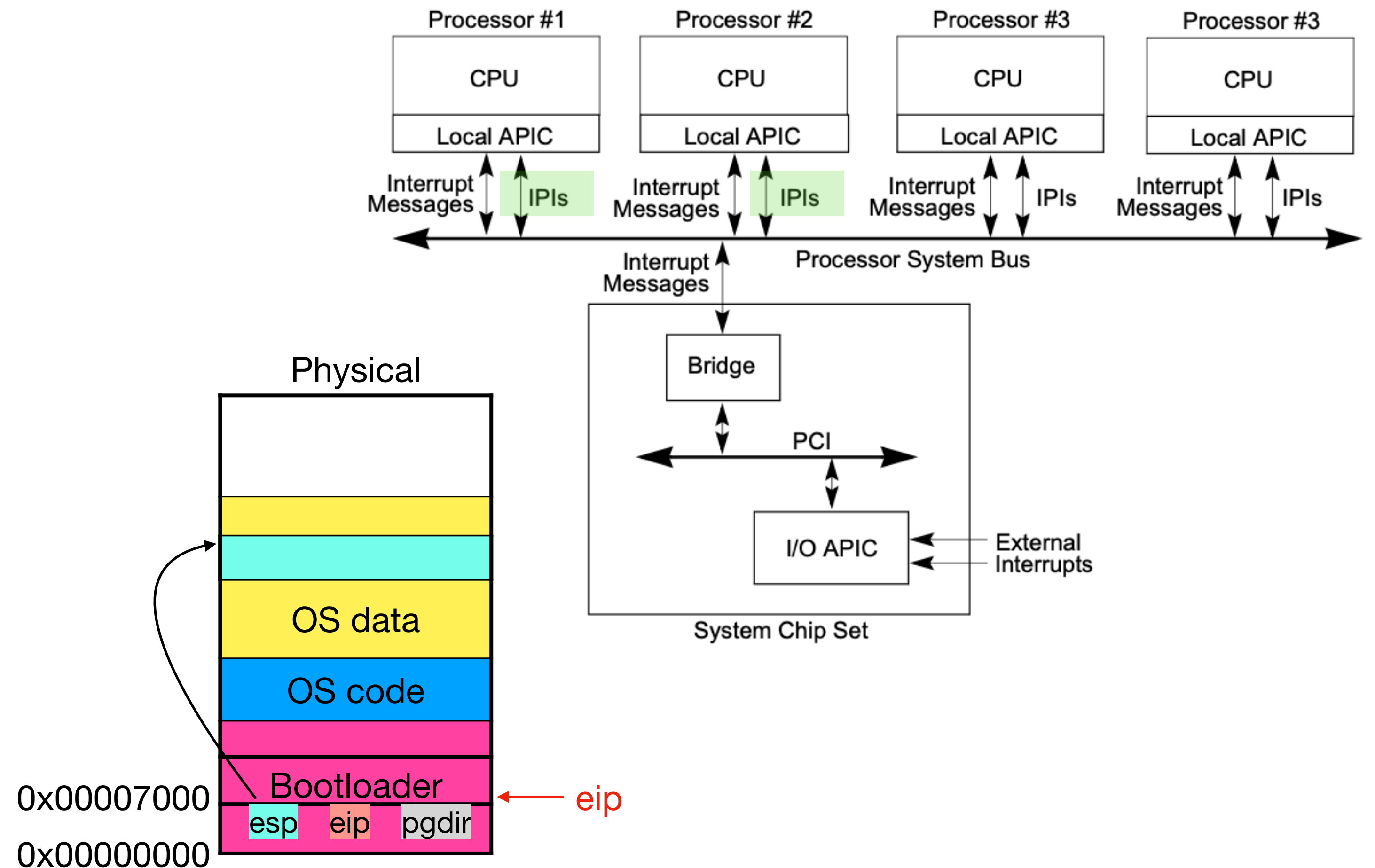
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        *(void**) (void) (code-8) = mpenter;  
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        ..  
        // wait for cpu to finish mpmain()  
        while(c->started == 0);  
    }  
}
```



Boot up sequence (8): Send **inter-processor interrupt** to start other CPU

```
static void startothers(void) {  
    ..  
    for(c = cpus; c < cpus+ncpu; c++){  
        ..  
        lapicstartap(c->apicid, V2P(code));  
        // wait for cpu to finish mpmain()  
        while(c->started == 0);  
    }  
}
```



Boot up sequence (9): AP performs similar steps as BSP's boot

- Disable interrupts
- Switch to 32 bit mode
- Setup segment selectors

```
.code16                                                    entryother.S
.globl start
start:
    cli
    ..
    # Complete the transition to 32-bit protected mode
    ljmpl    $(SEG_KCODE<<3), $(start32)

.code32
start32:
    movw     $(SEG_KDATA<<3), %ax
    movw     %ax, %ds
    ..
```

Boot up sequence (10): AP transitions to virtual addresses

```
start32:
```

```
..
```

```
# Use entrypgdir as our initial page table
```

```
movl    (start-12), %eax
```

```
movl    %eax, %cr3
```

```
# Turn on paging.
```

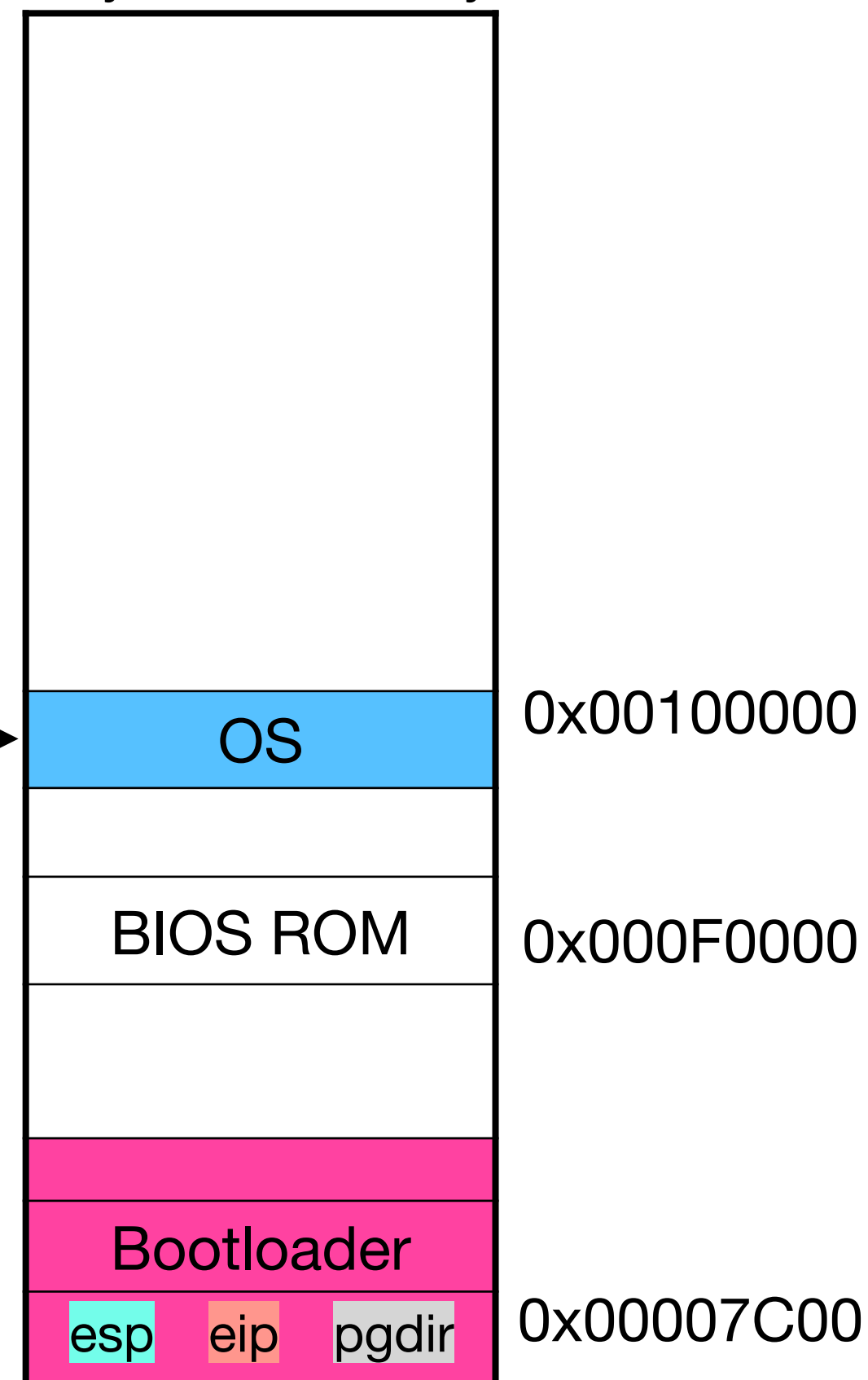
```
movl    %cr0, %eax
```

```
orl     $(CR0_PE|CR0_PG|CR0_WP), %eax
```

```
movl    %eax, %cr0
```

eip →

Physical memory



Boot up sequence (10): AP transitions to virtual addresses

```
start32:
```

```
..
```

```
# Use entrypgdir as our initial page table
```

```
movl    (start-12), %eax
```

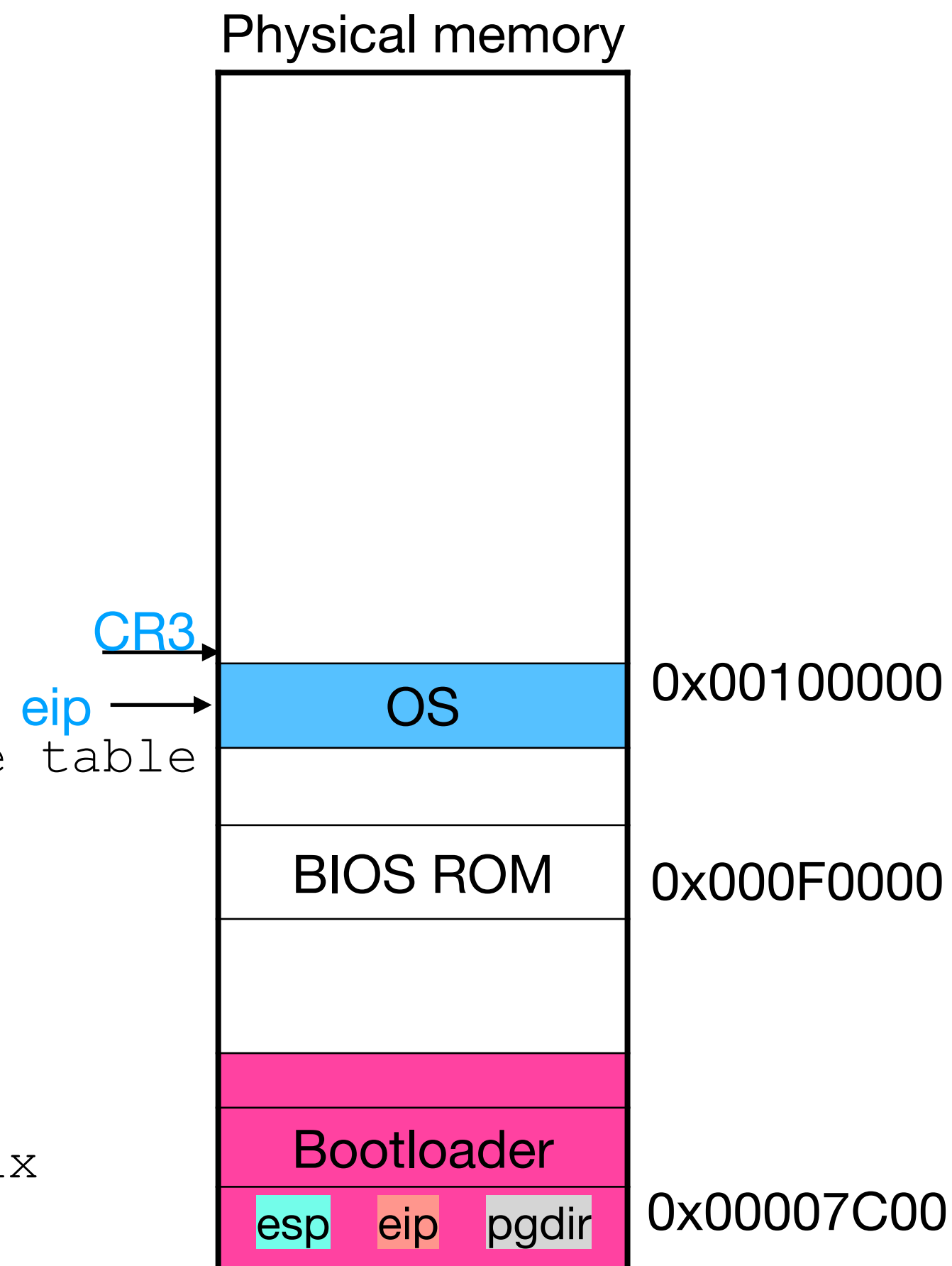
```
movl    %eax, %cr3
```

```
# Turn on paging.
```

```
movl    %cr0, %eax
```

```
orl     $(CR0_PE|CR0_PG|CR0_WP), %eax
```

```
movl    %eax, %cr0
```

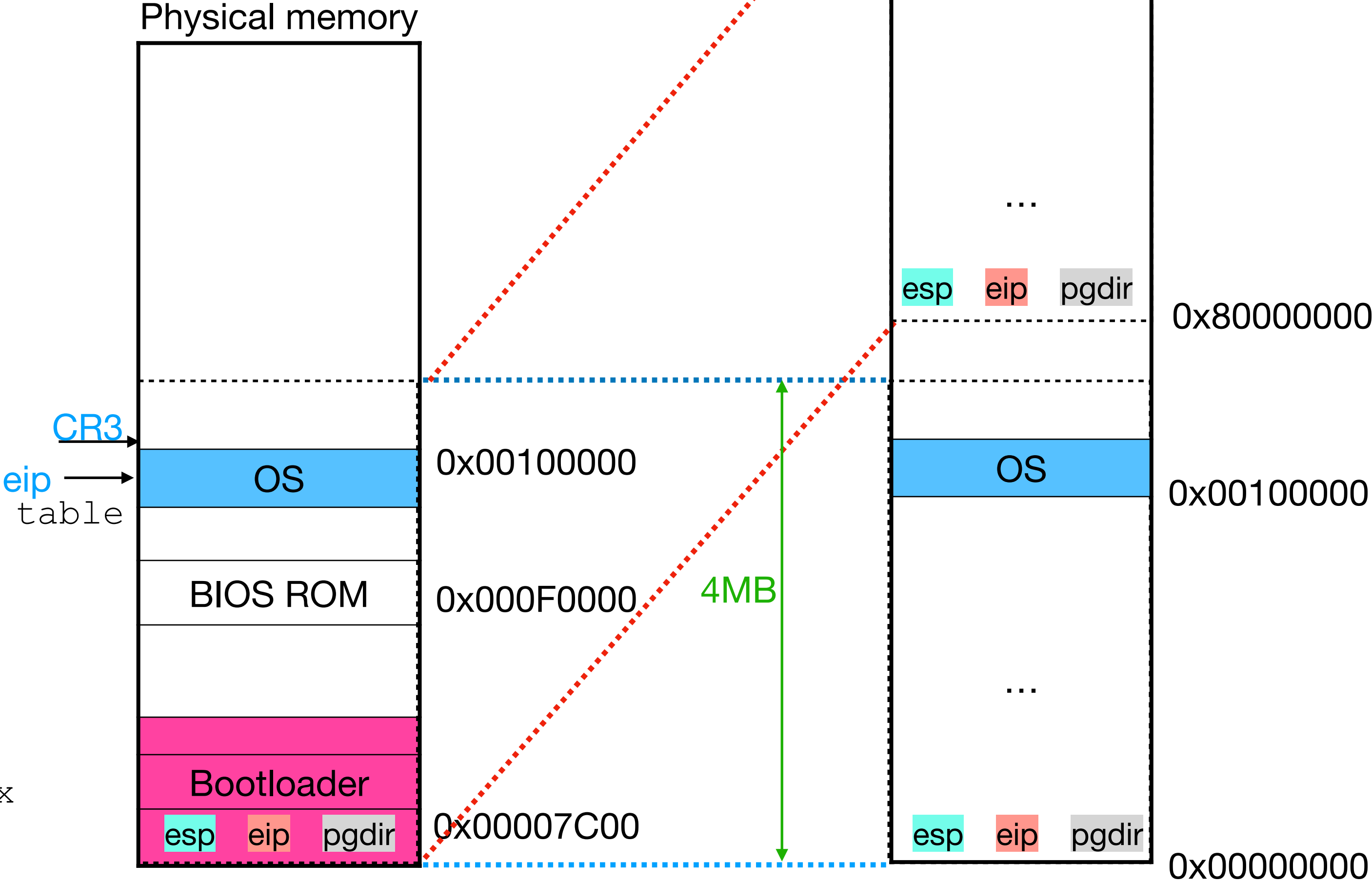


Boot up sequence (10): AP transitions to virtual addresses

```
start32:
    ..

    # Use entrypgdir as our initial page table
    movl    (start-12), %eax
    movl    %eax, %cr3

    # Turn on paging.
    movl    %cr0, %eax
    orl     $(CR0_PE|CR0_PG|CR0_WP), %eax
    movl    %eax, %cr0
```

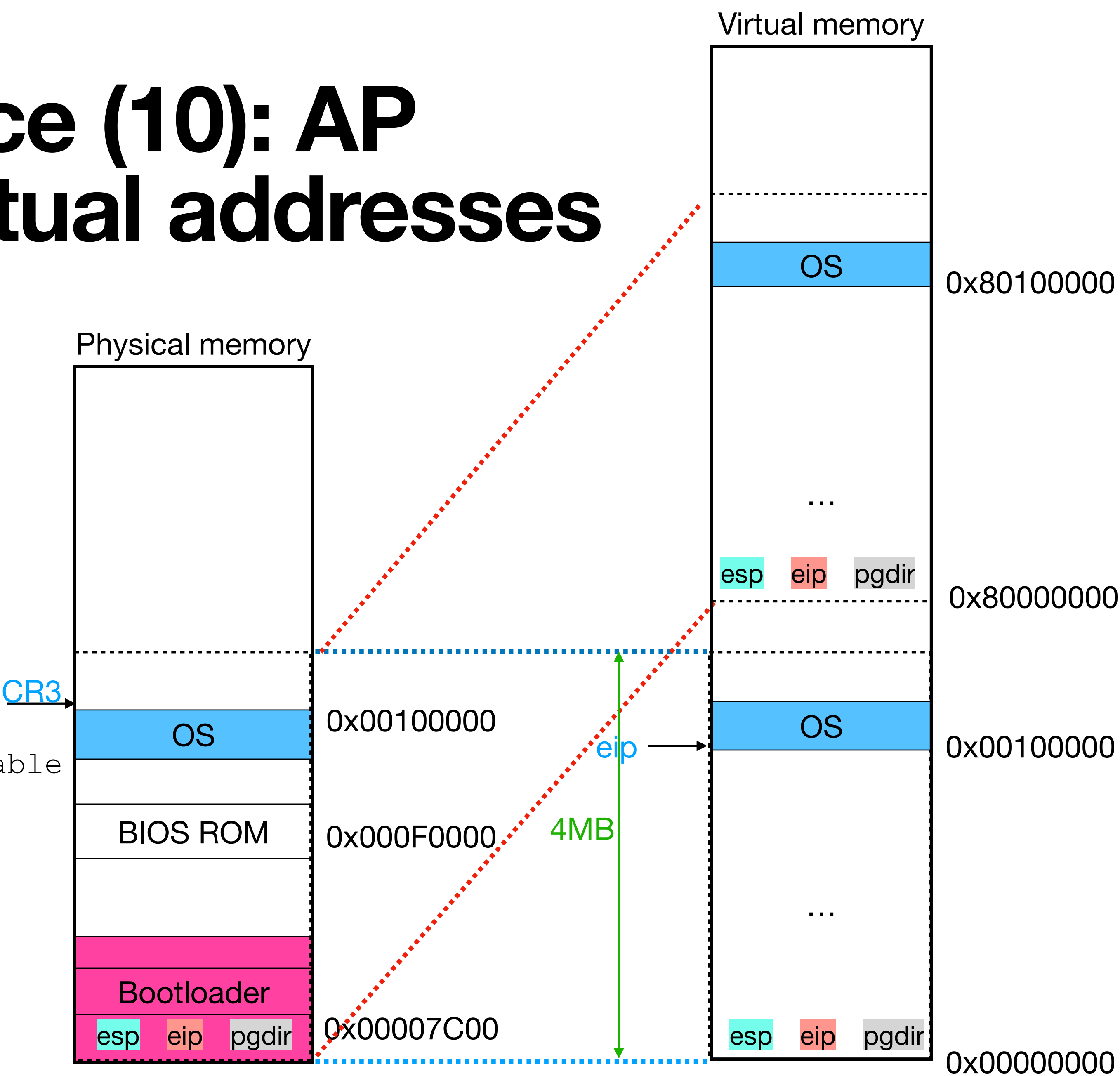


Boot up sequence (10): AP transitions to virtual addresses

```
start32:
    ..

    # Use entrypgdir as our initial page table
    movl    (start-12), %eax
    movl    %eax, %cr3

    # Turn on paging.
    movl    %cr0, %eax
    orl     $(CR0_PE|CR0_PG|CR0_WP), %eax
    movl    %eax, %cr0
```



Boot up sequence (10): AP transitions to virtual addresses

- Use the initial page table

```
start32:
```

```
..
```

```
# Use entrypgdir as our initial page table
```

```
movl    (start-12), %eax
```

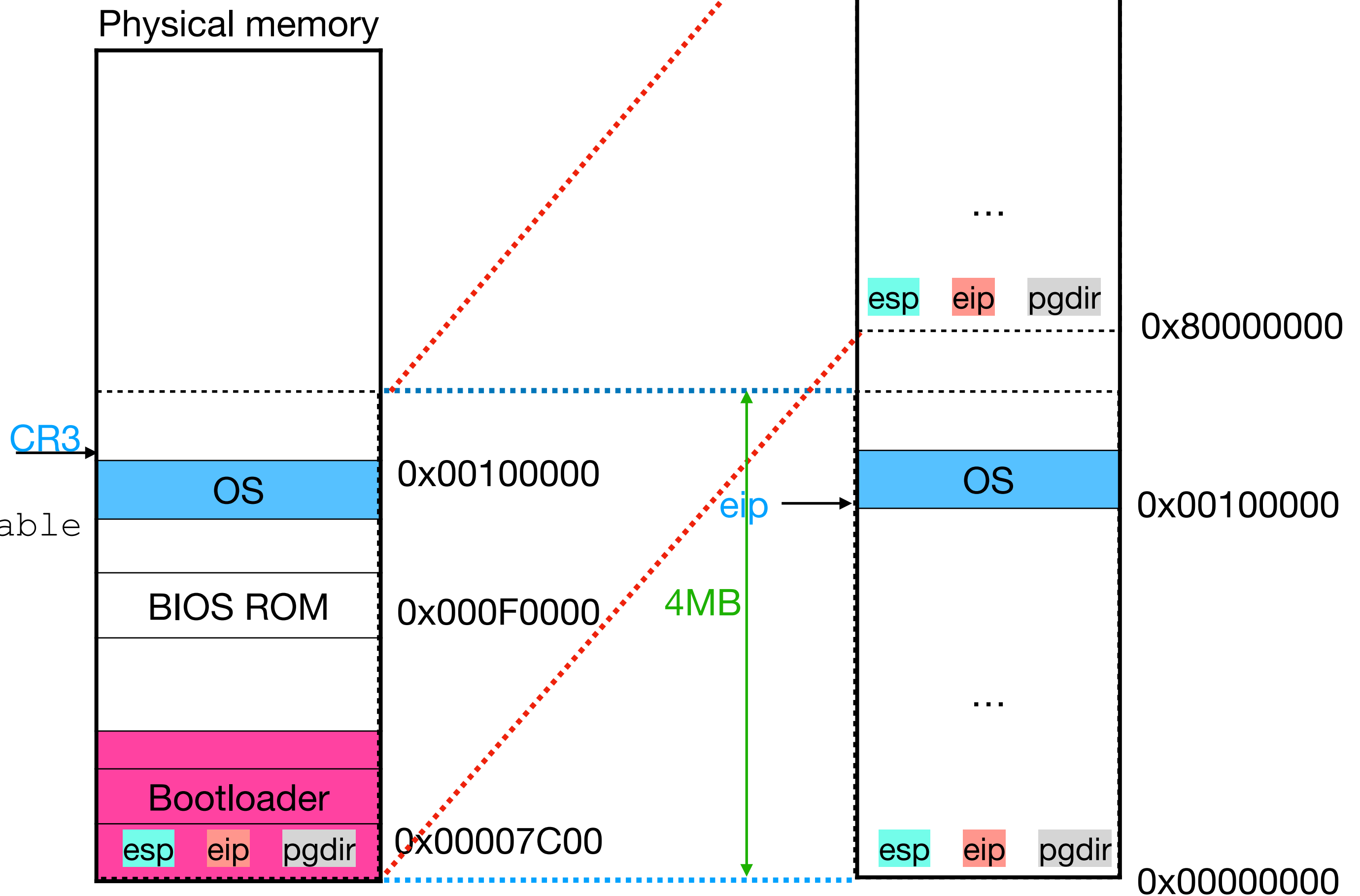
```
movl    %eax, %cr3
```

```
# Turn on paging.
```

```
movl    %cr0, %eax
```

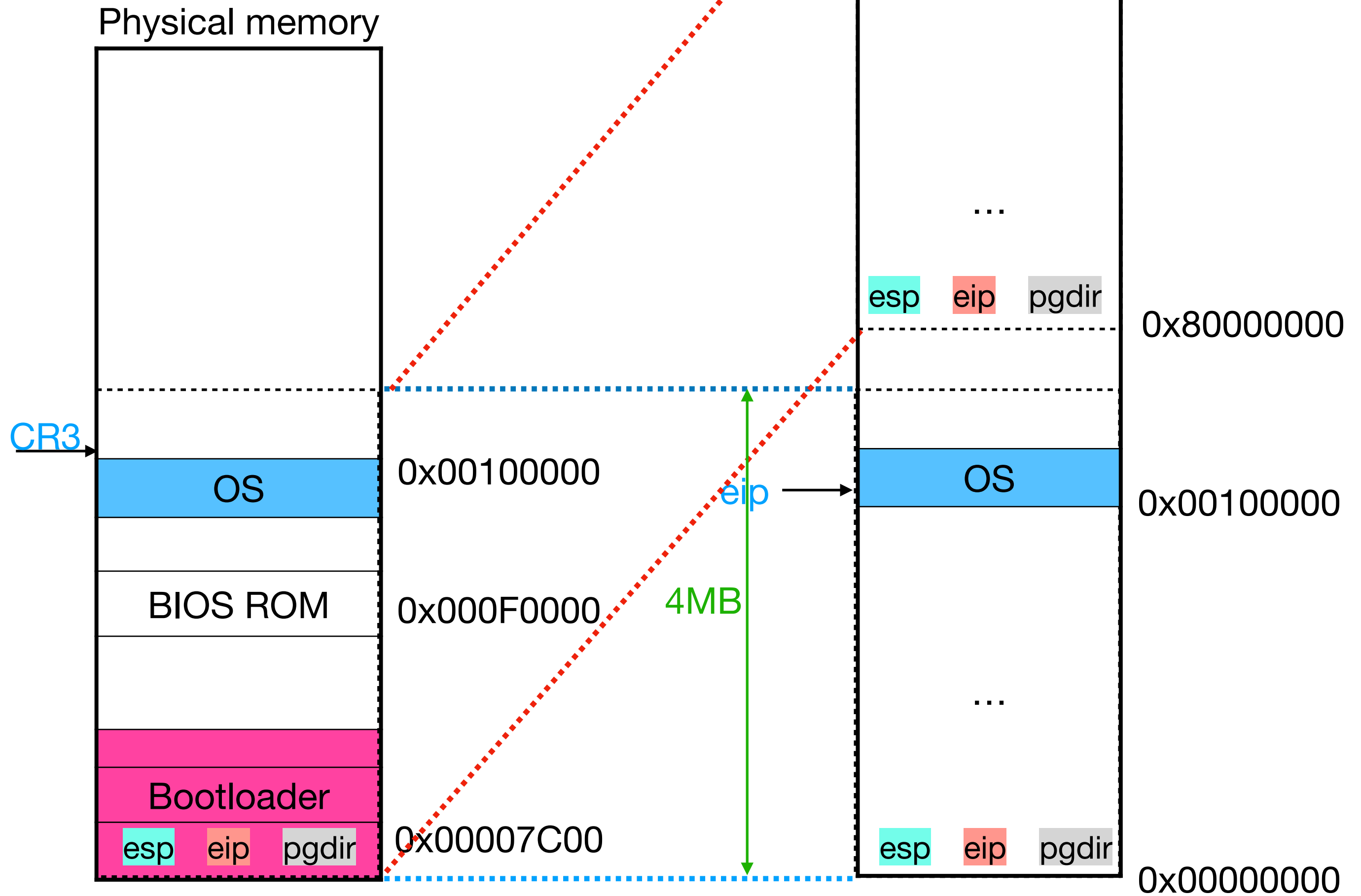
```
orl     $(CR0_PE|CR0_PG|CR0_WP), %eax
```

```
movl    %eax, %cr0
```



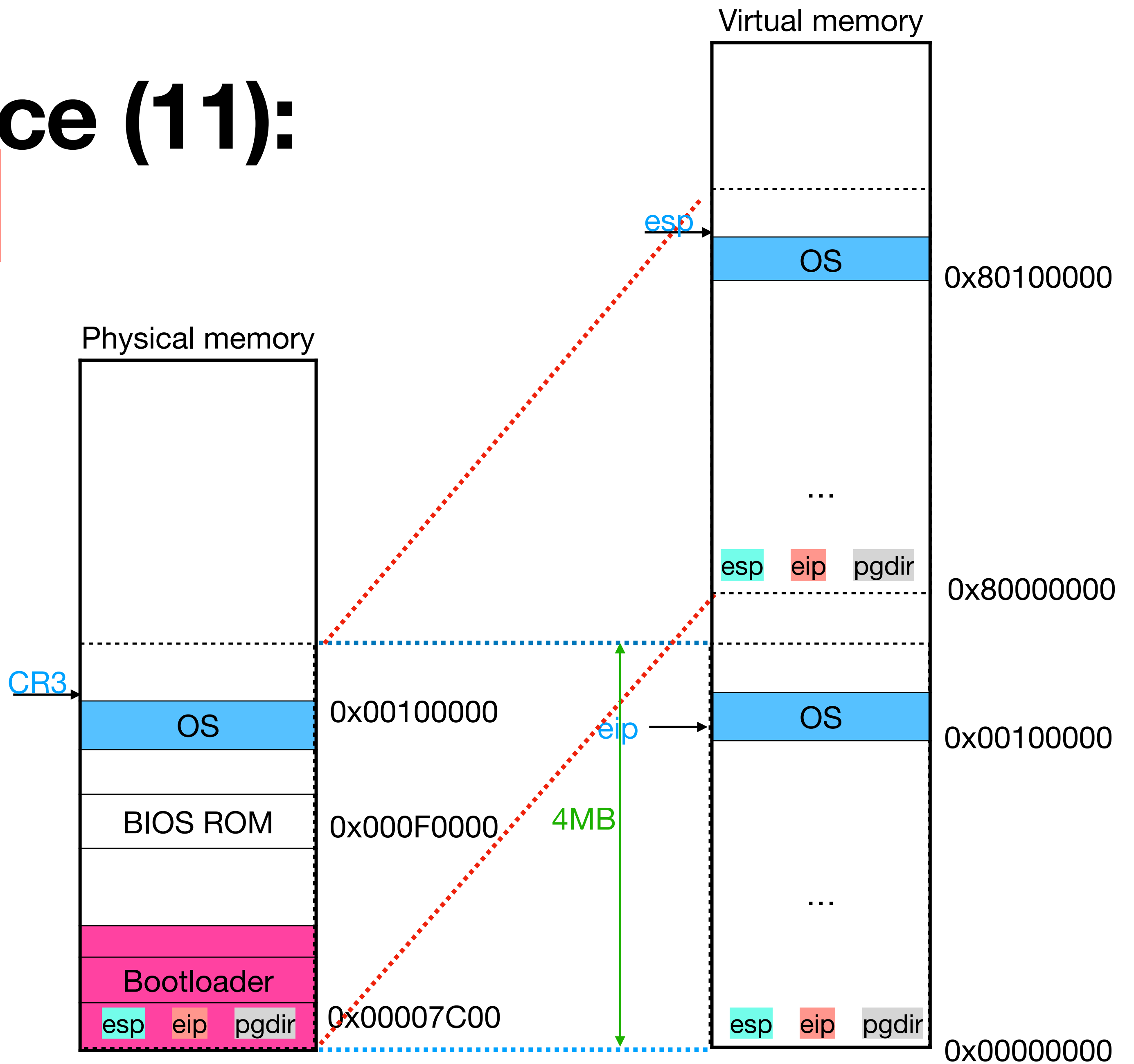
Boot up sequence (11): AP sets **esp**, **eip**

```
.code32
start32:
    ..
    # Switch to the stack
    #   allocated by startothers()
    movl    (start-4), %esp
    # Call mpenter()
    call    *(start-8)
```



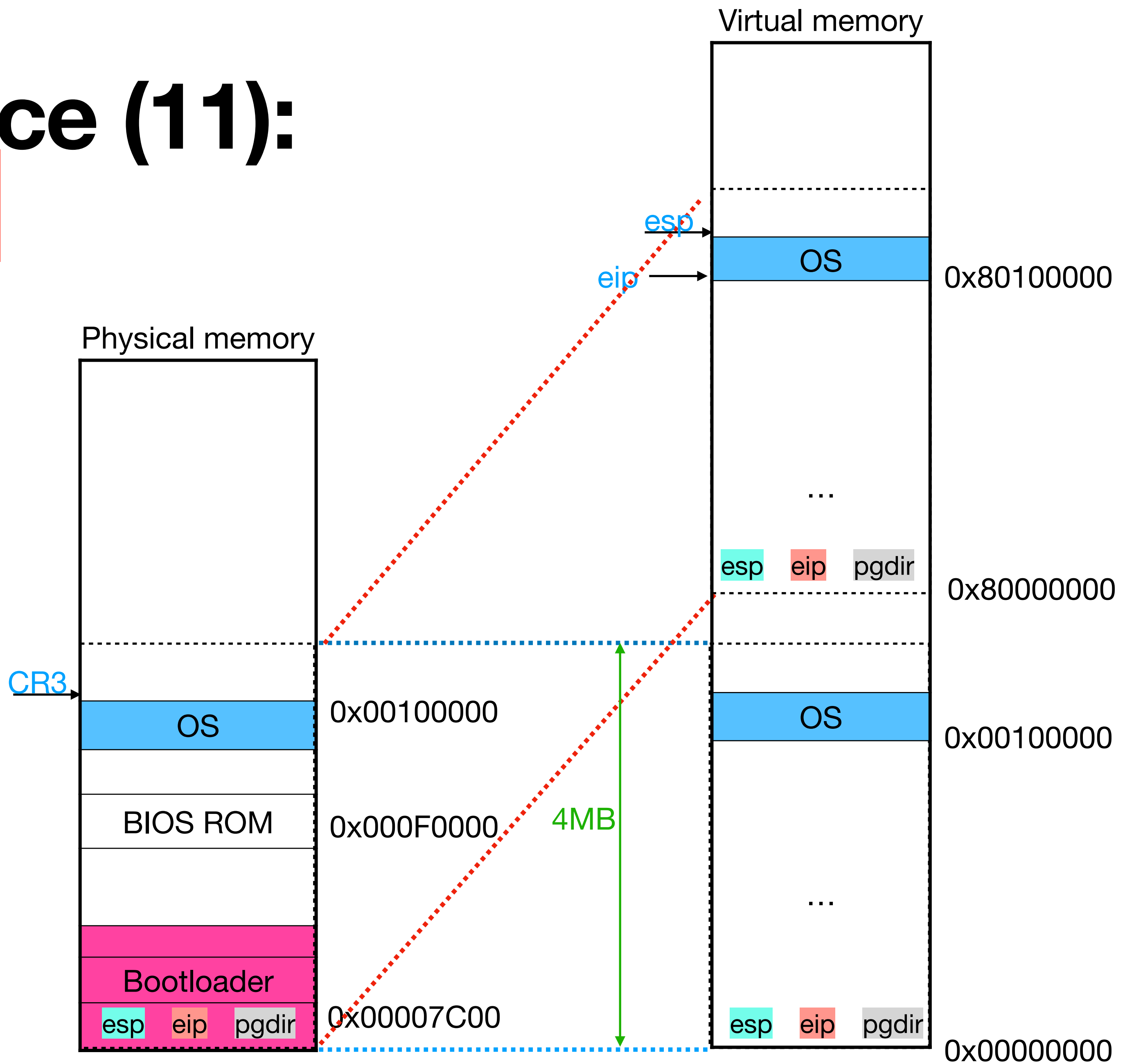
Boot up sequence (11): AP sets **esp**, **eip**

```
.code32
start32:
    ..
    # Switch to the stack
    #   allocated by startothers()
    movl    (start-4), %esp
    # Call mpenter()
    call    *(start-8)
```



Boot up sequence (11): AP sets **esp**, **eip**

```
.code32
start32:
    ..
    # Switch to the stack
    #   allocated by startothers()
    movl    (start-4), %esp
    # Call mpenter()
    call    *(start-8)
```



Boot up sequence (12): Additional CPU completes boot

```
static void mpenter(void) {
    switchkvm();
    seginit();
    lapicinit();
    mpmain();
}

static void mpmain(void) {
    idtinit();
    xchg(&(mycpu()->started), 1);
    scheduler();
}
```

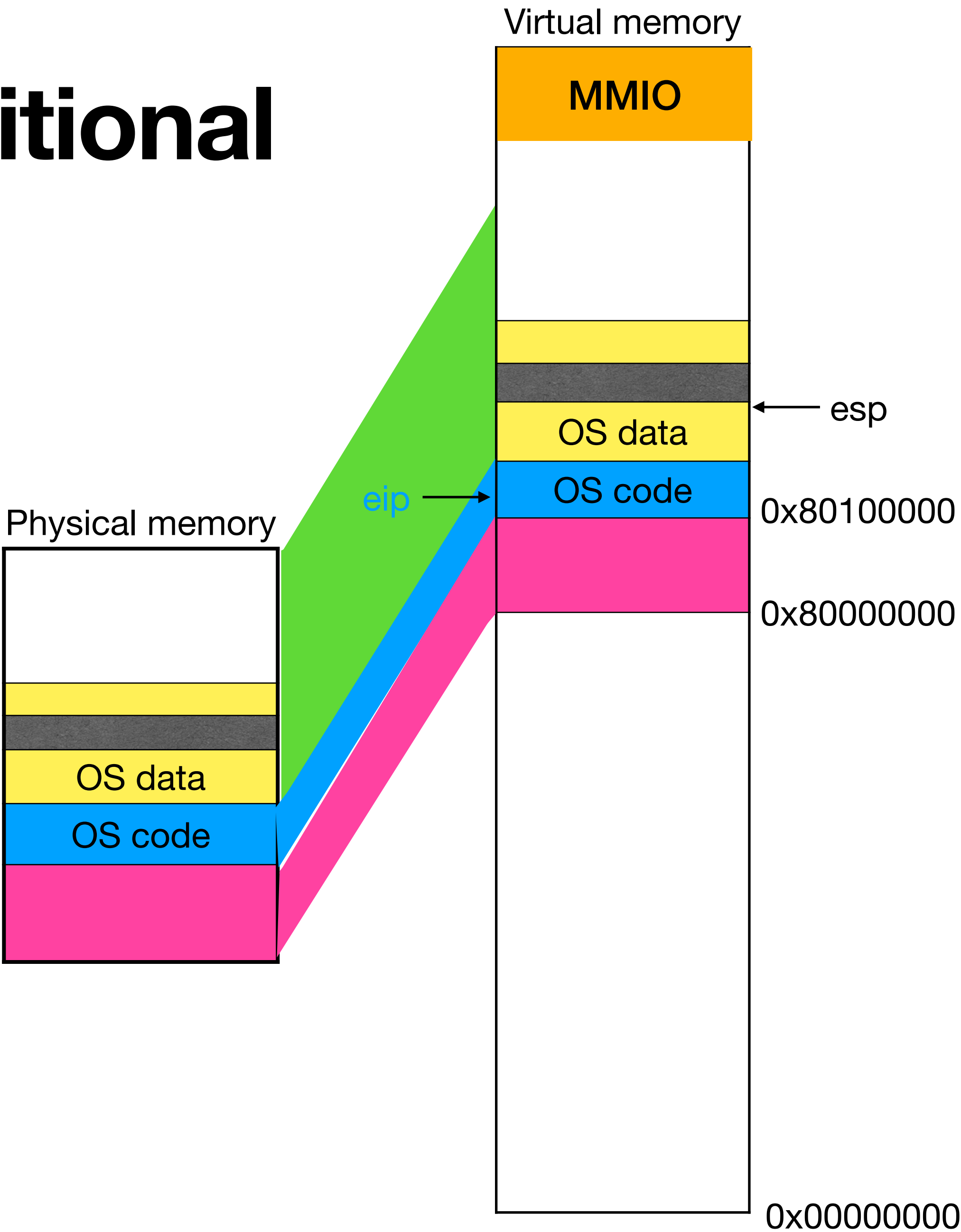
Switch to full page table

Initialize GDT

Setup IDTR

Signal main CPU that we have started

Start running processes



Boot up sequence (12): Additional CPU completes boot

```
static void mpenter(void) {
    switchkvm();
    seginit();
    lapicinit();
    mpmain();
}

static void mpmain(void) {
    idtinit();
    xchg(&(mycpu()->started), 1);
    scheduler();
}
```

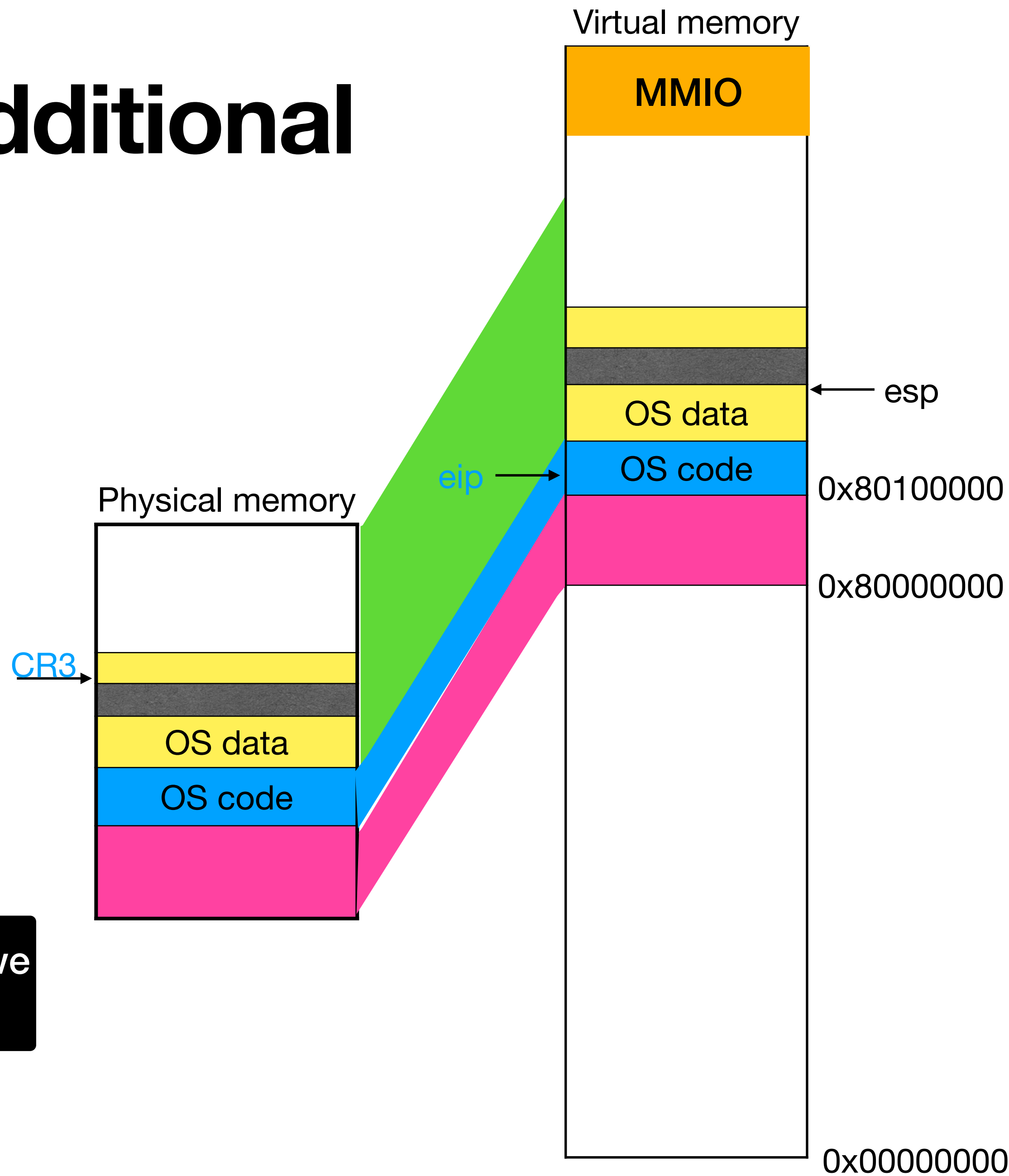
Switch to full page table

Initialize GDT

Setup IDTR

Signal main CPU that we have started

Start running processes



Boot up sequence (12): Additional CPU completes boot

```
static void mpenter(void) {
    switchkvm();
    seginit();
    lapicinit();
    mpmain();
}

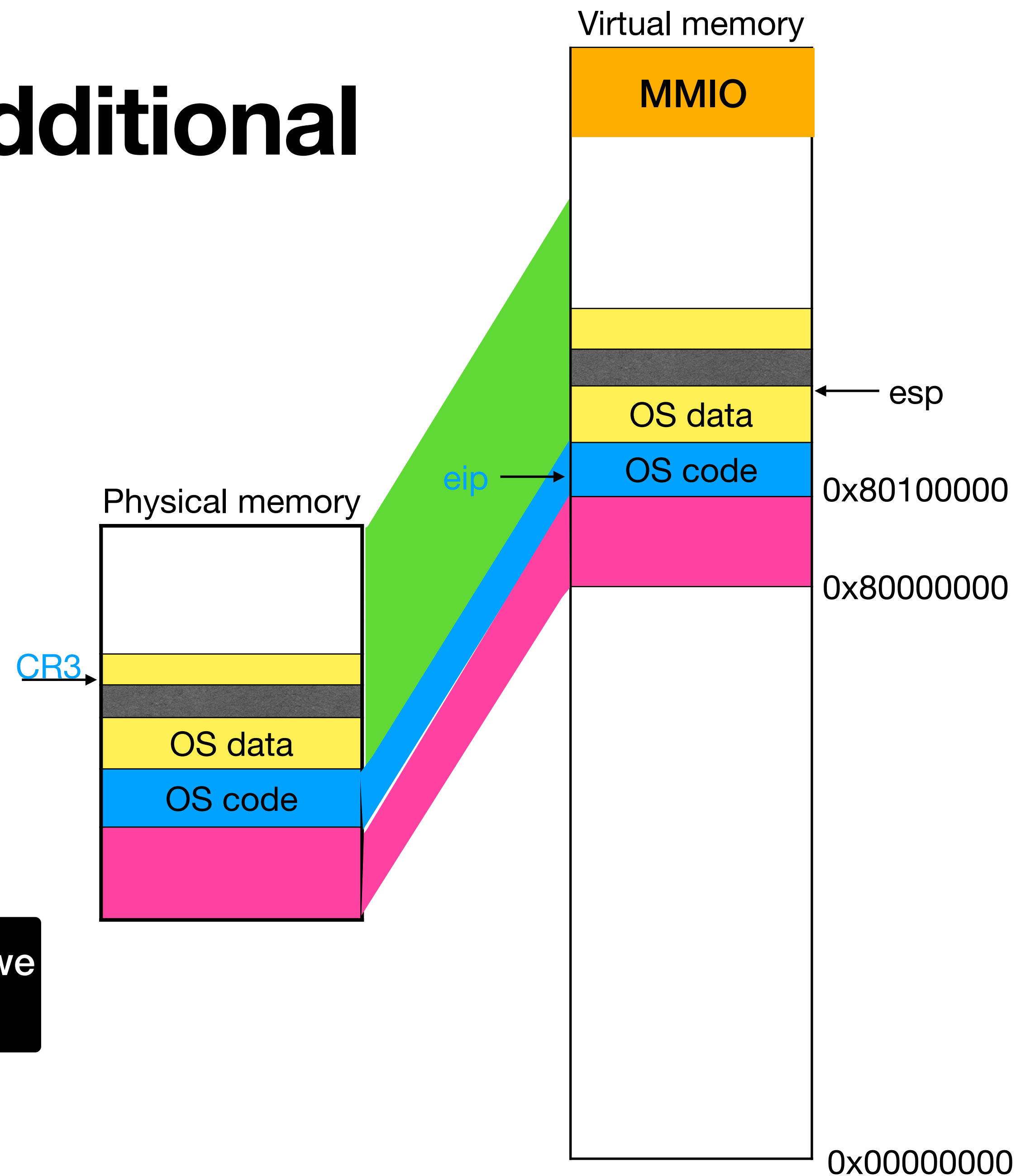
static void mpmain(void) {
    idtinit();
    xchg(&(mycpu()->started), 1);
    scheduler();
}
```

Annotations for `mpenter()`:

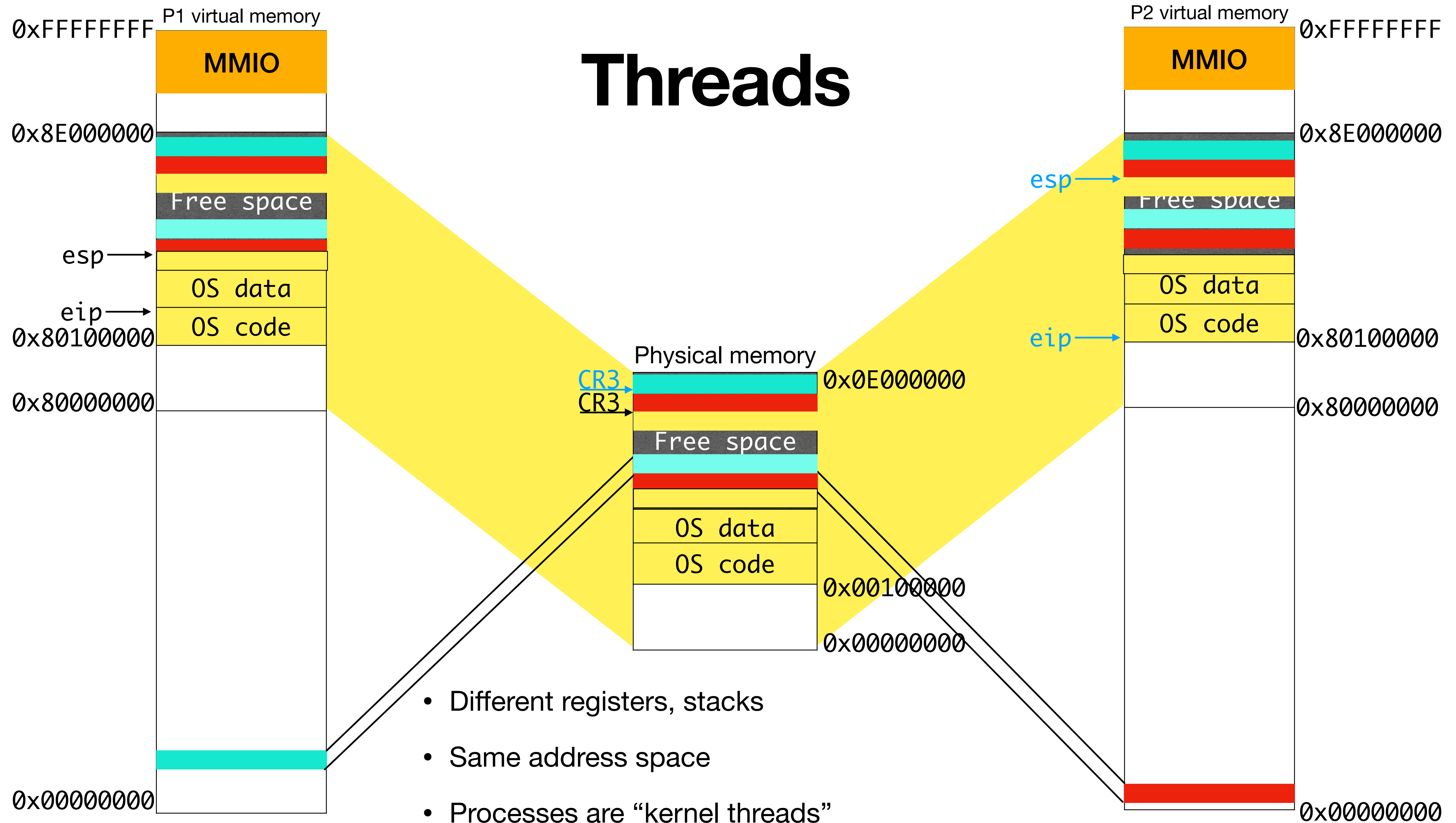
- `switchkvm();` → Switch to full page table
- `seginit();` → Initialize GDT

Annotations for `mpmain()`:

- `idtinit();` → Setup IDTR
- `xchg(&(mycpu()->started), 1);` → Signal main CPU that we have started
- `scheduler();` → Start running processes

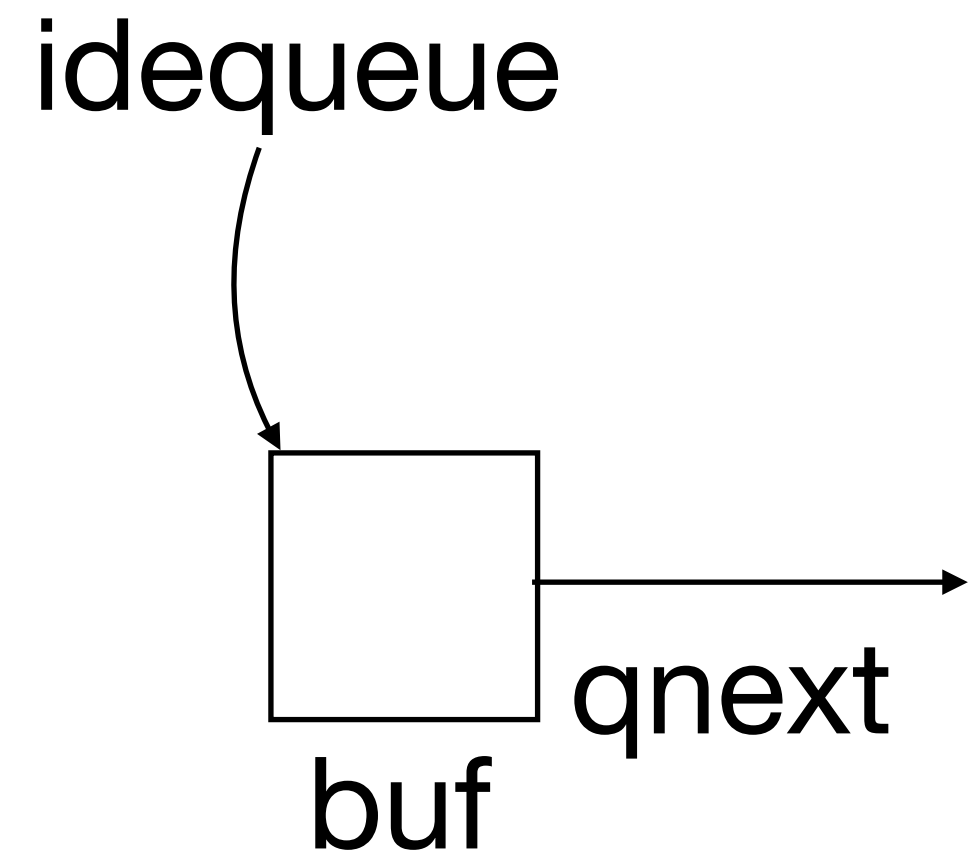


Threads



Race conditions

Example: kernel threads reading a block in ide.c

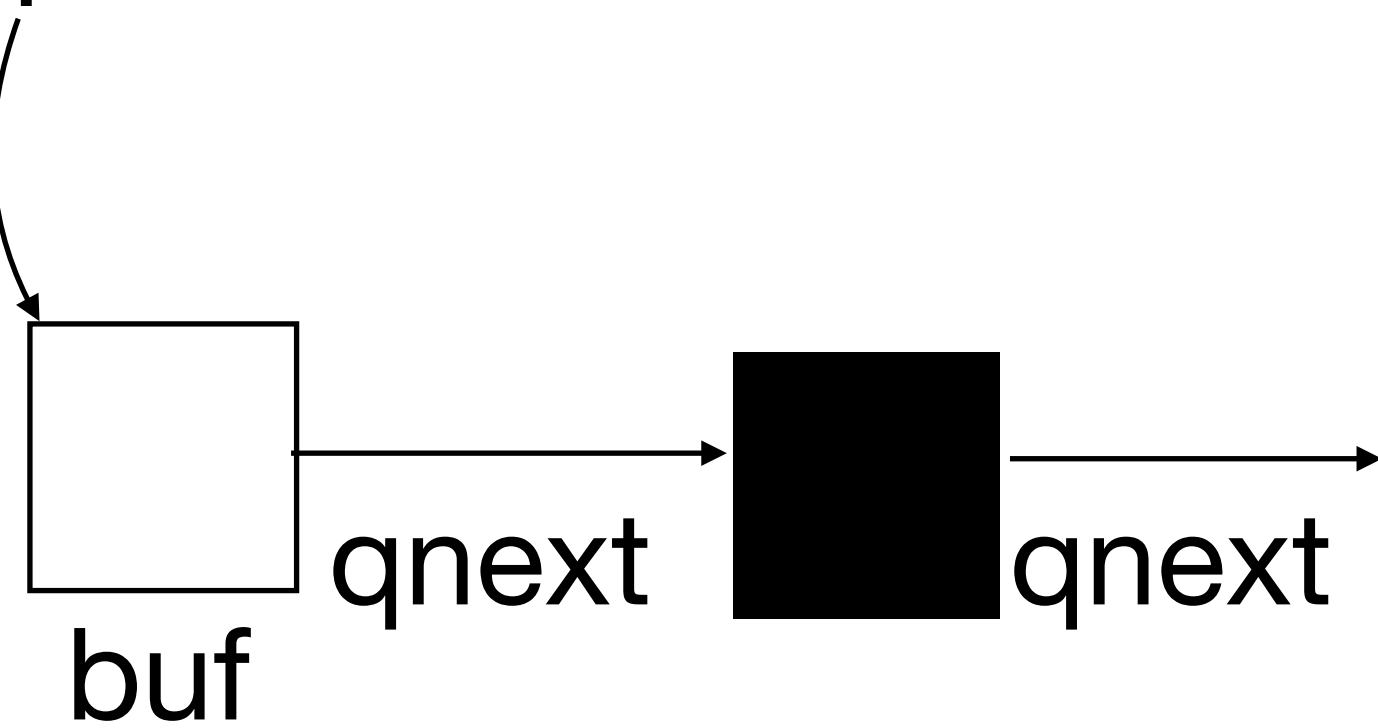


```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```

Race conditions

Example: kernel threads reading a block in ide.c

idequeue



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

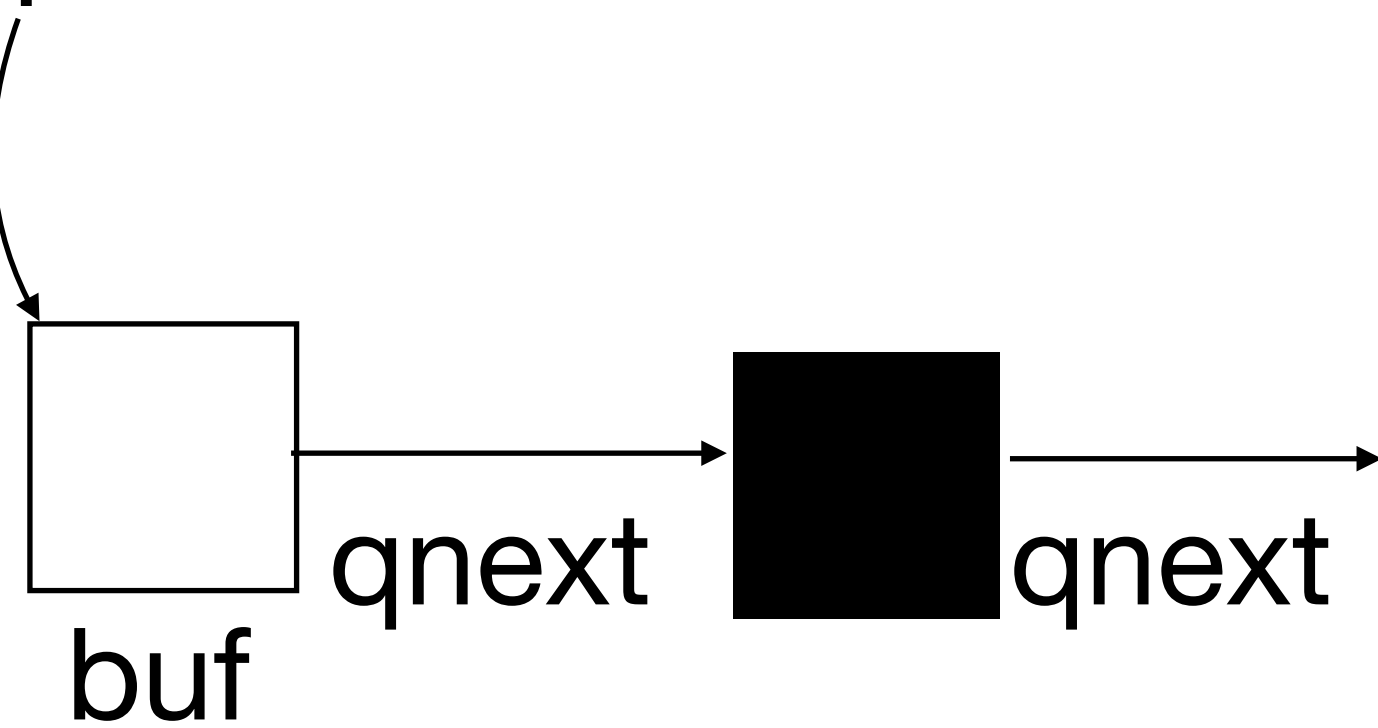
```
..
```

```
}
```

Race conditions

Example: kernel threads reading a block in ide.c

idequeue



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

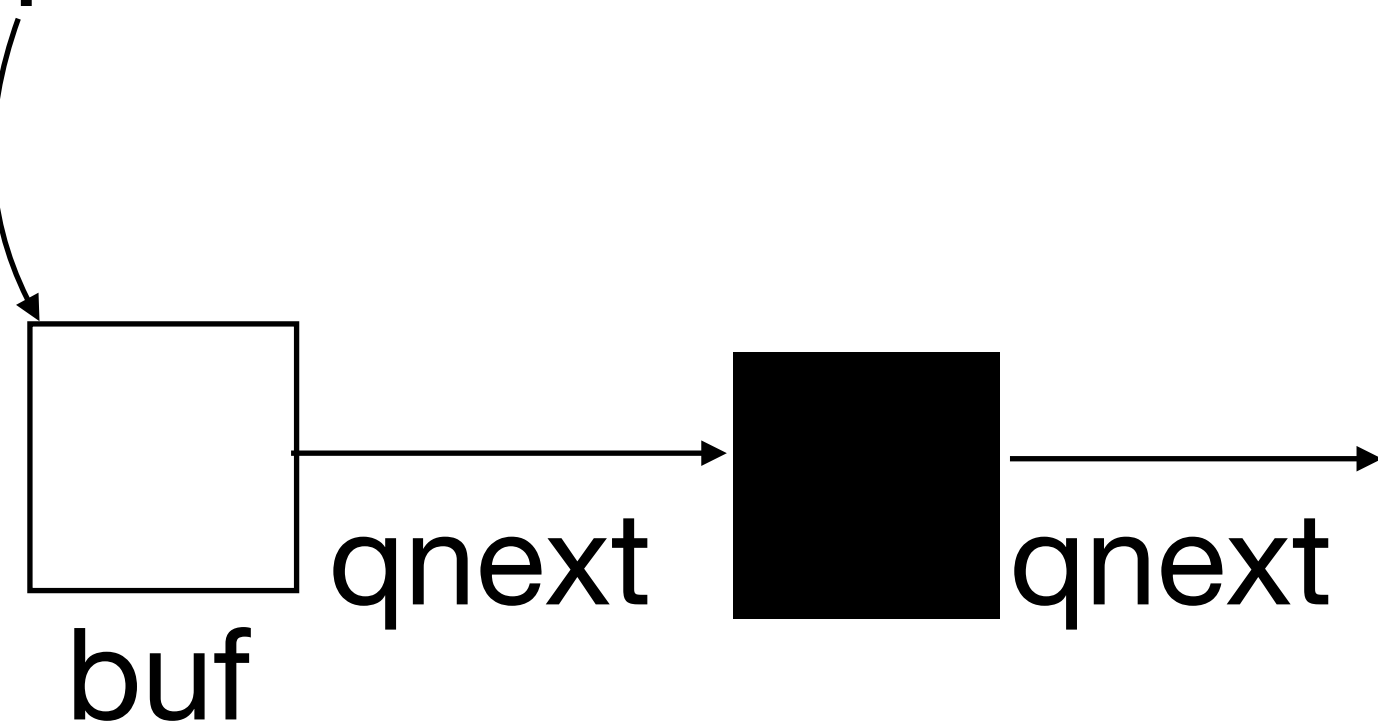
```
..
```

```
}
```


Race conditions

Example: kernel threads reading a block in ide.c

idequeue



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

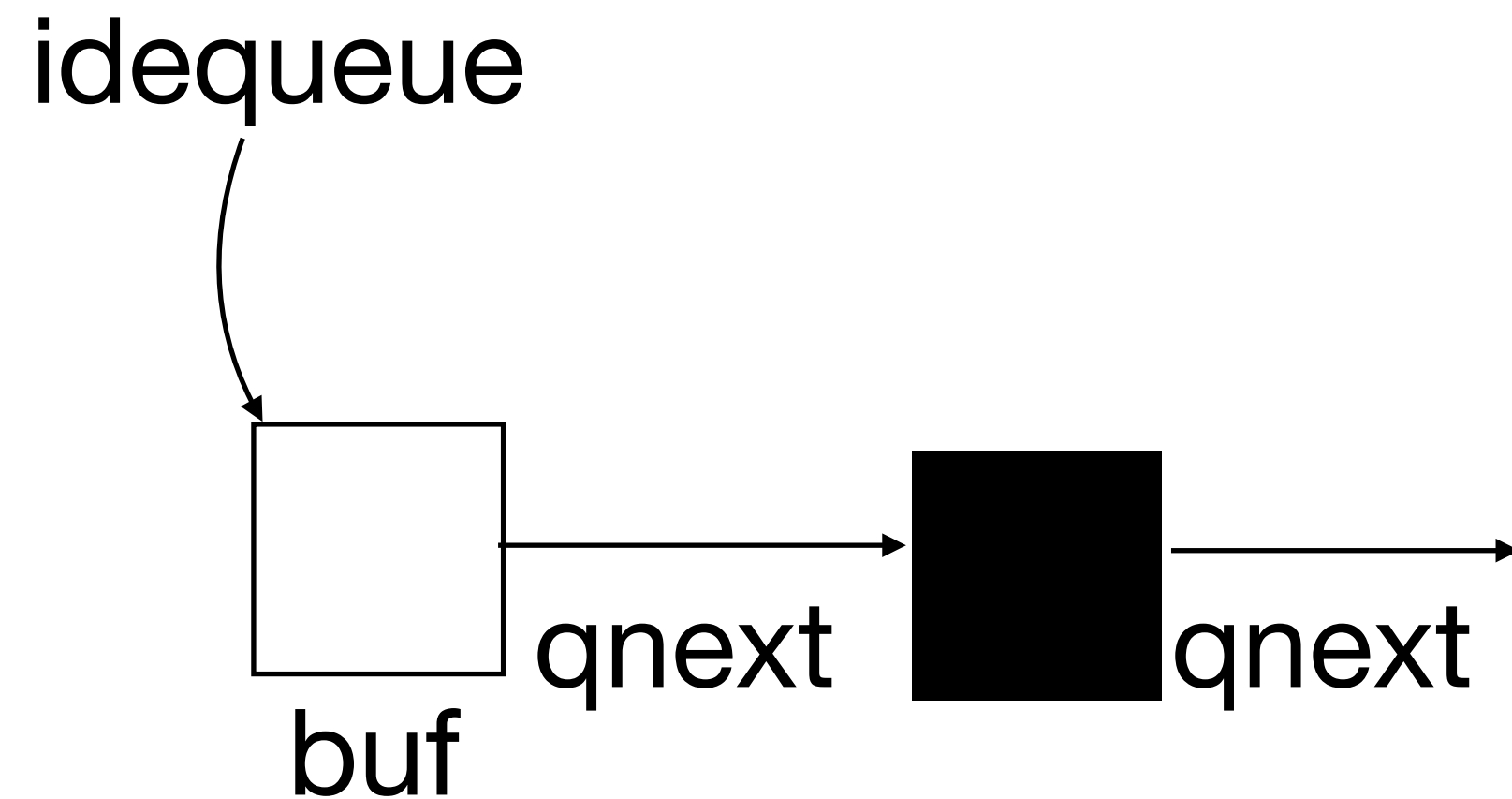
```
    *pp = b;
```

```
..
```

```
}
```

Race conditions

Example: kernel threads reading a block in `ide.c`



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

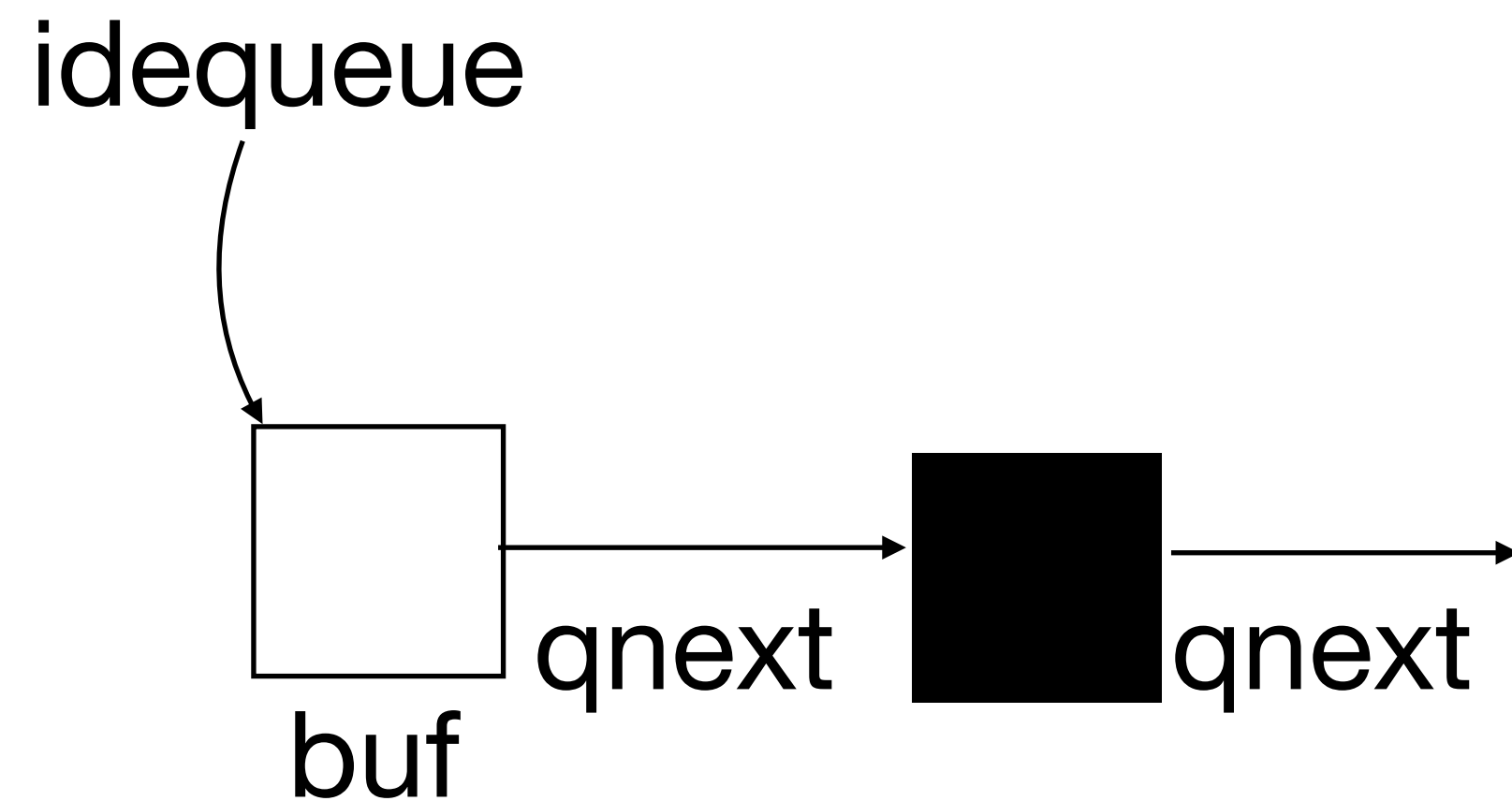
```
..
```

```
}
```

Context switch

Race conditions

Example: kernel threads reading a block in `ide.c`

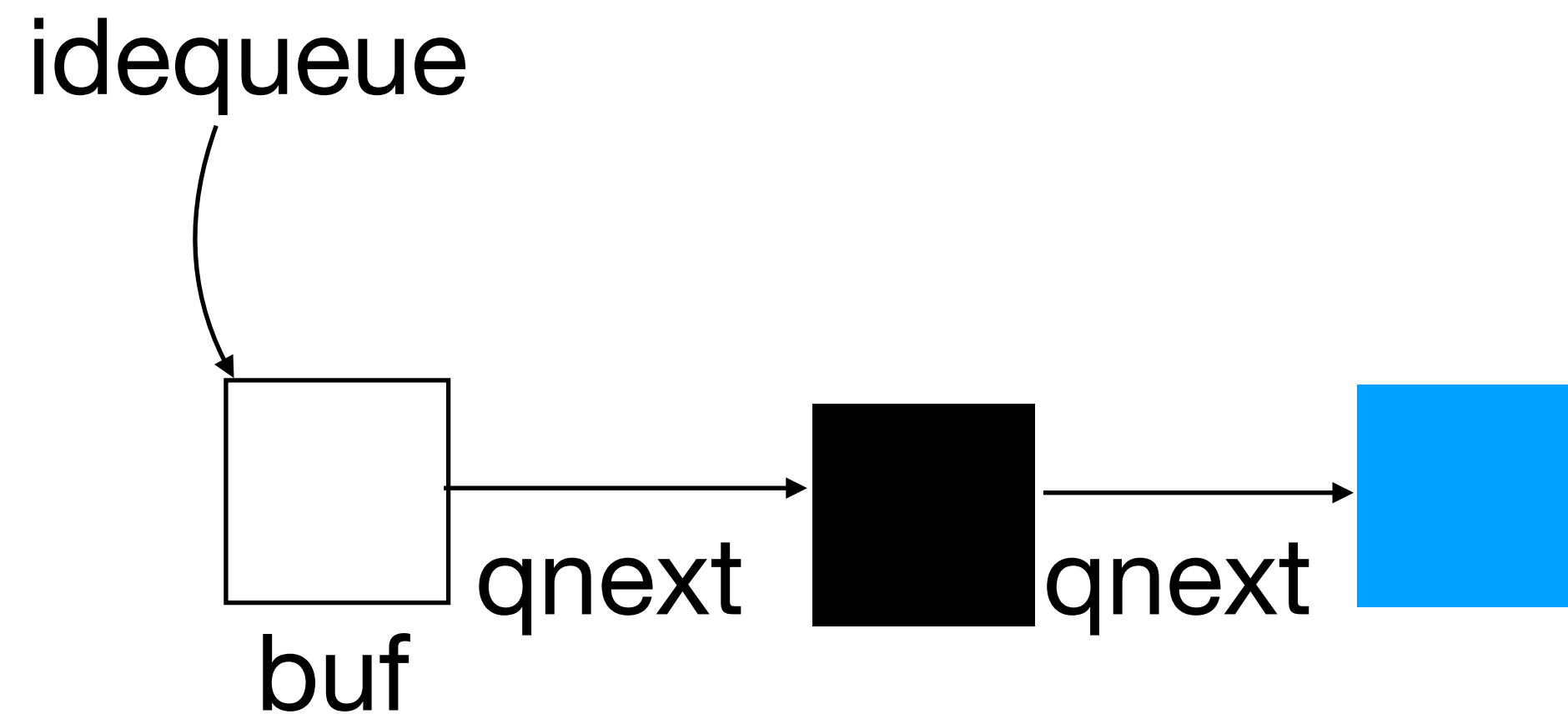


Context switch

```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```


Race conditions

Example: kernel threads reading a block in `ide.c`

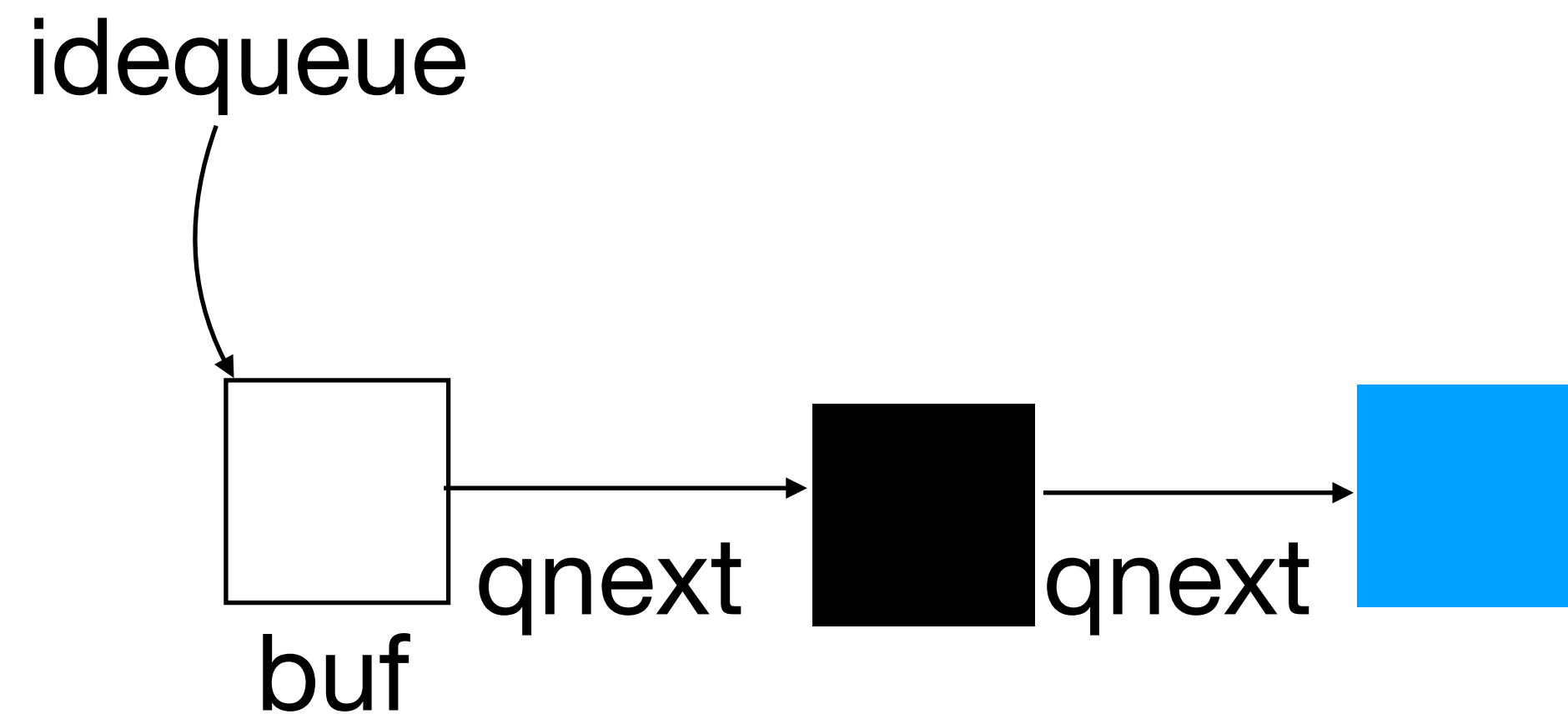


Context switch

```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```


Race conditions

Example: kernel threads reading a block in `ide.c`



Context switch

```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

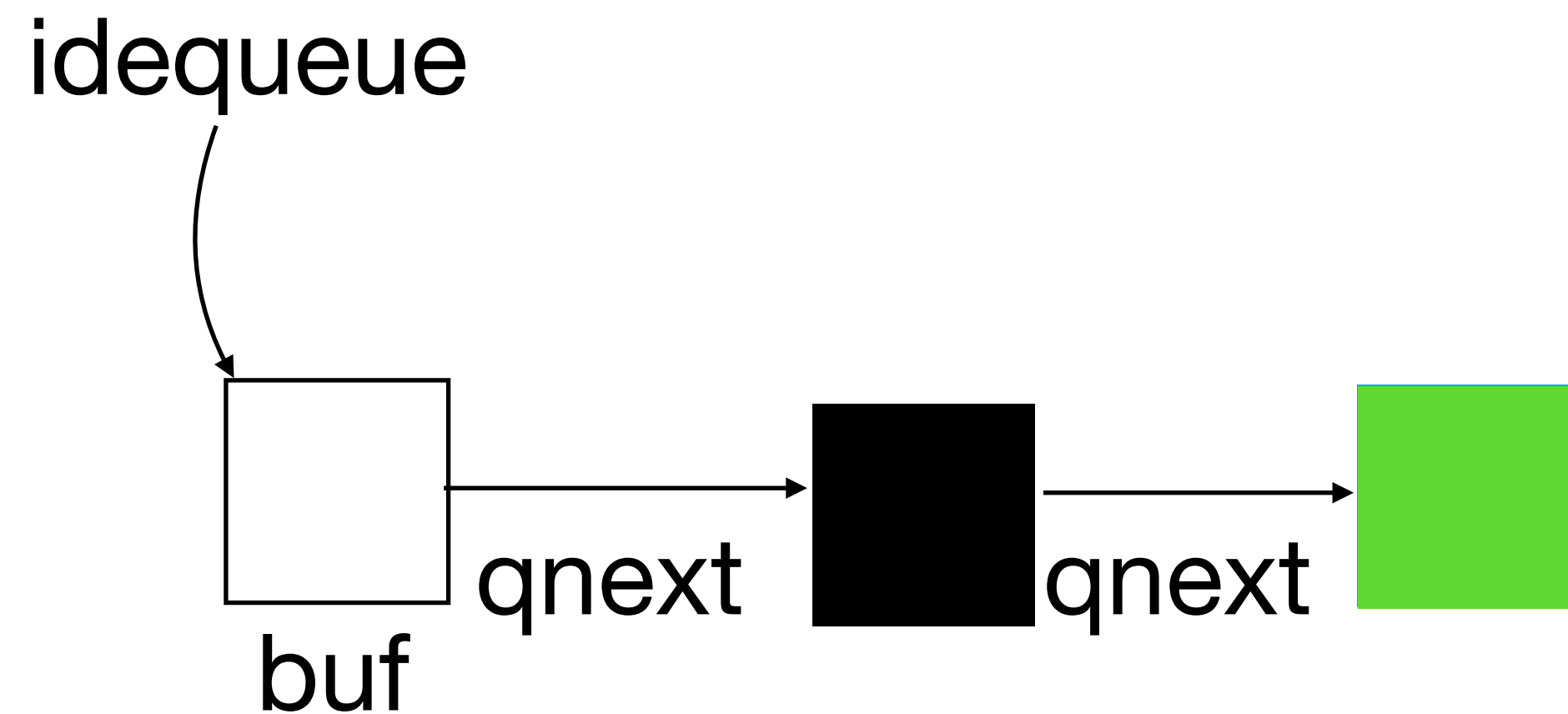
```
    *pp = b;
```

```
..
```

```
}
```

Race conditions

Example: kernel threads reading a block in `ide.c`

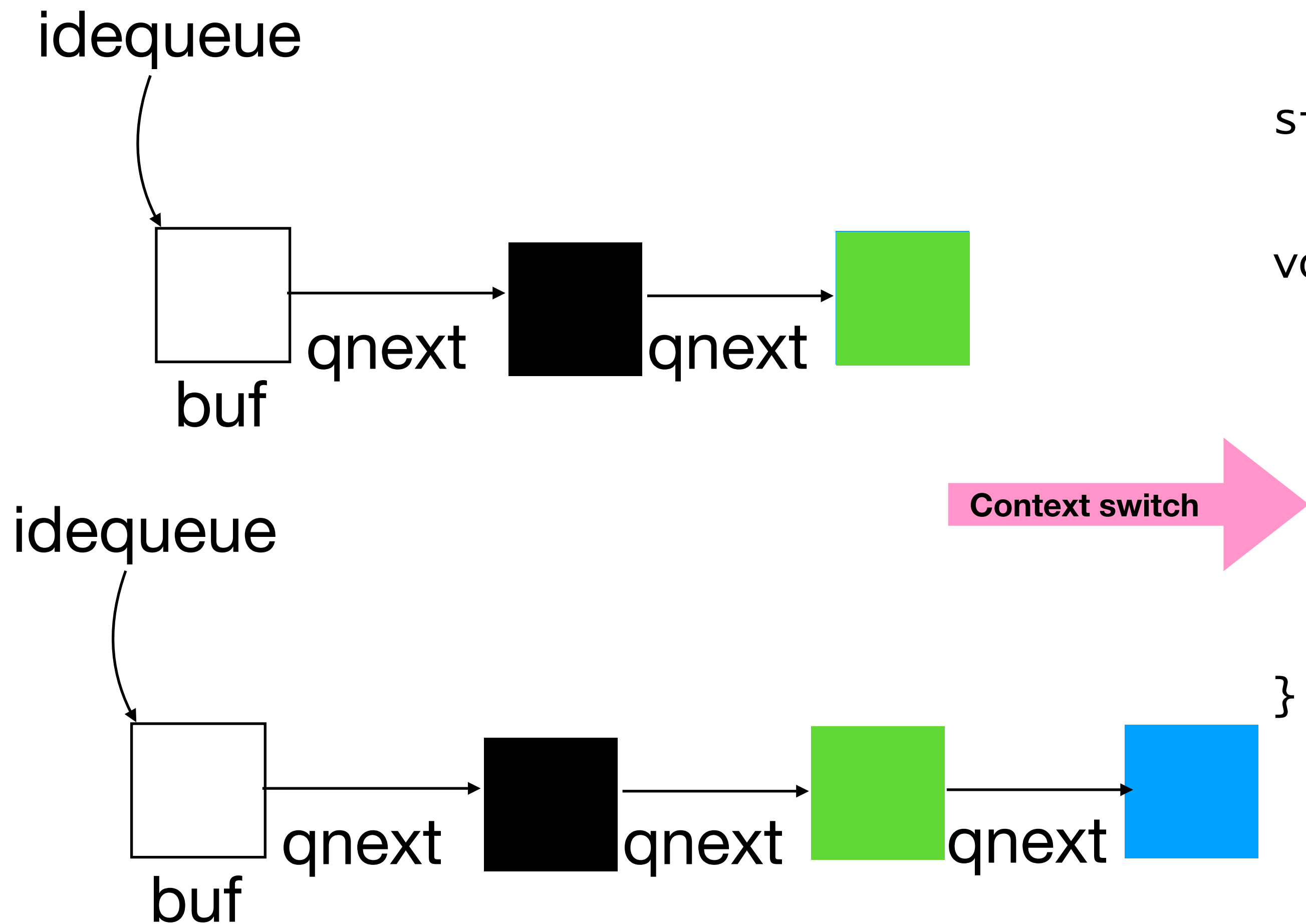


Context switch

```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};  
  
static struct buf *idequeue;  
  
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
}
```

Race conditions

Example: kernel threads reading a block in `ide.c`



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

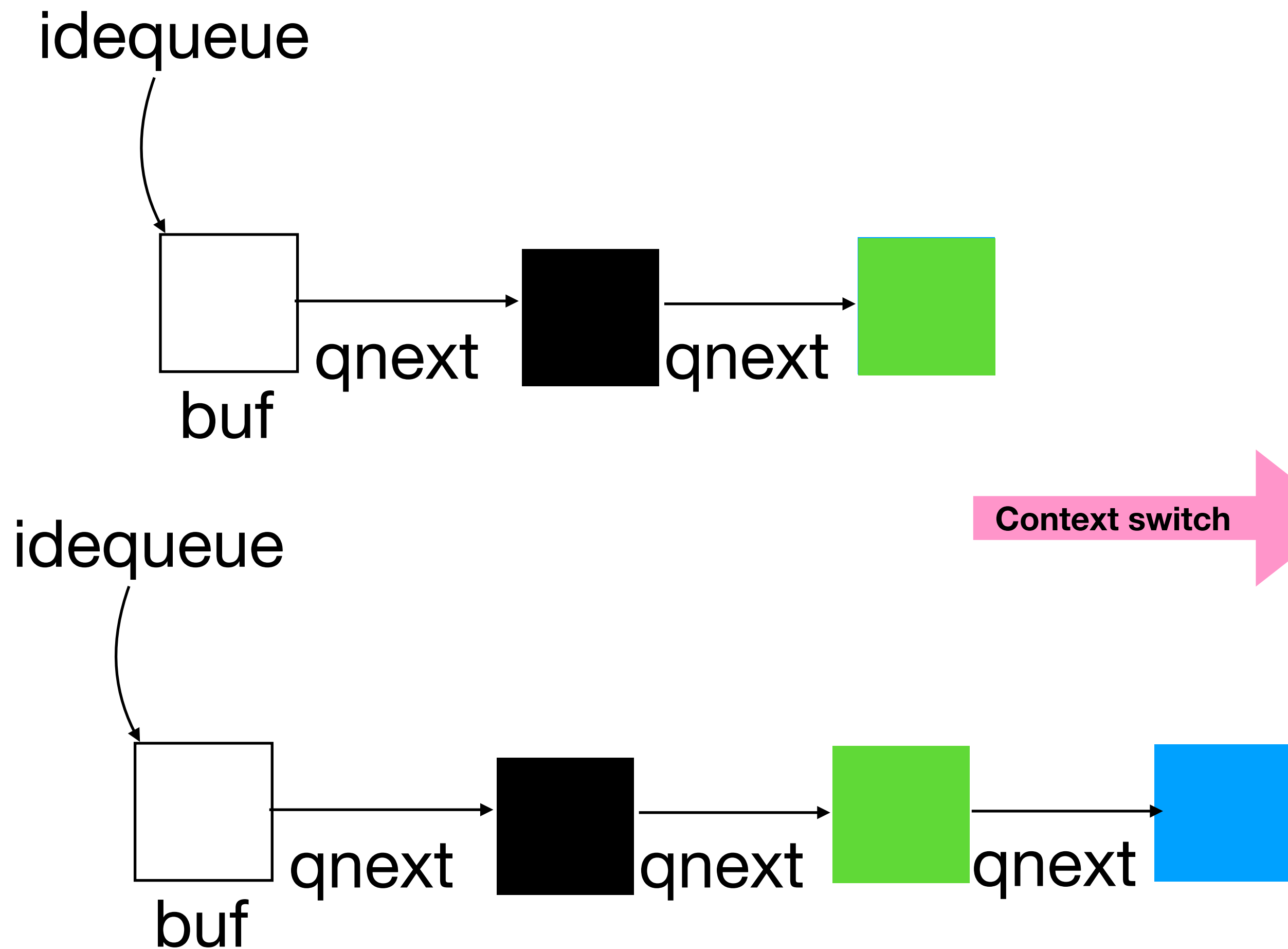
```
    *pp = b;
```

```
..
```

```
}
```


Race conditions

Example: kernel threads reading a block in `ide.c`



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

```
..
```

```
}
```

Lock implementation

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    acquire();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
  
    ..  
    release();  
}
```

Lock implementation

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    acquire();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
    release();  
}
```


Lock implementation

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    acquire();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
    release();  
}
```

```
void acquire() {  
    pushcli();  
}  
void pushcli(void) {  
    int eflags = readeflags();  
    cli();  
    if(cpu->ncli == 0)  
        cpu->intena = eflags & FL_IF;  
    cpu->ncli += 1;  
}
```

Lock implementation

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    acquire();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
    release();  
}
```

```
void acquire() {  
    pushcli();  
}  
void pushcli(void) {  
    int eflags = readeflags();  
    cli();  
    if(cpu->ncli == 0)  
        cpu->intena = eflags & FL_IF;  
    cpu->ncli += 1;  
}  
void release() {  
    popcli();  
}  
void popcli(void) {  
    cpu->ncli--;  
    if(cpu->ncli == 0 && cpu->intena)  
        sti();  
}
```


Lock implementation

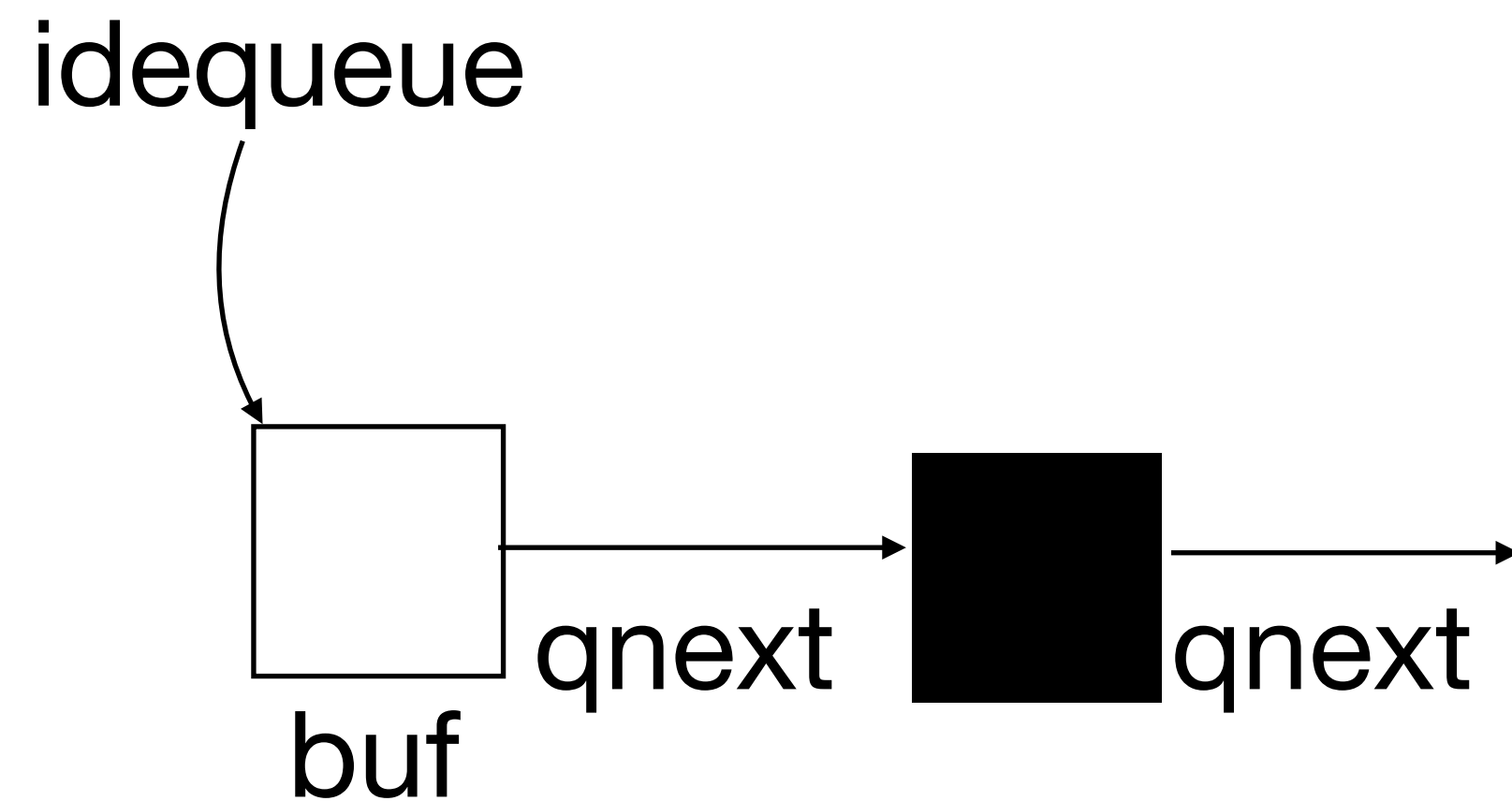
```
void iderw(struct buf *b) {  
    struct buf **pp;  
    acquire();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    ..  
    release();  
}
```

- Timer interrupt and hence context switch cannot happen between acquire and release

```
void acquire() {  
    pushcli();  
}  
void pushcli(void) {  
    int eflags = readeflags();  
    cli();  
    if(cpu->ncli == 0)  
        cpu->intena = eflags & FL_IF;  
    cpu->ncli += 1;  
}  
void release() {  
    popcli();  
}  
void popcli(void) {  
    cpu->ncli--;  
    if(cpu->ncli == 0 && cpu->intena)  
        sti();  
}
```

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {
```

```
..
```

```
    struct buf *qnext; // disk queue
```

```
    uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

```
void iderw(struct buf *b) {
```

```
    struct buf **pp;
```

```
    b->qnext = 0;
```

```
    cli();
```

```
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
    *pp = b;
```

```
    sti();
```

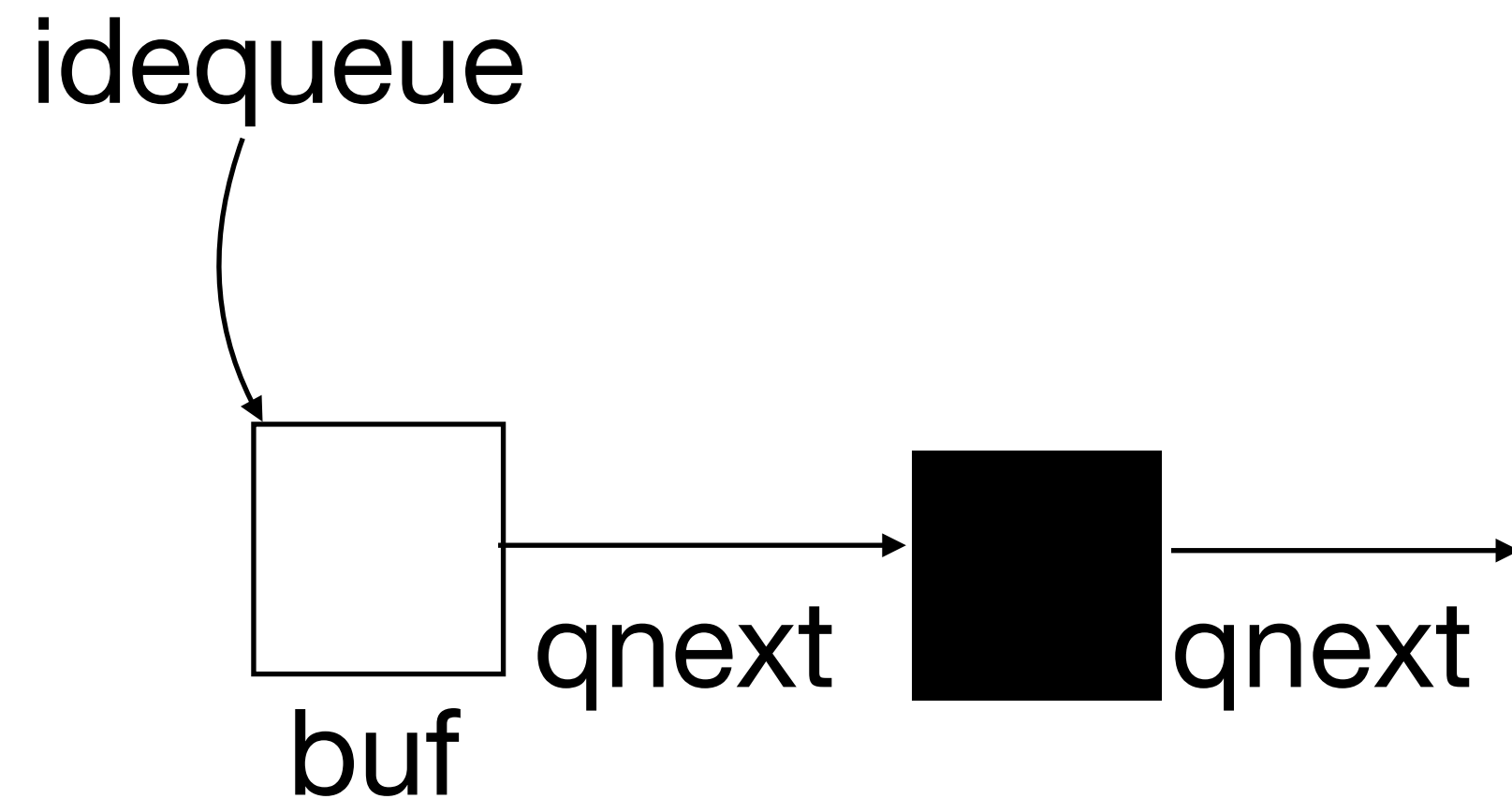
```
..
```

```
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

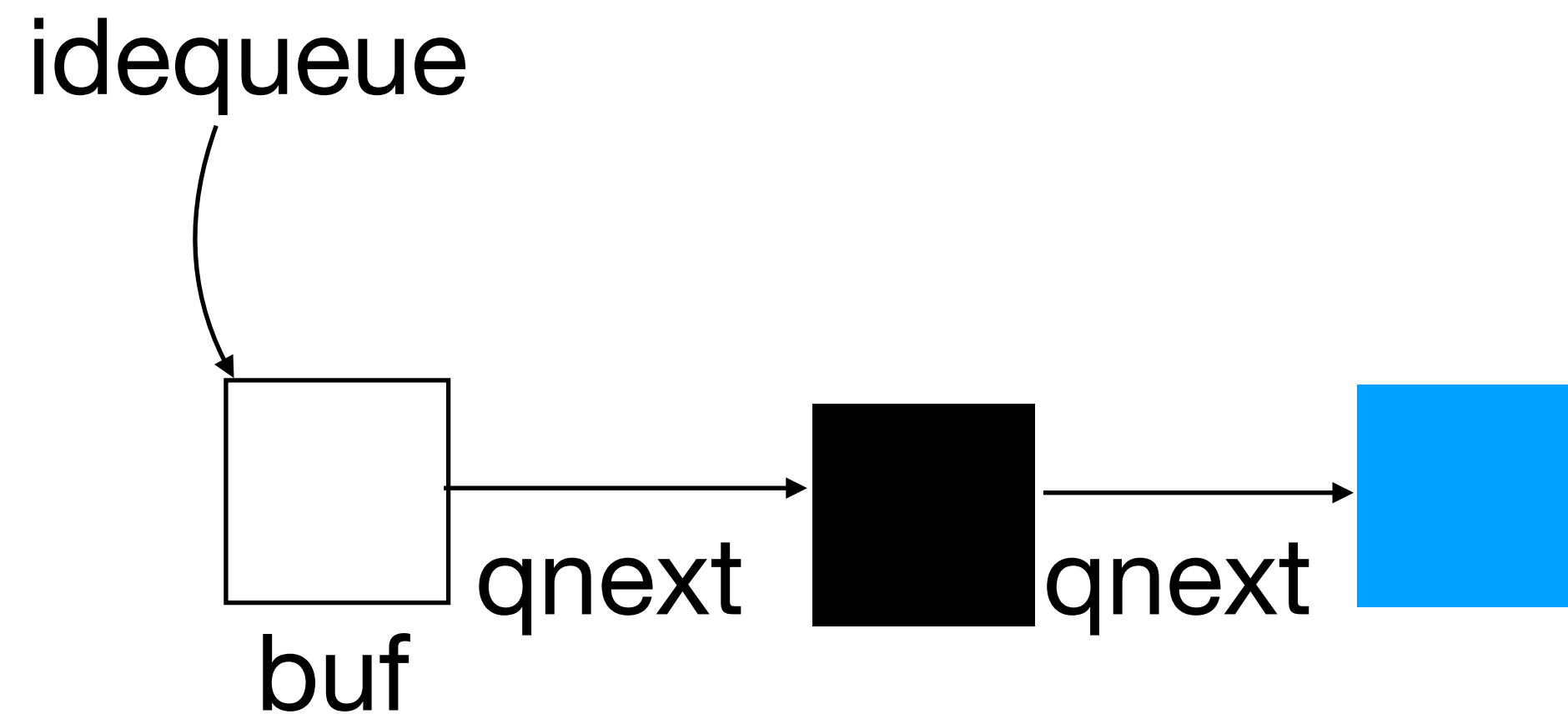
CPU 1

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    cli();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    sti();  
    ..  
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

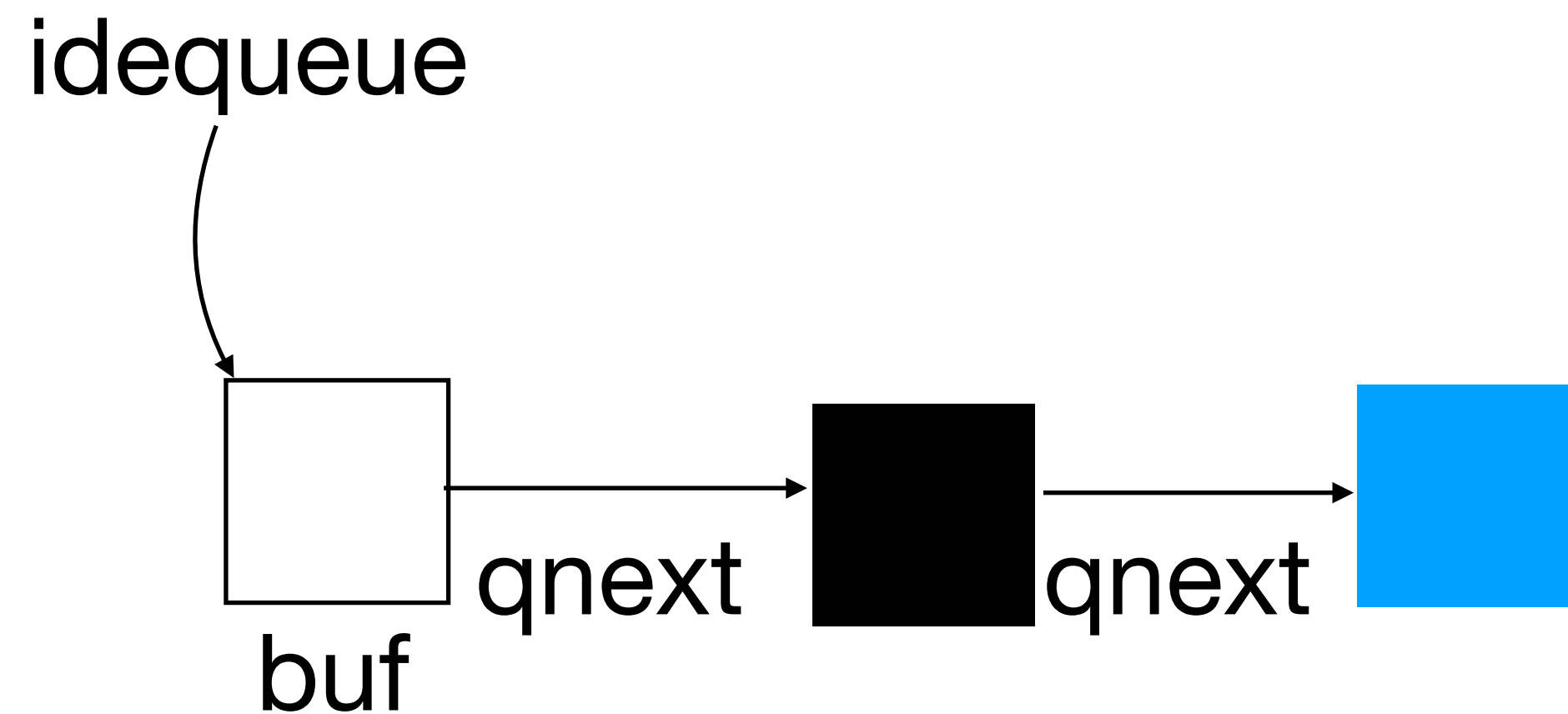
CPU 1

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    cli();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    sti();  
    ..  
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

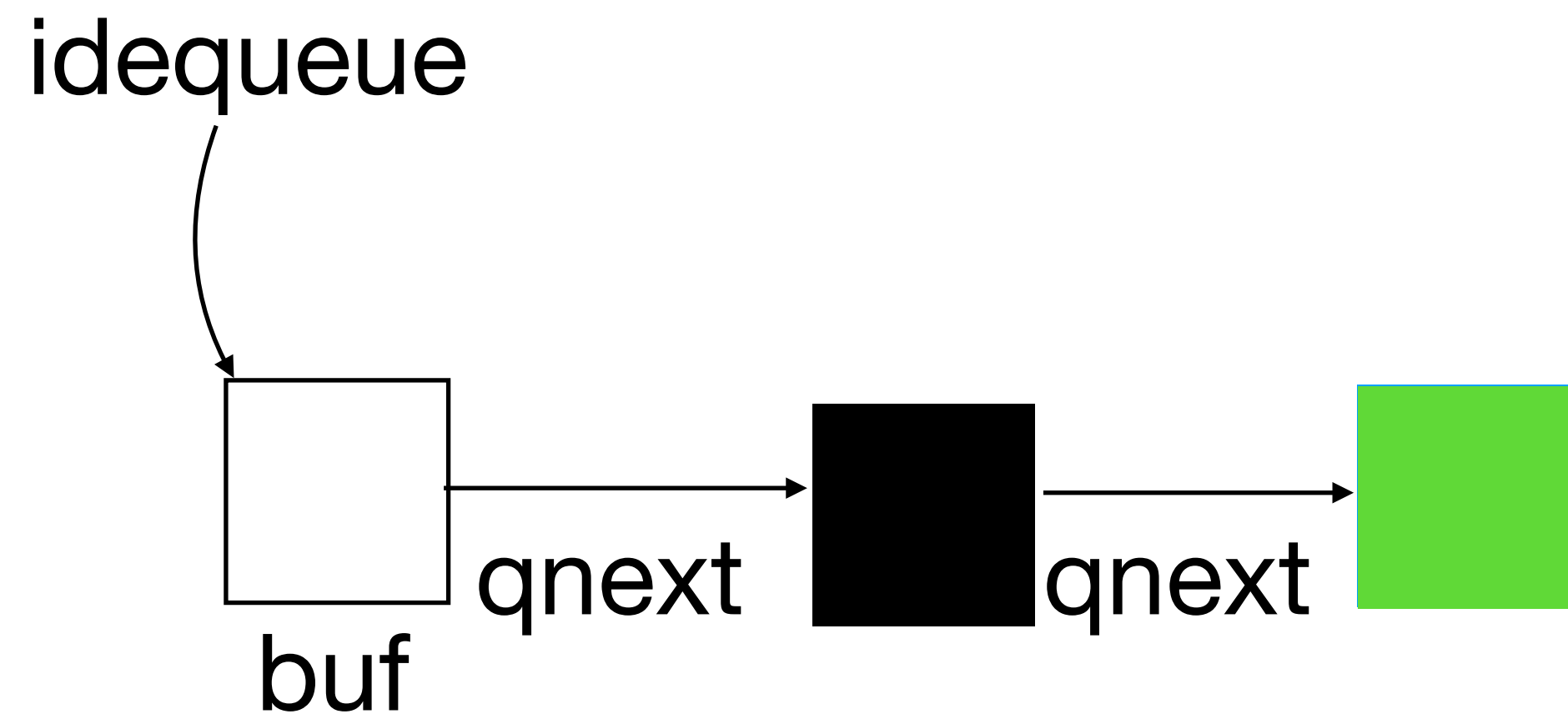
CPU 1

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    cli();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    sti();  
    ..  
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {  
    ..  
    struct buf *qnext; // disk queue  
    uchar data[BSIZE];  
};
```

```
static struct buf *idequeue;
```

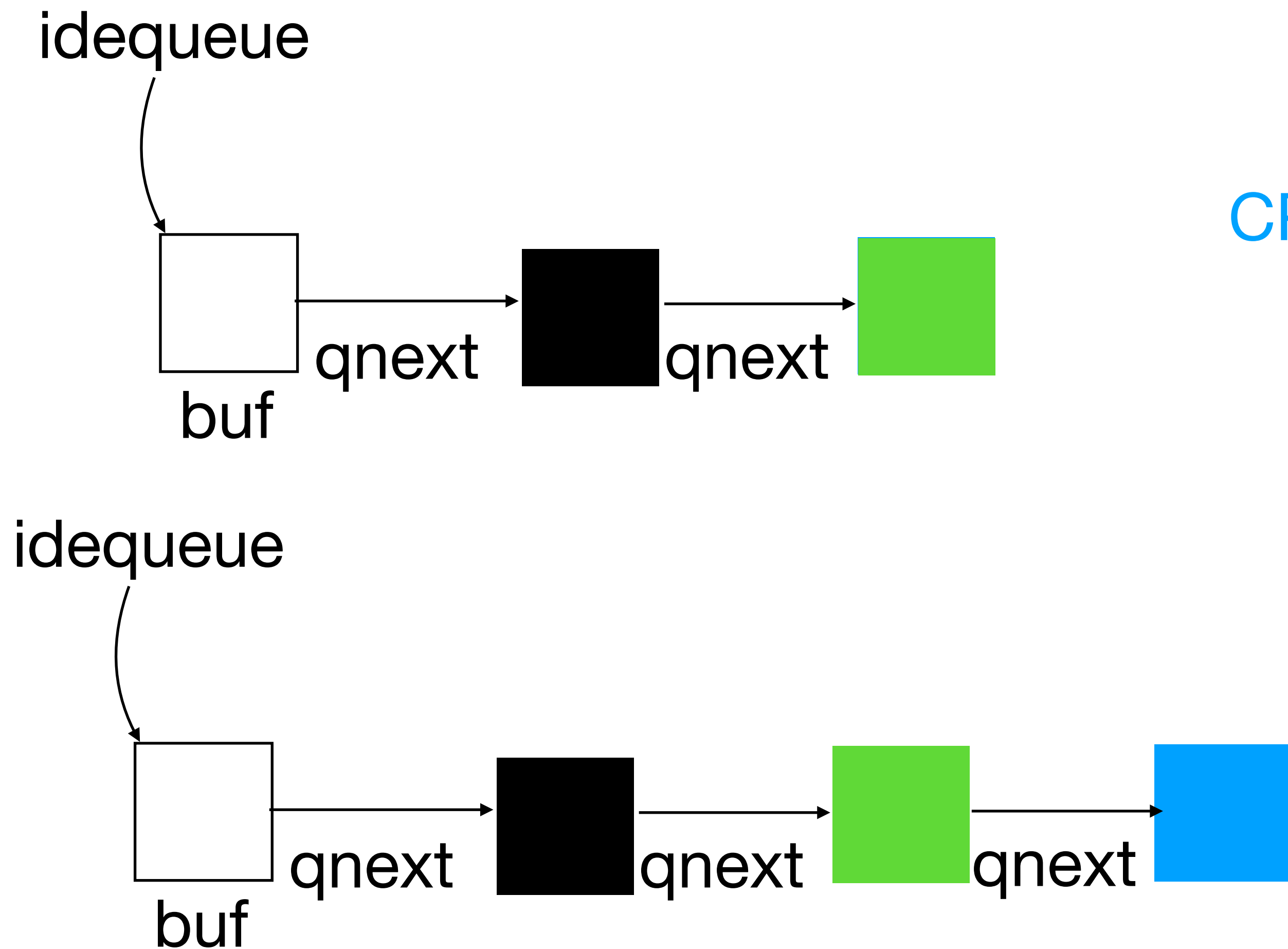
CPU 1

```
void iderw(struct buf *b) {  
    struct buf **pp;  
    b->qnext = 0;  
    cli();  
    for(pp=&idequeue; *pp; pp=&(*pp)->qnext);  
    *pp = b;  
    sti();  
    ..  
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {
```

```
..
```

```
struct buf *qnext; // disk queue
```

```
uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

CPU 1

```
void iderw(struct buf *b) {
```

```
struct buf **pp;
```

```
b->qnext = 0;
```

```
cli();
```

```
for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
*pp = b;
```

```
sti();
```

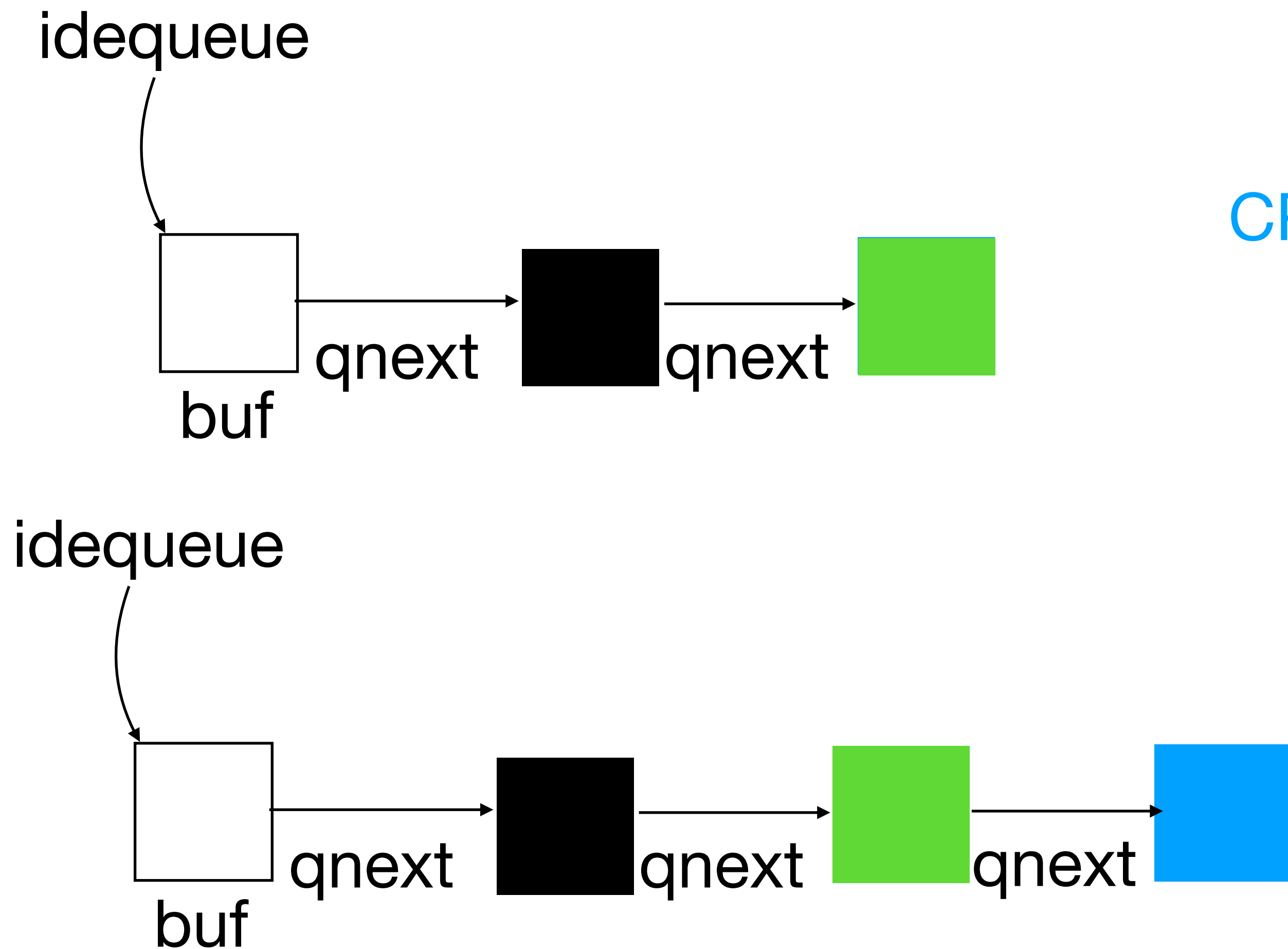
```
..
```

```
}
```

CPU 2

Race conditions

Example: kernel threads reading a block in ide.c



```
struct buf {
```

```
..
```

```
struct buf *qnext; // disk queue
```

```
uchar data[BSIZE];
```

```
};
```

```
static struct buf *idequeue;
```

CPU 1

```
void iderw(struct buf *b) {
```

```
struct buf **pp;
```

```
b->qnext = 0;
```

```
cli();
```

```
for(pp=&idequeue; *pp; pp=&(*pp)->qnext);
```

```
*pp = b;
```

```
sti();
```

```
..
```

```
}
```

CPU 2

Race conditions and critical sections

Similar races can happen in user threads.

- Example: 01/threads.c

Thread 1	Thread 2
Read counter = 0	
Write counter = 1	
	Read counter = 1
	Writer counter = 2
Read counter = 2	
	Read counter = 2
	Writer counter = 3
Writer counter = 3	

Read counter, writer counter
needs to happen atomically

Critical section: “counter++”
09/threads-safe.c

Problems with disabling interrupts

- Does not work on multiple processor
- For user-level code:
 - After acquiring lock, threads goes into infinite loop
 - OS lost control of the CPU

Spin locks

Call to lock spins
while waiting for the
other thread to
unlock

```
1  typedef struct __lock_t { int flag; } lock_t;
2
3  void init(lock_t *mutex) {
4      // 0 -> lock is available, 1 -> held
5      mutex->flag = 0;
6  }
7
8  void lock(lock_t *mutex) {
9      while (mutex->flag == 1)    // TEST the flag
10         ; // spin-wait (do nothing)
11     mutex->flag = 1;           // now SET it!
12 }
13
14 void unlock(lock_t *mutex) {
15     mutex->flag = 0;
16 }
```


Spin locks

Call to lock spins
while waiting for the
other thread to
unlock

```
1  typedef struct __lock_t { int flag; } lock_t;
2
3  void init(lock_t *mutex) {
4      // 0 -> lock is available, 1 -> held
5      mutex->flag = 0;
6  }
7
8  void lock(lock_t *mutex) {
9      while (mutex->flag == 1) // TEST the flag
10         ; // spin-wait (do nothing)
11     mutex->flag = 1; // now SET it!
12 }
13
14 void unlock(lock_t *mutex) {
15     mutex->flag = 0;
16 }
```

CPU 2



Spin locks

Call to lock spins
while waiting for the
other thread to
unlock

CPU 1



CPU 2

```
1  typedef struct __lock_t { int flag; } lock_t;
2
3  void init(lock_t *mutex) {
4      // 0 -> lock is available, 1 -> held
5      mutex->flag = 0;
6  }
7
8  void lock(lock_t *mutex) {
9      while (mutex->flag == 1) // TEST the flag
10         ; // spin-wait (do nothing)
11     mutex->flag = 1; // now SET it!
12 }
13
14 void unlock(lock_t *mutex) {
15     mutex->flag = 0;
16 }
```

Spin locks

Call to lock spins while waiting for the other thread to unlock

CPU 1



CPU 2



```
1  typedef struct __lock_t { int flag; } lock_t;
2
3  void init(lock_t *mutex) {
4      // 0 -> lock is available, 1 -> held
5      mutex->flag = 0;
6  }
7
8  void lock(lock_t *mutex) {
9      while (mutex->flag == 1) // TEST the flag
10         ; // spin-wait (do nothing)
11     mutex->flag = 1; // now SET it!
12 }
13
14 void unlock(lock_t *mutex) {
15     mutex->flag = 0;
16 }
```


Spin locks

Call to lock spins while waiting for the other thread to unlock

CPU 1



CPU 2



```
1  typedef struct __lock_t { int flag; } lock_t;
2
3  void init(lock_t *mutex) {
4      // 0 -> lock is available, 1 -> held
5      mutex->flag = 0;
6  }
7
8  void lock(lock_t *mutex) {
9      while (mutex->flag == 1) // TEST the flag
10         ; // spin-wait (do nothing)
11     mutex->flag = 1; // now SET it!
12 }
13
14 void unlock(lock_t *mutex) {
15     mutex->flag = 0;
16 }
```

Write to two different flags to avoid races?

```
int flag[2];

void init() {
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock
}

void lock() {
    flag[self] = 1; // self: thread ID of caller
    while (flag[1 - self] == 1) ; // spin-wait
}

void unlock() {
    flag[self] = 0; // simply undo your intent
}
```


Write to two different flags to avoid races?

```
int flag[2];

void init() {
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock
}
```

CPU 1

CPU 2

```
void lock() {
    flag[self] = 1; // self: thread ID of caller
    while (flag[1 - self] == 1) ; // spin-wait
}
```

```
void unlock() {
    flag[self] = 0; // simply undo your intent
}
```

Deadlock

flag[0] = 1	
	flag[1] = 1
	while(flag[0] == 1);
while(flag[1] == 1);	

Safety and liveness

- Safety: Bad things never happen
 - Two threads shall never *simultaneously* acquire the lock
- Liveness: Good things eventually happen
 - Some thread (trying to lock) *eventually* gets to acquire the lock

Trivial useless locks

Safe but not live

```
void lock() {  
    while(1);  
}
```

Live but not safe

```
void lock() {  
    return;  
}
```


Peterson's algorithm*

```
int flag[2];
int turn = 0; // whose turn? (thread 0 or 1?)

void init() {
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock
}

void lock() {
    flag[self] = 1; // self: thread ID of caller
    turn = self; // make it my turn
    while ((flag[1 - self] == 1) && (turn == 1 - self)) ; // spin-wait
}

void unlock() {
    flag[self] = 0; // simply undo your intent
}
```

Peterson's algorithm*

turn breaks the tie

<code>while(flag[1] == 1 && turn == 1);</code>	<code>while(flag[0] == 1 && turn == 0);</code>
--	--

```
int flag[2];  
int turn = 0; // whose turn? (thread 0 or 1?)
```

```
void init() {  
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock  
}
```

```
void lock() {  
    flag[self] = 1; // self: thread ID of caller  
    turn = self; // make it my turn  
    while ((flag[1 - self] == 1) && (turn == 1 - self)) ; // spin-wait  
}
```

```
void unlock() {  
    flag[self] = 0; // simply undo your intent  
}
```

Peterson's algorithm*

```
void lock() {  
    flag[0] = 1;  
    turn = 0;    // B  
    while ((flag[1] == 1) && (turn == 1)); //C  
}
```

```
void lock() {  
    flag[1] = 1;  
    turn = 1;    // E  
    while ((flag[0] == 1) && (turn == 0)); // F  
}
```


Peterson's algorithm*

```
void lock() {  
    flag[0] = 1;  
    turn = 0;    // B  
    while ((flag[1] == 1) && (turn == 1)); //C  
}
```

```
void lock() {  
    flag[1] = 1;  
    turn = 1;    // E  
    while ((flag[0] == 1) && (turn == 0)); // F  
}
```

B: turn = 0
E: turn = 1
C: turn != 1
F: turn != 0

Peterson's algorithm*

```
void lock() {
    flag[0] = 1;
    turn = 0;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 1;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 0	B: turn = 0
E: turn = 1	E: turn = 1
C: turn != 1	F: turn != 0
F: turn != 0	C: turn != 1

Peterson's algorithm*

```
void lock() {
    flag[0] = 1;
    turn = 0;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 1;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 0	B: turn = 0	B: turn = 0
E: turn = 1	E: turn = 1	C: turn != 1
C: turn != 1	F: turn != 0	E: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0

Peterson's algorithm*

```
void lock() {
    flag[0] = 1;
    turn = 0;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 1;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 0	B: turn = 0	B: turn = 0	E: turn = 1
E: turn = 1	E: turn = 1	C: turn != 1	B: turn = 0
C: turn != 1	F: turn != 0	E: turn = 1	F: turn != 0
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1

Peterson's algorithm*

```
void lock() {
    flag[0] = 1;
    turn = 0;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 1;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 0	B: turn = 0	B: turn = 0	E: turn = 1	E: turn = 1
E: turn = 1	E: turn = 1	C: turn != 1	B: turn = 0	B: turn = 0
C: turn != 1	F: turn != 0	E: turn = 1	F: turn != 0	C: turn != 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0

Peterson's algorithm*

```
void lock() {
    flag[0] = 1;
    turn = 0;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 1;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 0	B: turn = 0	B: turn = 0	E: turn = 1	E: turn = 1	E: turn = 1
E: turn = 1	E: turn = 1	C: turn != 1	B: turn = 0	B: turn = 0	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 1	F: turn != 0	C: turn != 1	B: turn = 0
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1

Peterson's algorithm

```
int flag[2];
int turn = 0; // whose turn? (thread 0 or 1?)

void init() {
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock
}

void lock() {
    flag[self] = 1; // self: thread ID of caller

    turn = 1 - self; // make it other thread's turn
    while ((flag[1 - self] == 1) && (turn == 1 - self)) ; // spin-wait
}

void unlock() {
    flag[self] = 0; // simply undo your intent
}
```

Peterson's algorithm

```
void lock() {  
    flag[0] = 1;  
    turn = 1;    // B  
    while ((flag[1] == 1) && (turn == 1)); //C  
}
```

```
void lock() {  
    flag[1] = 1;  
    turn = 0;    // E  
    while ((flag[0] == 1) && (turn == 0)); // F  
}
```


Peterson's algorithm

```
void lock() {  
    flag[0] = 1;  
    turn = 1;    // B  
    while ((flag[1] == 1) && (turn == 1)); //C  
}
```

```
void lock() {  
    flag[1] = 1;  
    turn = 0;    // E  
    while ((flag[0] == 1) && (turn == 0)); // F  
}
```

B: turn = 1
E: turn = 0
C: turn != 1
F: turn != 0

Peterson's algorithm

```
void lock() {
    flag[0] = 1;
    turn = 1;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 0;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 1	B: turn = 1
E: turn = 0	E: turn = 0
C: turn != 1	F: turn != 0
F: turn != 0	C: turn != 1

Peterson's algorithm

```
void lock() {
    flag[0] = 1;
    turn = 1;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 0;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 1	B: turn = 1	B: turn = 1
E: turn = 0	E: turn = 0	C: turn != 1
C: turn != 1	F: turn != 0	E: turn = 0
F: turn != 0	C: turn != 1	F: turn != 0
		C: turn != 1

Peterson's algorithm

```
void lock() {
    flag[0] = 1;
    turn = 1;    // B
    while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
    flag[1] = 1;
    turn = 0;    // E
    while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1	

Peterson's algorithm

```
void lock() {
  flag[0] = 1;
  turn = 1;    // B
  while ((flag[1] == 1) && (turn == 1)); //C
}
```

```
void lock() {
  flag[1] = 1;
  turn = 0;    // E
  while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0
		C: turn != 1		

Peterson's algorithm

```
void lock() {
  flag[0] = 1;
  turn = 1;    // B
  while ((flag[1] == 1) && (turn == 1)); //C
}
```

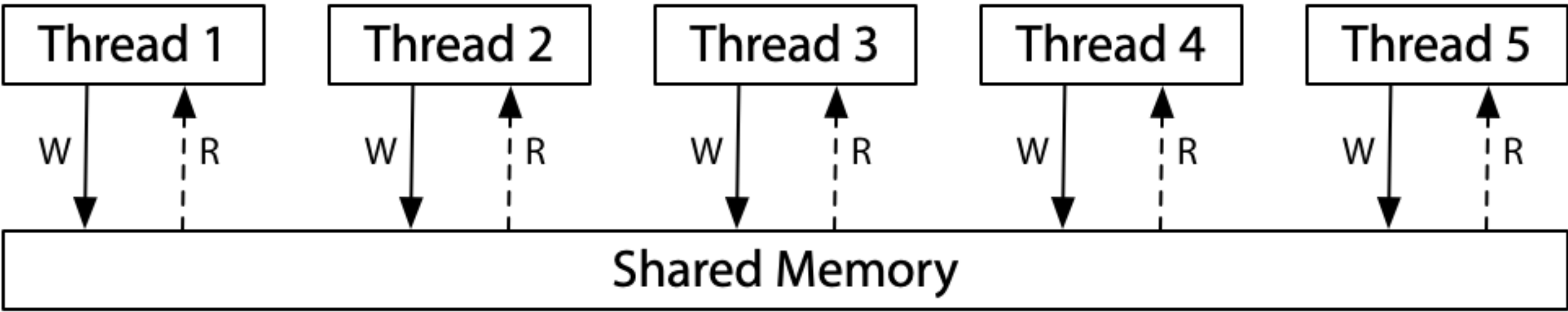
```
void lock() {
  flag[1] = 1;
  turn = 0;    // E
  while ((flag[0] == 1) && (turn == 0)); // F
}
```

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1	B: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1			F: turn != 0

Peterson algorithm breaks on x86 peterson-breaks.c

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1	B: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1			F: turn != 0

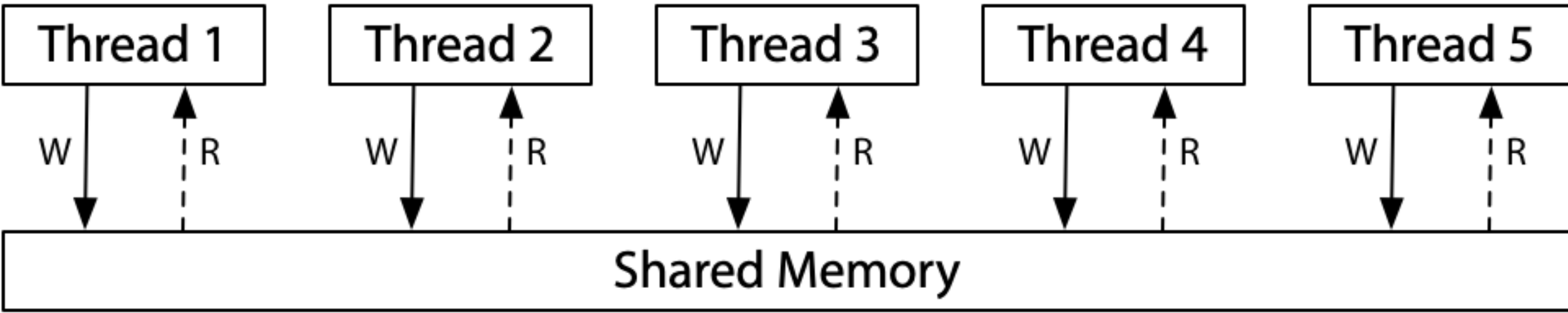
- Correctness analysis assumed “sequential consistency”:



Peterson algorithm breaks on x86 peterson-breaks.c

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1	B: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1			F: turn != 0

- Correctness analysis assumed “sequential consistency”:
 - CPU is executing one instruction after another

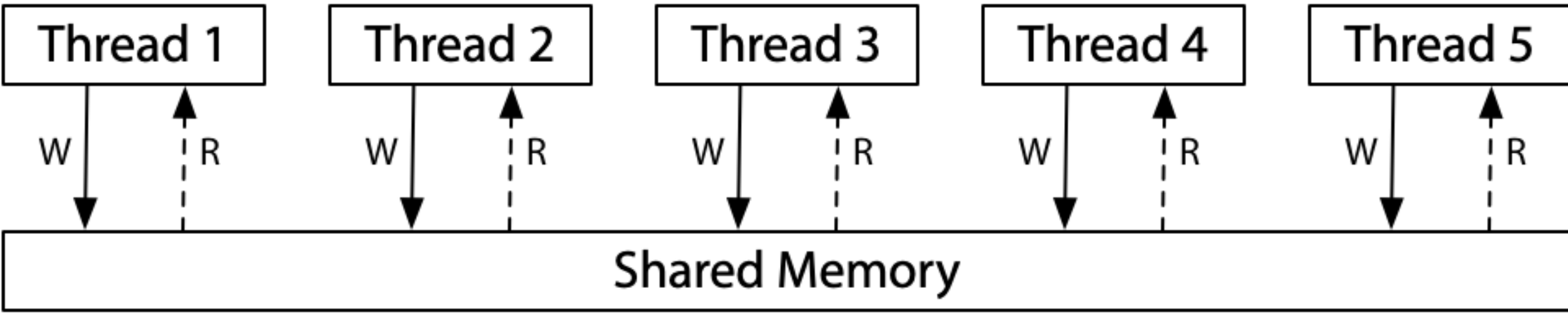


Peterson algorithm breaks on x86

peterson-breaks.c

B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1	B: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1			F: turn != 0

- Correctness analysis assumed “sequential consistency”:
 - CPU is executing one instruction after another
 - All reads and writes are served by shared memory one-at-a-time. No caches!

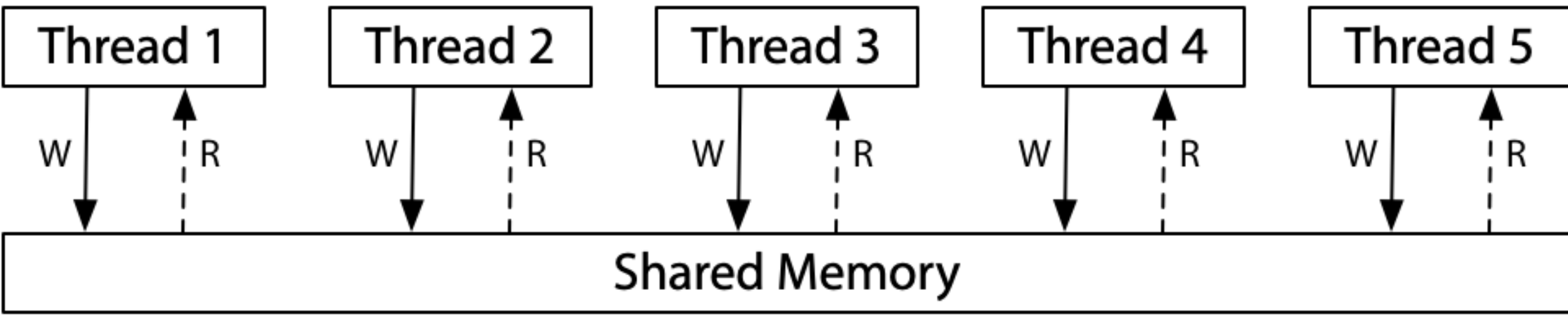


Peterson algorithm breaks on x86

peterson-breaks.c

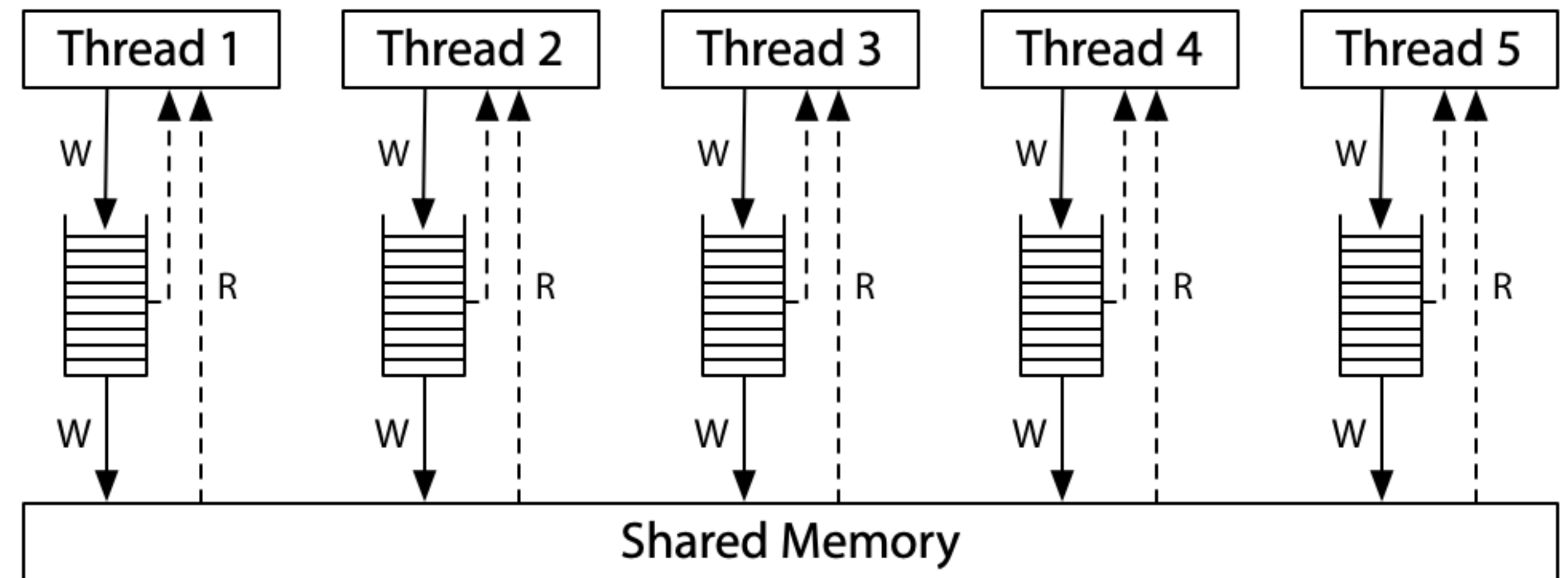
B: turn = 1	B: turn = 1	B: turn = 1	E: turn = 0	E: turn = 0	E: turn = 0
E: turn = 0	E: turn = 0	C: turn != 1	B: turn = 1	B: turn = 1	F: turn != 0
C: turn != 1	F: turn != 0	E: turn = 0	F: turn != 0	C: turn != 1	B: turn = 1
F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1	F: turn != 0	C: turn != 1
		C: turn != 1			F: turn != 0

- Correctness analysis assumed “sequential consistency”:
 - CPU is executing one instruction after another
 - All reads and writes are served by shared memory one-at-a-time. No caches!
 - Parallel executions are just interleavings of sequential executions



CPU's do not follow sequential consistency!

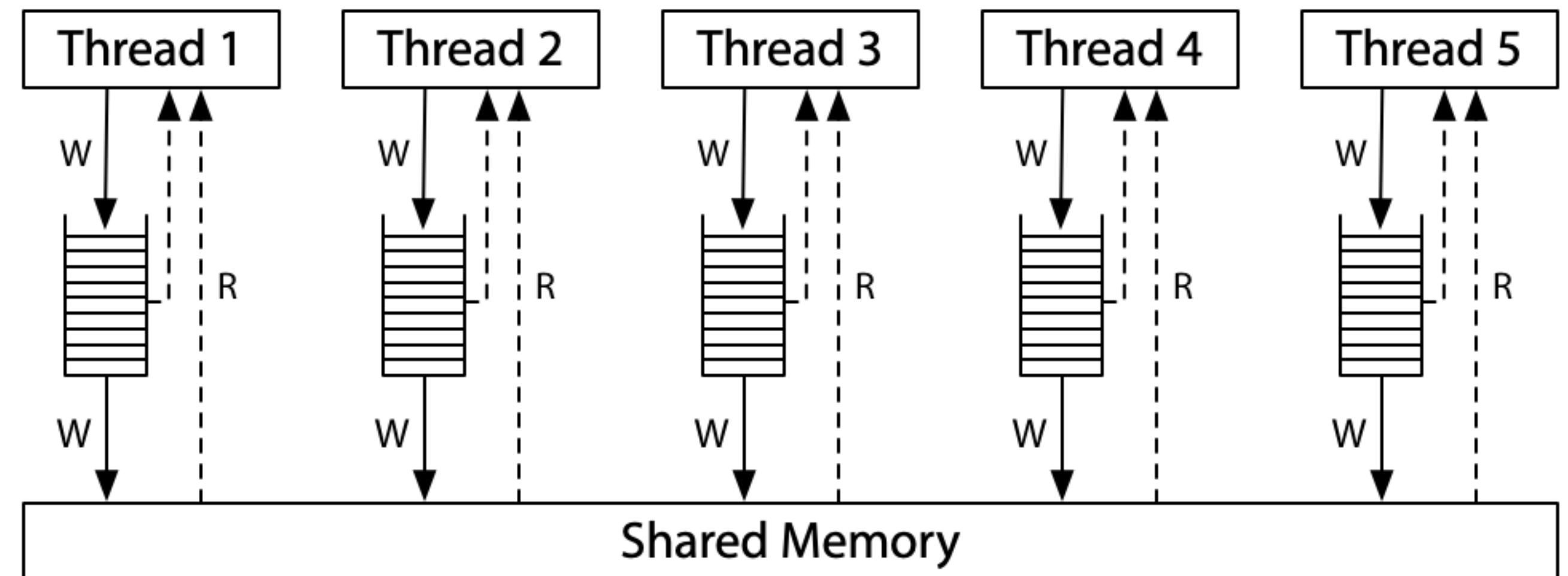
x86 follows “Total Store Order” (TSO). Intel SDM Vol 3A, Section 8.2



CPU's do not follow sequential consistency!

x86 follows “Total Store Order” (TSO). Intel SDM Vol 3A, Section 8.2

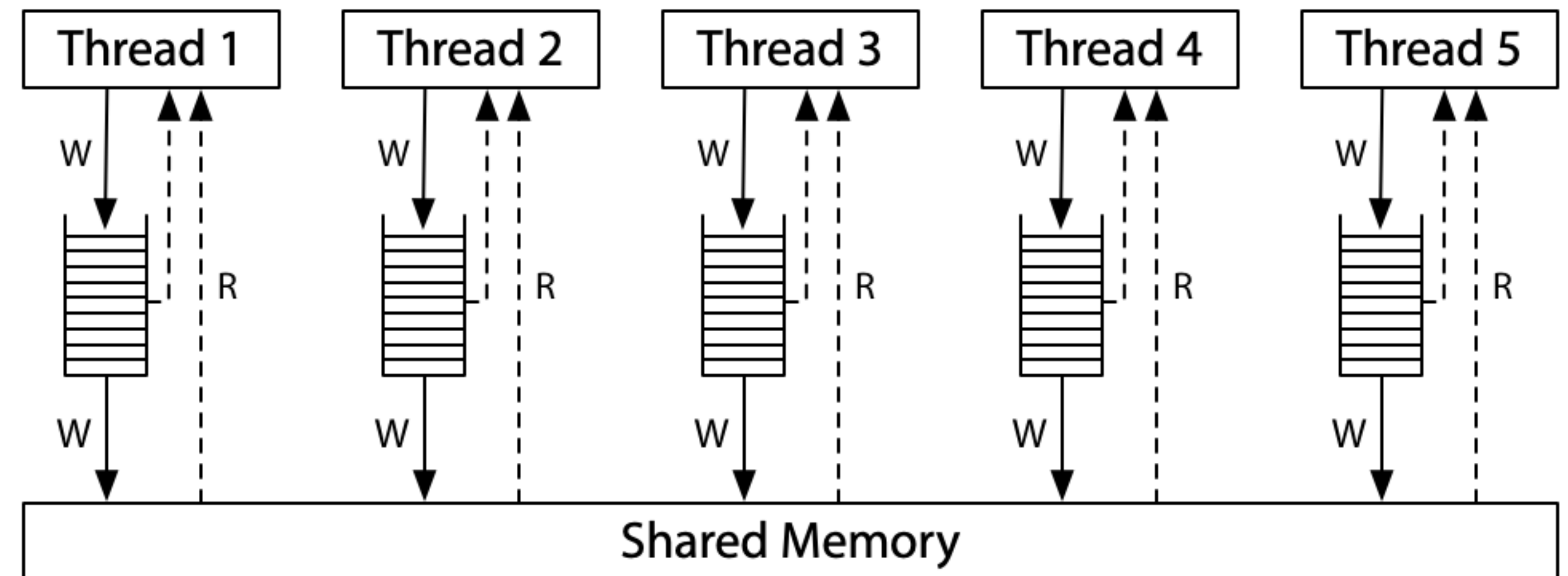
- CPUs can run instructions out of order



CPU's do not follow sequential consistency!

x86 follows “Total Store Order” (TSO). Intel SDM Vol 3A, Section 8.2

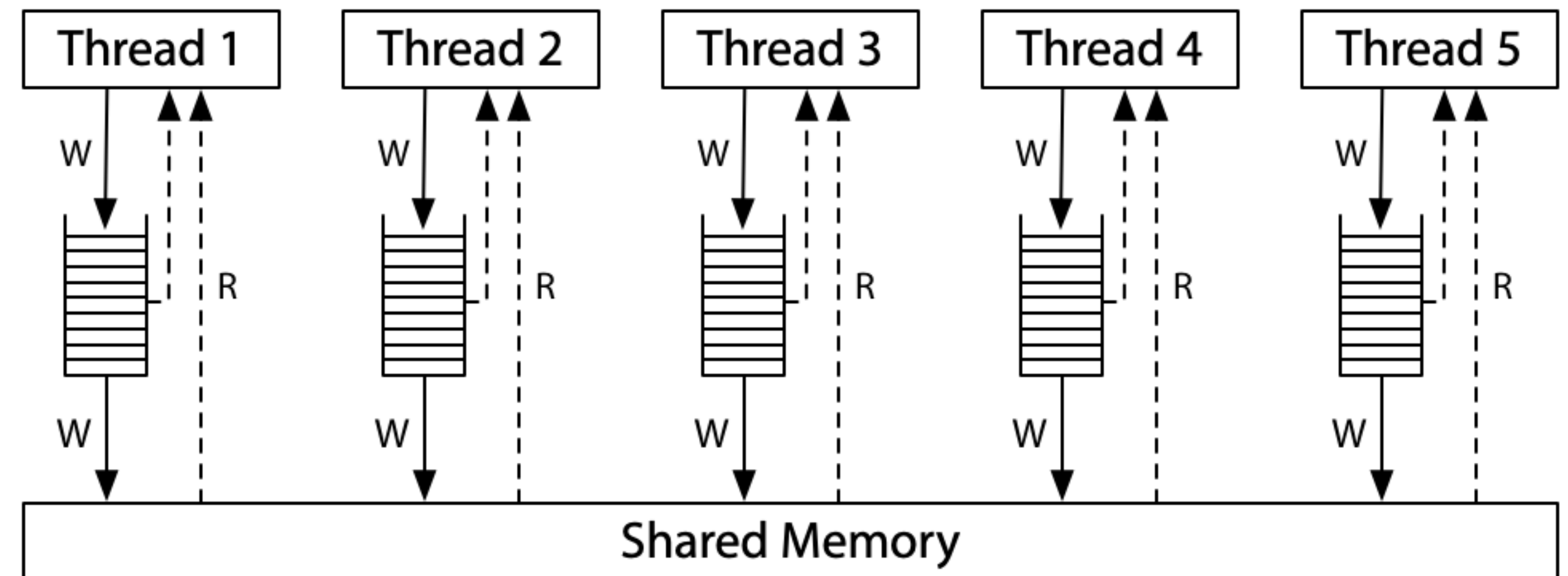
- CPUs can run instructions out of order
- Writes go to CPU's FIFO store buffer



CPU's do not follow sequential consistency!

x86 follows “Total Store Order” (TSO). Intel SDM Vol 3A, Section 8.2

- CPUs can run instructions out of order
- Writes go to CPU's FIFO store buffer
- Reads first check the local store buffer

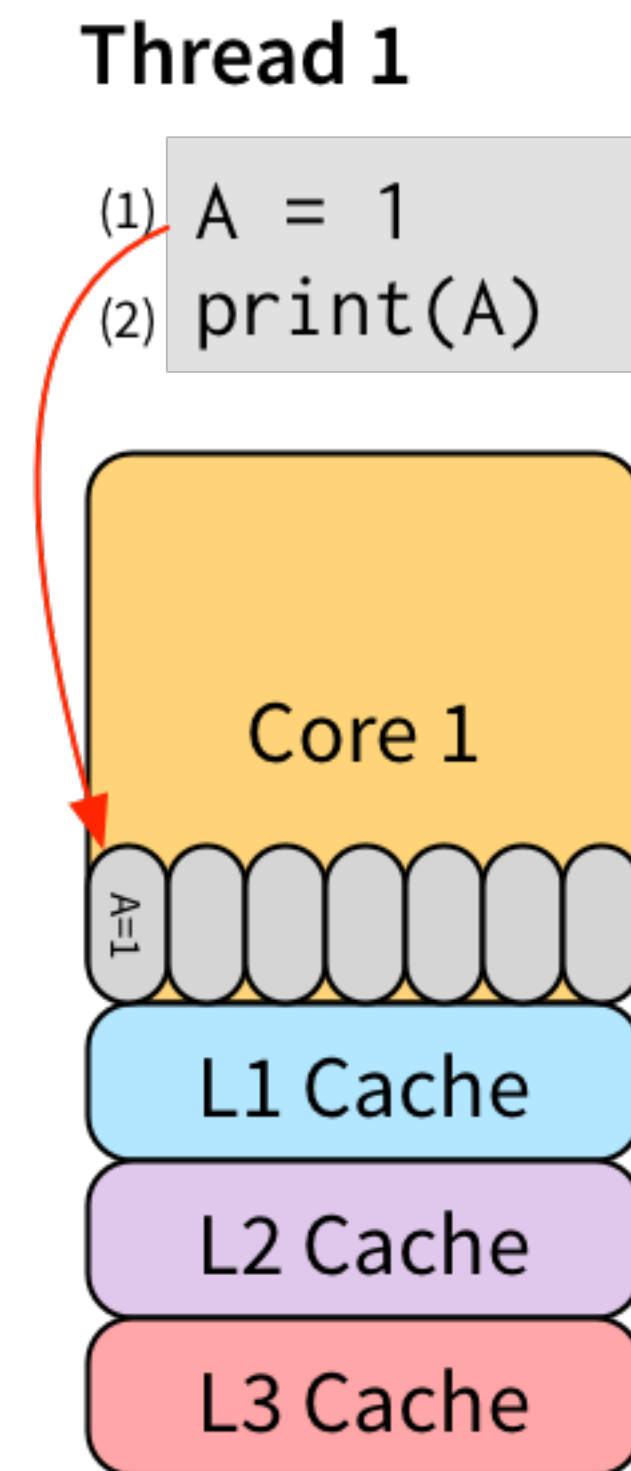


x86 follows “Total Store Order” (TSO). Intel SDM Vol 3A, Section 8.2

-
- The diagram illustrates a shared-memory multiprocessor system. At the top, five boxes represent threads: Thread 1, Thread 2, Thread 3, Thread 4, and Thread 5. Each thread has a local memory stack, depicted as a vertical rectangle with horizontal lines. Below each thread's stack is a solid arrow pointing downwards, labeled 'W', representing a write operation to the shared memory. To the right of each stack are two dashed arrows pointing upwards, labeled 'R', representing read operations from the shared memory. At the bottom of the diagram is a long horizontal rectangle labeled 'Shared Memory'. Vertical lines connect the bottom of each thread's stack to the top of the shared memory block, indicating the communication path for both writes and reads.

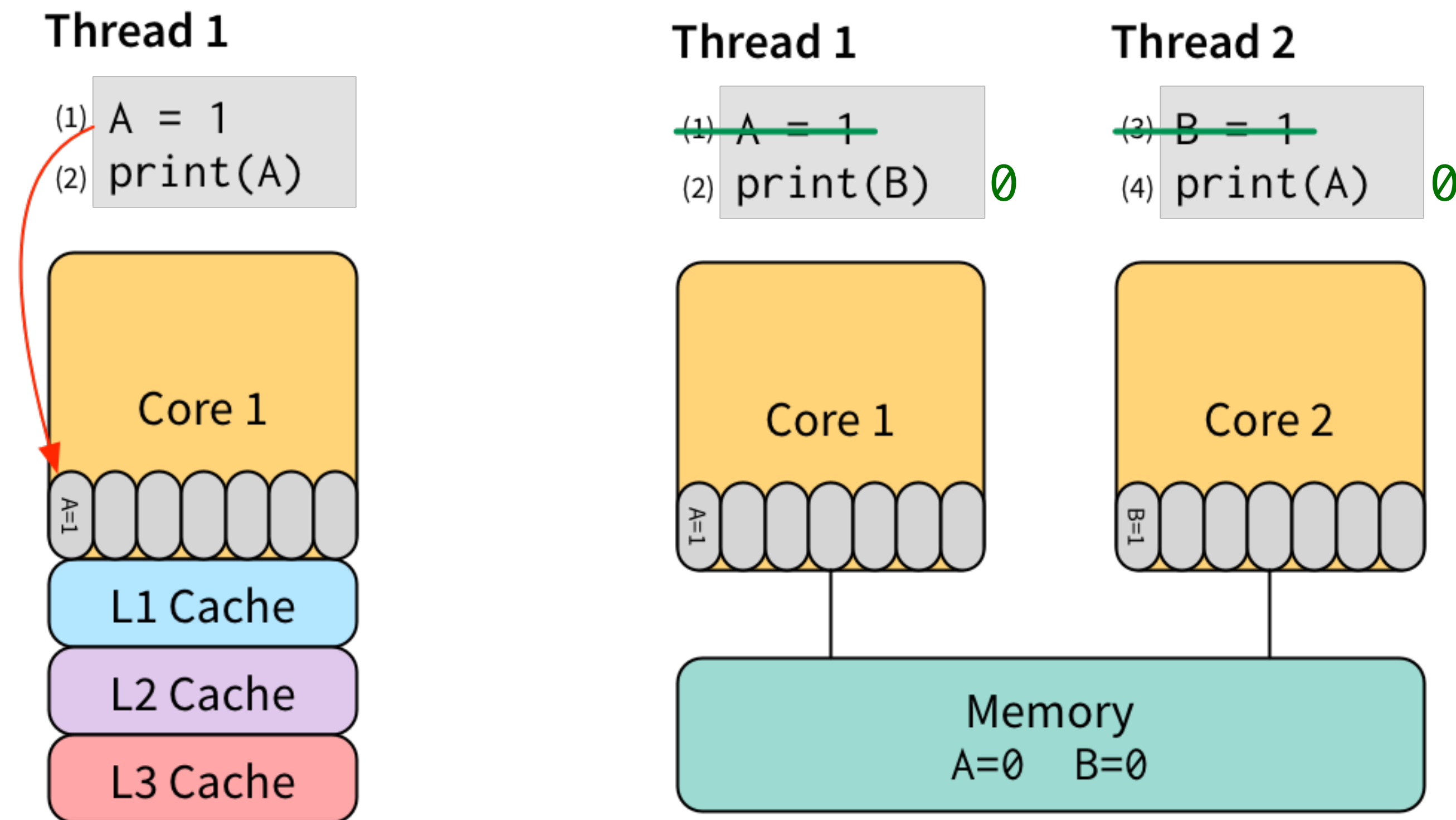
Example weird behaviours from weak memory models

Memory models define observable behaviours



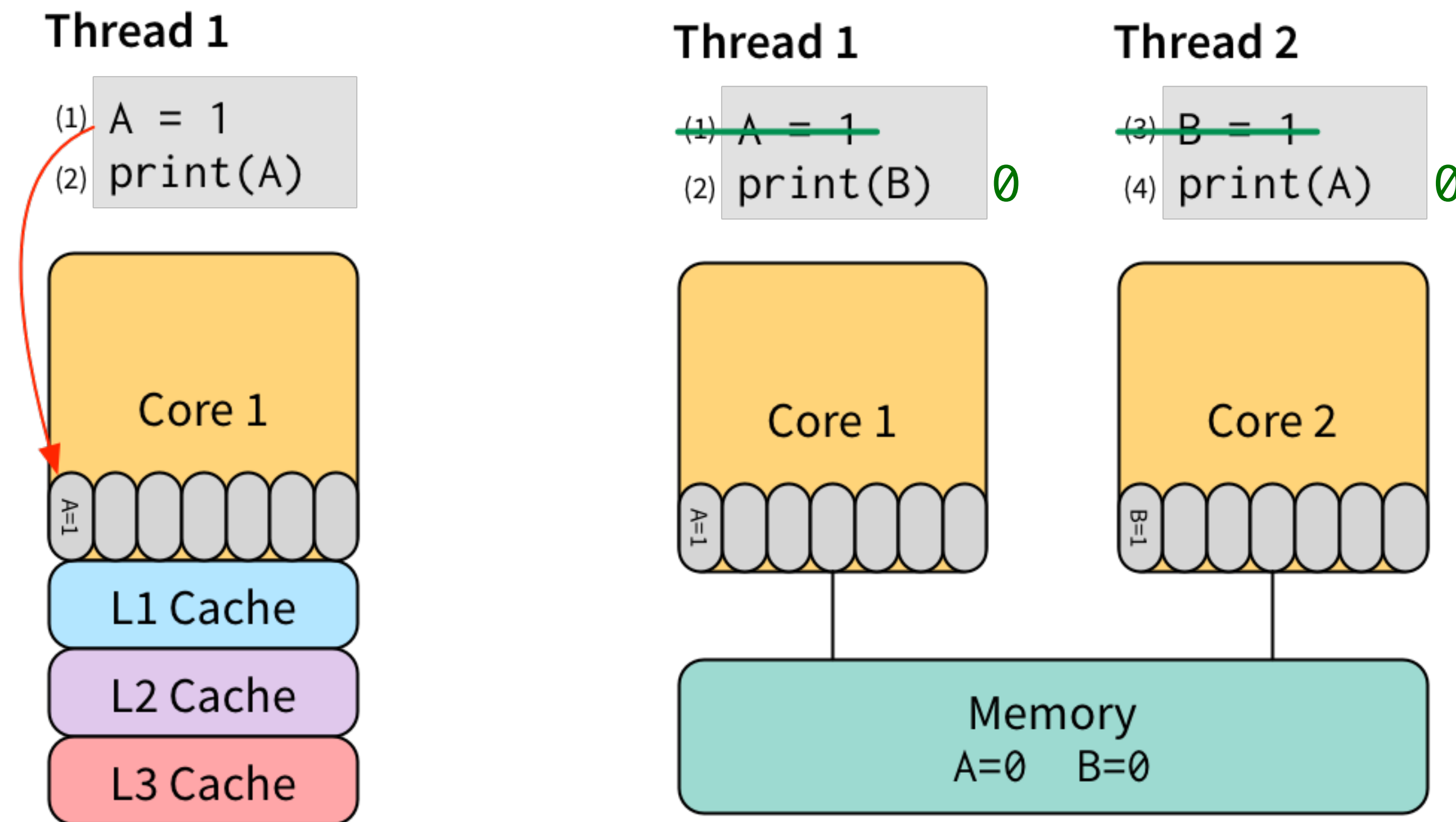
Example weird behaviours from weak memory models

Memory models define observable behaviours



Example weird behaviours from weak memory models

Memory models define observable behaviours



No interleaving of sequential executions can lead to this outcome!

Peterson algorithm breaks on x86

peterson-breaks.c

CPU 1	CPU 2
mov \$1, flag[0]	mov \$1, flag[1]
mov \$1, turn	mov \$0, turn
mov flag[1], %eax	mov flag[0], %ebx
mov turn, %ecx	mov turn, %ecx
flag[1] is 1 and turn is 1	flag[0] is 1 and turn is 1

Peterson algorithm breaks on x86

peterson-breaks.c

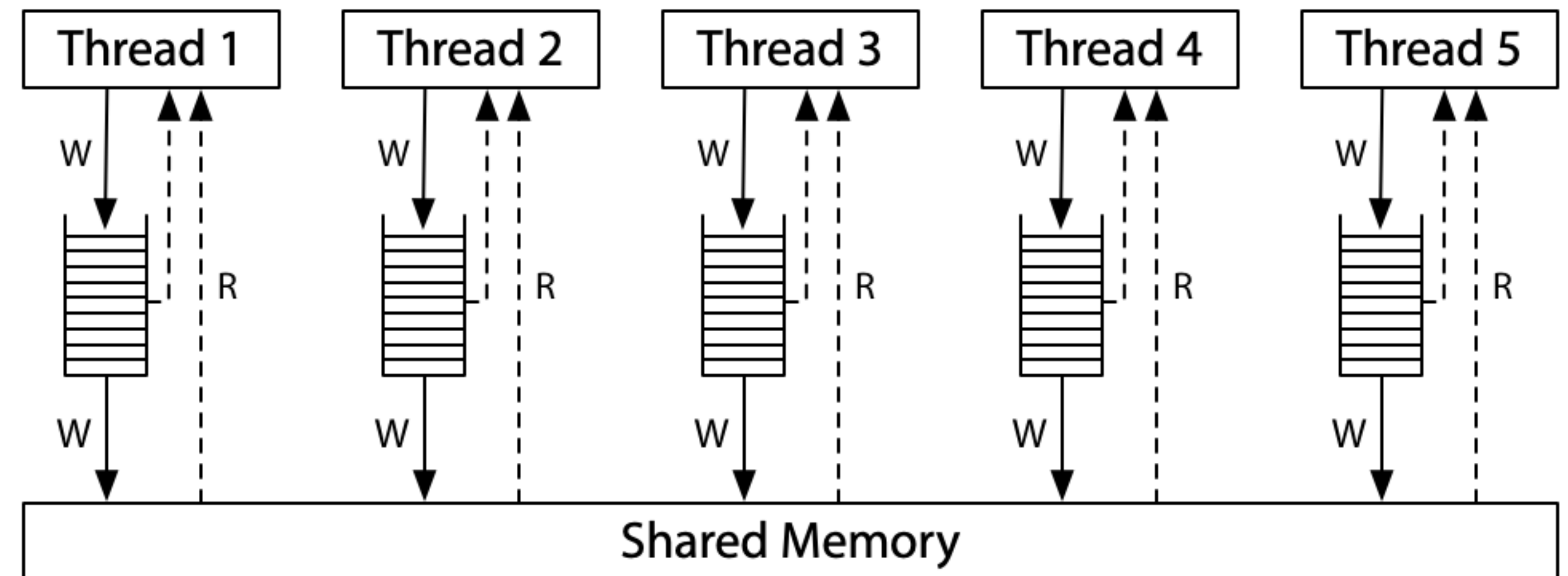
CPU 1	CPU 2
mov \$1, flag[0]	mov \$1, flag[1]
mov \$1, turn	mov \$0, turn
mov flag[1], %eax	mov flag[0], %ebx
mov turn, %ecx	mov turn, %ecx
flag[1] is 1 and turn is 1	flag[0] is 1 and turn is 1

CPU 1	CPU 2
mov \$1, flag[0]	mov \$1, flag[1]
mov \$1, turn	mov \$0, turn
mov flag[1], %eax	mov flag[0], %ebx
mov turn, %ecx	mov turn, %ecx
flag[1] is 0 and turn is 1	flag[0] is 0 and turn is 1

flag[*]=1 is in local store buffer at the time of read

Fences

- MFENCE: finish all pending reads and writes before continuing. Flush the store buffer before continuing



Fixing Peterson's algorithm

```
int flag[2];
int turn = 0; // whose turn? (thread 0 or 1?)

void init() {
    flag[0] = flag[1] = 0; // indicates that you want to hold the lock
}

void lock() {
    flag[self] = 1; // self: thread ID of caller
    turn = 1 - self; // make it other thread's turn

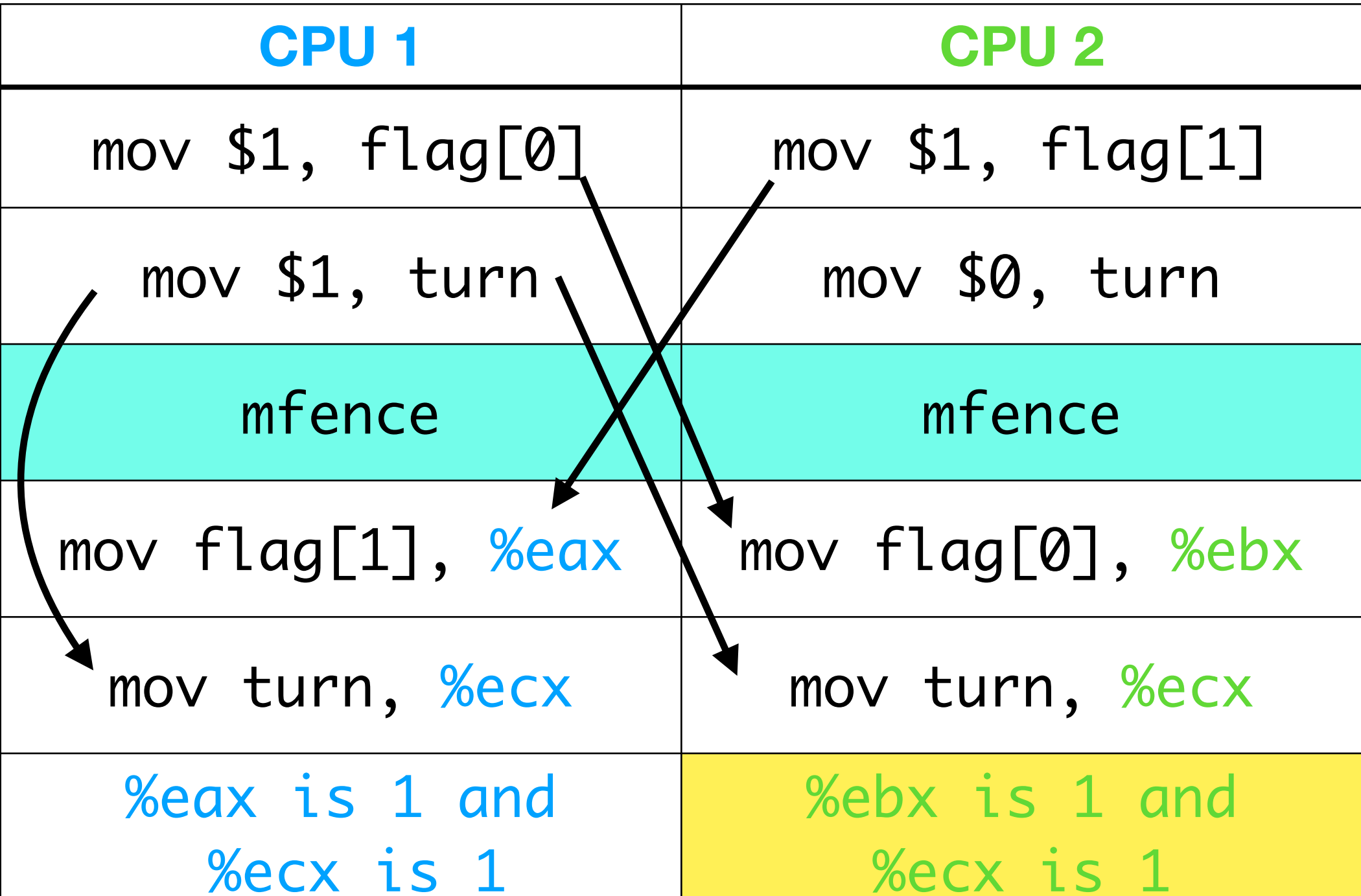
    asm volatile("mfence");
    while ((flag[1 - self] == 1) && (turn == 1 - self)) ; // spin-wait
}

void unlock() {
    flag[self] = 0; // simply undo your intent
}
```

Why does it work?

- mfence forces writes to flag[0] and flag[1] to main memory
- Reads of flag[0] and flag[1] are served from main memory

CPU 1	CPU 2
mov \$1, flag[0]	mov \$1, flag[1]
mov \$1, turn	mov \$0, turn
mfence	mfence
mov flag[1], %eax	mov flag[0], %ebx
mov turn, %ecx	mov turn, %ecx
%eax is 1 and %ecx is 1	%ebx is 1 and %ecx is 1



flag[*]=1 **cannot** be in local store buffer at the time of read

Compiler can reorder instructions!

```
void do() {  
  
    flag[self] = 1;  
    turn = 1 - self;  
    asm volatile("mfence");  
  
    while ((flag[1 - self] == 1) && (turn == 1 - self));  
    counter++; // critical section  
    flag[self] = 0;  
  
}
```


Compiler can reorder instructions!

```
void do() {
```

```
    flag[self] = 1;
```

```
    turn = 1 - self;
```

```
    asm volatile("mfence");
```

Independent variables from
compiler's point of view!



```
    while ((flag[1 - self] == 1) && (turn == 1 - self));
```

```
    counter++; // critical section
```

```
    flag[self] = 0;
```

```
}
```

Compiler can reorder instructions!

```
void do() {
```

```
    flag[self] = 1;
```

```
    turn = 1 - self;
```

```
    asm volatile("mfence");
```

Independent variables from
compiler's point of view!

```
    while ((flag[1 - self] == 1) && (turn == 1 - self));
```

```
    counter ++; // critical section
```

```
    flag[self] = 0;
```

```
}
```


Compiler can reorder instructions!

- Compiler might also remove instructions. Make shared variables volatile!

```
void do() {
```

```
flag[self] = 1;  
    turn = 1 - self;  
    asm volatile("mfence");
```

Independent variables from
compiler's point of view!

```
    while ((flag[1 - self] == 1) && (turn == 1 - self));  
    counter++; // critical section  
    flag[self] = 0;
```

```
}
```

Just put mfence everywhere?

- Works fine. But mfence is expensive!

```
time ./mfence-perf
```

```
void do() {  
  
    flag[self] = 1;  
    turn = 1 - self;  
    asm volatile("mfence");  
  
    while ((flag[1 - self] == 1) && (turn == 1 - self));  
  
    asm volatile("mfence");  
    counter++; // critical section  
    asm volatile("mfence");  
  
    flag[self] = 0;  
}
```

Software barriers

```
time ./swfence-perf  
peterson-fence
```

```
volatile int flag[2], turn, counter;
```

```
void do() {  
    flag[self] = 1;  
    turn = 1 - self;
```

```
    __sync_synchronize( ); // software + hardware barrier  
    while ((flag[1 - self] == 1) && (turn == 1 - self));
```

```
    asm volatile ("" ::: "memory"); // software barrier  
    counter ++; // critical section
```

```
    asm volatile ("" ::: "memory"); // software barrier  
    flag[self] = 0;
```

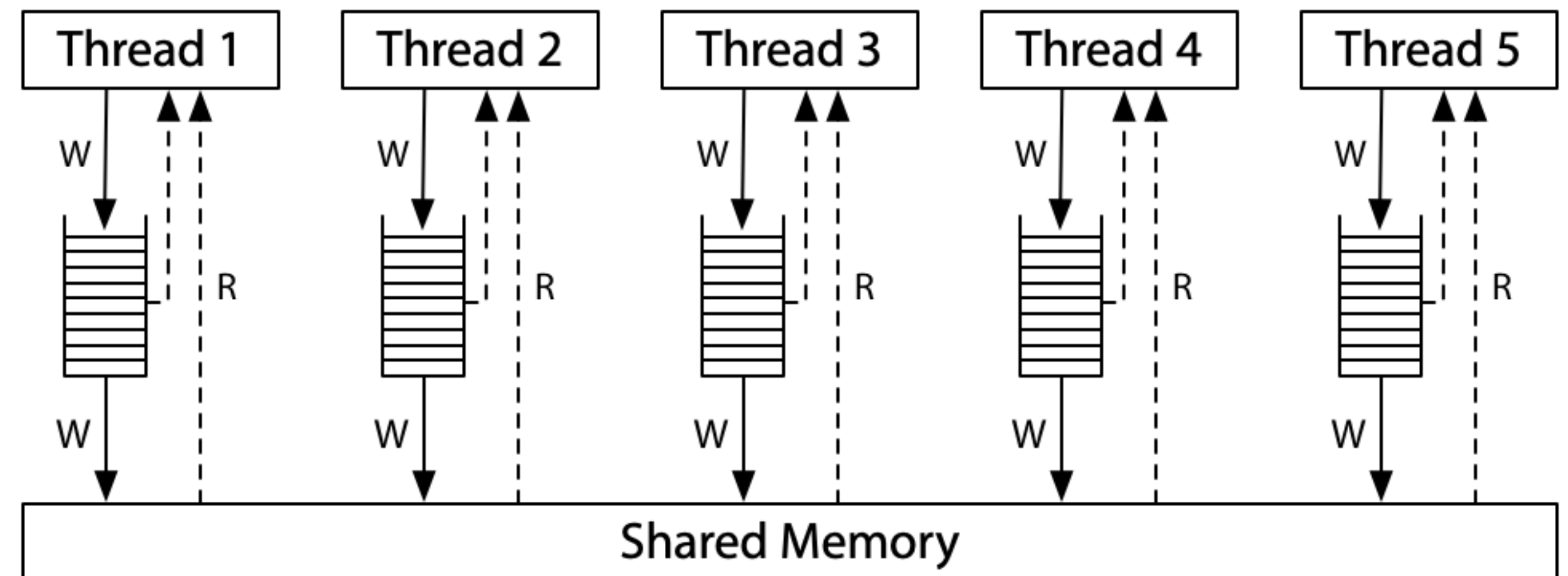
```
}
```


Why do we NOT need a hardware barrier when releasing?

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```
flag[self] = 0;
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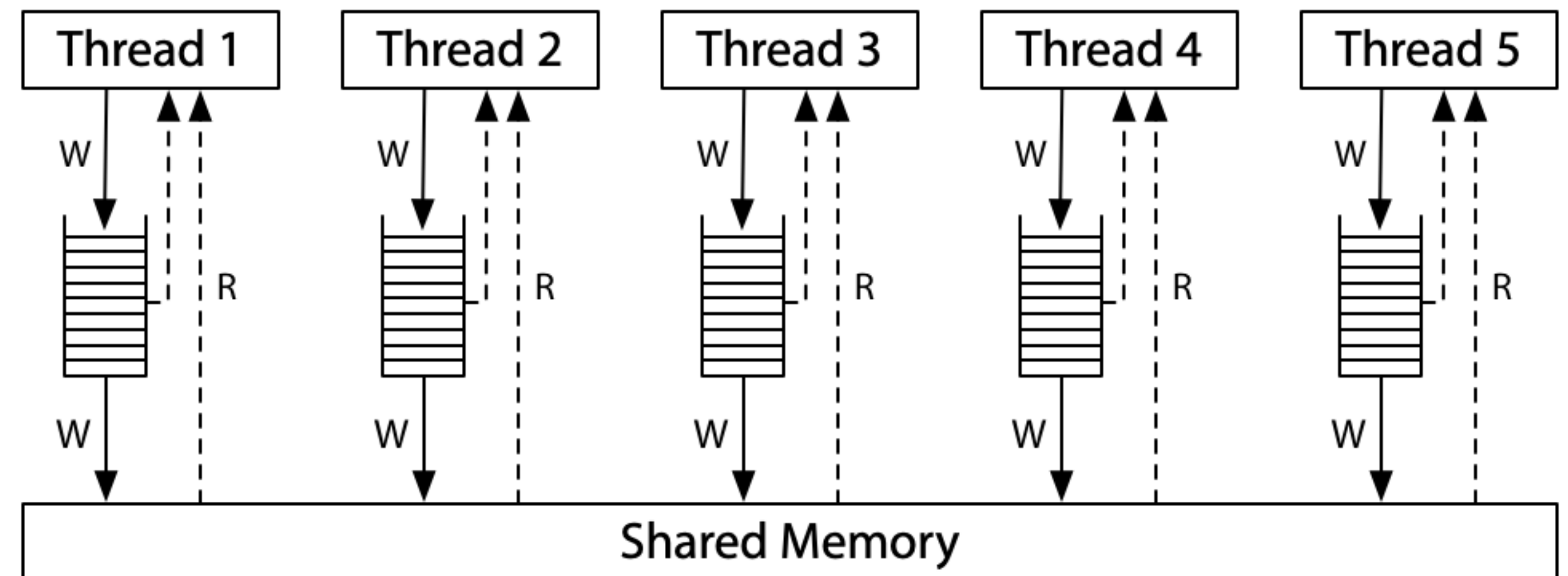
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- Store buffer is FIFO!



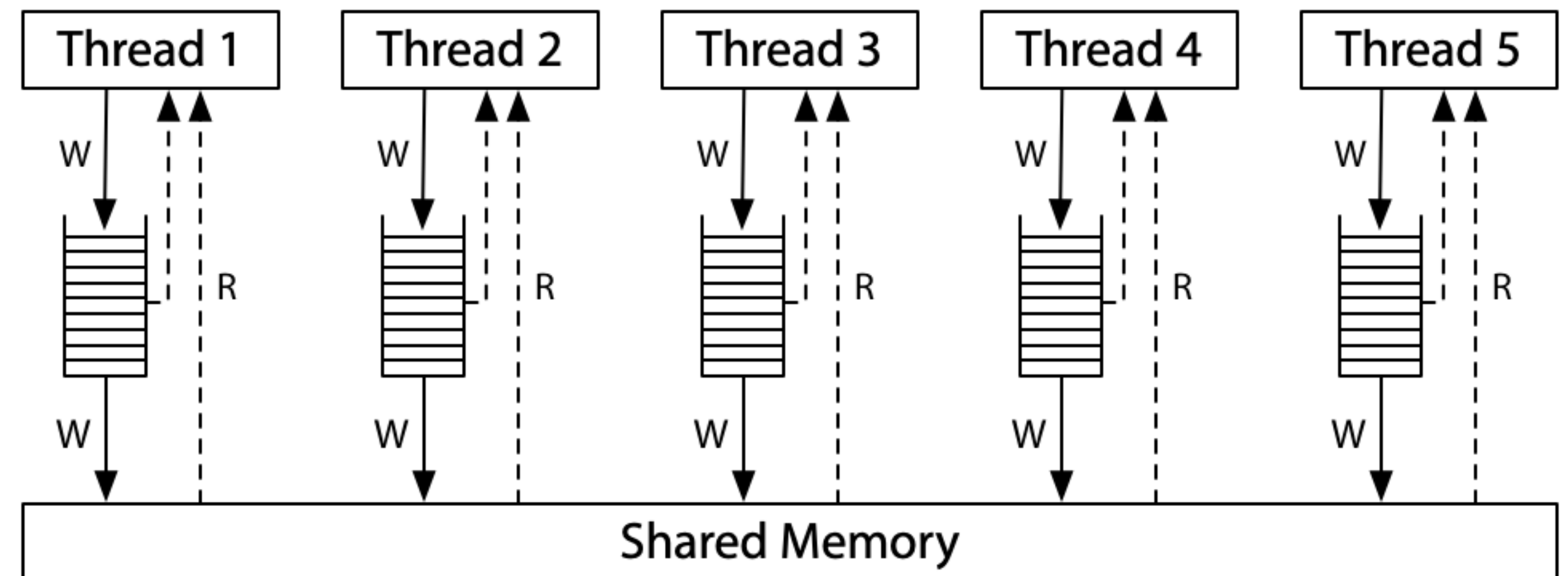
Why do we NOT need a hardware barrier when releasing?

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```
asm volatile ("" ::: "memory"); // software barrier
```

```
flag[self] = 0;
```

- Store buffer is FIFO!
- If “flag[self] = 0” is visible, then write to counter is also visible!

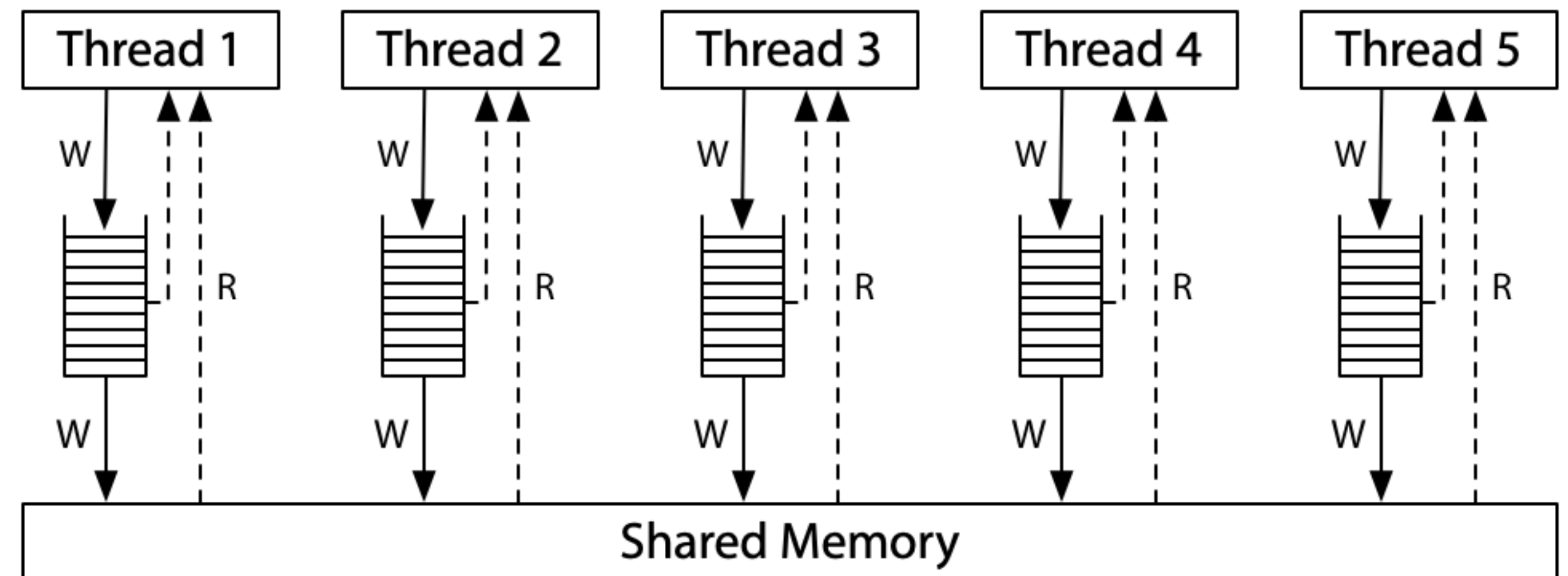


Why do we NOT need a hardware barrier when releasing?

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counter++; // critical section
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asm volatile ("" ::: "memory"); // software barrier  
flag[self] = 0;
```

- Store buffer is FIFO!
- If “flag[self] = 0” is visible, then write to counter is also visible!
- Point of long debate in Linux kernel mailing list (1999). Improved some benchmark performance by 4%!



x86 synchronising instructions

- LOCK <instruction>:
 - MFENCE +
 - Only one CPU can run a locked instruction at a time

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x86 synchronising instructions

- LOCK <instruction>:
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 - Only one CPU can run a locked instruction at a time
 - Example: atomic.c. Note that atomic.s has “LOCK addl \$1, acounter”
 - If compiler cannot generate the corresponding LOCK instruction, it uses compare-exchange operation
 - “acounter *= 23” translates to a loop over “imull \$23, %eax, %ecx; lock cmpxchgl %ecx, acounter”

Spin locks in xv6

```
void acquire(struct spinlock *lk)
{
    pushcli(); // disable interrupts to avoid deadlock.

    // The xchg is atomic.
    while(xchg(&lk->locked, 1) != 0);
    __sync_synchronize();
    asm volatile ("" ::: "memory");
    ..
}
```

```
static inline uint xchg(volatile uint *addr, uint newval) {
    uint result;

    // The + in "+m" denotes a read-modify-write operand
    asm volatile("lock; xchgl %0, %1" :
        "+m" (*addr), "=a" (result) :
        "1" (newval) :
        "cc");
    return result;
}
```

```
void release(struct spinlock *lk)
{
    ..
    __sync_synchronize();
    asm volatile ("" ::: "memory");

    // Release the lock, equivalent to lk->locked = 0.
    asm volatile("movl $0, %0" : "+m" (lk->locked) : );

    popcli();
}
```

Other metrics

- Safety: Bad things never happen
 - Two threads shall never *simultaneously* acquire the lock
- Liveness: Good things eventually happen
 - Some thread (trying to lock) eventually gets to acquire the lock
- Fairness
 - Do some threads (CPUs) get unfair advantage over others?
 - Starvation freedom: Is it guaranteed that all threads will eventually get the lock?

Spin locks in xv6 are not starvation free

```
void acquire(struct spinlock *lk)
{
    pushcli(); // disable interrupts to avoid deadlock.

    // The xchg is atomic.
    while(xchg(&lk->locked, 1) != 0);
    __sync_synchronize();

    ..
}
```

```
static inline uint xchg(volatile uint *addr, uint newval) {
    uint result;

    // The + in "+m" denotes a read-modify-write operand
    asm volatile("lock; xchgl %0, %1" :
                 "+m" (*addr), "=a" (result) :
                 "1" (newval) :
                 "cc");
    return result;
}
```

```
void release(struct spinlock *lk)
{
    ..

    __sync_synchronize();

    // Release the lock, equivalent to lk->locked = 0.
    asm volatile("movl $0, %0" : "+m" (lk->locked) : );

    popcli();
}
```


Ticket locks

```
1  typedef struct __lock_t {
2      int ticket;
3      int turn;
4  } lock_t;
5
6  void lock_init(lock_t *lock) {
7      lock->ticket = 0;
8      lock->turn   = 0;
9  }
10
11 void lock(lock_t *lock) {
12     int myturn = FetchAndAdd(&lock->ticket);
13     while (lock->turn != myturn)
14         ; // spin
15 }
16
17 void unlock(lock_t *lock) {
18     lock->turn = lock->turn + 1;
19 }
```

Atomic instruction



What if critical sections are long?

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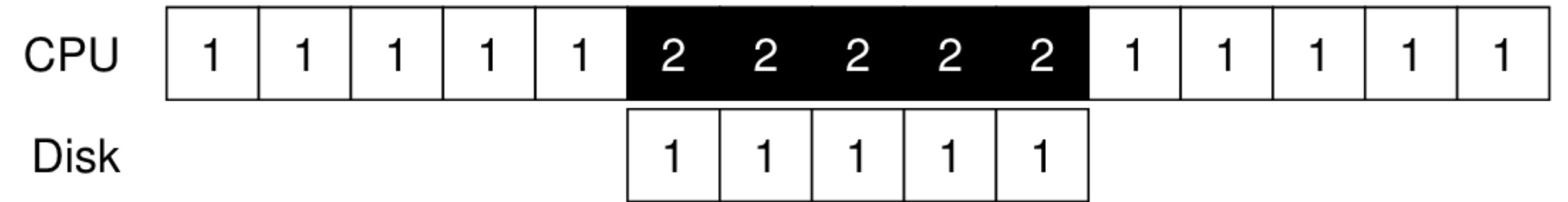
- Spin locks waste CPU time while unnecessary spinning
- Yield — let something else run while we are anyways waiting
- May not know the length of critical section while locking
 - Hybrid locks: spin for some time and then yield
 - Example: futex in linux (Figure 28.10 OSTEP book)

Yielding: condition variables

Case study: xv6 disk driver

```
void sleep(void *chan) {
    struct proc *p = myproc();
    // Go to sleep.
    p->chan = chan;
    p->state = SLEEPING;
    sched();
    // Tidy up.
    p->chan = 0;
}

void wakeup(void *chan) {
    struct proc *p;
    for(p = ptable.proc; p < &ptable.proc[NPROC]; p++)
        if(p->state == SLEEPING && p->chan == chan)
            p->state = RUNNABLE;
}
```



```
void ideintr(void) {
    struct buf *b = idequeue;
    if(!(b->flags & B_DIRTY) && idewait(1) >= 0)
        insl(0x1f0, b->data, BSIZE/4);
    b->flags |= B_VALID;
    b->flags &= ~B_DIRTY;
    wakeup(b);
}

void iderw(struct buf *b){
    if(idequeue == b)
        idestart(b);
    while((b->flags & (B_VALID|B_DIRTY)) != B_VALID){
        noop();
        sleep(b);
    }
}
```

Case study: wait/exit

```
void exit(void) {
    struct proc *curproc = myproc();
    struct proc *p;

    wakeup(curproc->parent);
}

void sleep(void *chan) {
    struct proc *p = myproc();
    p->chan = chan;
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    sched();
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void wakeup(void *chan) {
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```
int wait(void) {
    struct proc *p;
    struct proc *curproc = myproc();

    for(;;) {
        // Scan through table looking for exited children.
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++) {
            if(p->parent != curproc)
                continue;
            if(p->state == ZOMBIE) {
                // Found one.
                ...
                return pid;
            }
        }
        // Wait for children to exit
        sleep(curproc);
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```


Lost wakeup problem

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Use spin locks!

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void exit(void) {
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    struct proc *p;
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        // Scan through table looking for exited children.
        for(p = ptable.proc; p < &ptable.proc[NPROC]; p++){
            if(p->parent != curproc)
                continue;
            if(p->state == ZOMBIE){
                // Found one.
                ...
                return pid;
            }
        }
        // Wait for children to exit
        sleep(curproc, &ptable.lock);
    }
}
```

Use spin locks!

```
void exit(void) {
    struct proc *curproc = myproc();
    struct proc *p;
    acquire(&ptable.lock);
    wakeup(curproc->parent);
}

void sleep(void *chan, struct spinlock *lk) {
    struct proc *p = myproc();
    p->chan = chan;
    p->state = SLEEPING;
    sched();
}

void wakeup(void *chan) {
    struct proc *p;
    for(p = ptable.proc; p < &ptable.proc[NPROC]; p++)
        if(p->state == SLEEPING && p->chan == chan)
            p->state = RUNNABLE;
}
```

eip →

```
int wait(void) {
    struct proc *p;
    struct proc *curproc = myproc();
    acquire(&ptable.lock);
    for(;;) {
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    }
}
```


Using locks

- Lost wakeup problem:
 - wait-attempt1.c: signal before wait
 - wait-attempt2.c: check for done first. Can still signal before wait.
 - wait.c: Protect done flag by a mutex. Works!
pthread_cond_wait(&c, &m) forces programmer to think what they should protect by mutex!
- Producer consumer
 - mypipe.c: writer sleeps when pipe is full, reader sleeps when pipe is empty. Writer wakes up reader, reader wakes up writer
- Allocation:
 - alloc-attempt1.c: Threads could not allocate even though memory is available
 - alloc.c: Broadcast wakes up everyone. Thundering herd problem
 - Sleep/wakeup in xv6 is basically broadcast

Semaphores

- semlock.c: Use in place of locks by initialising to 1
- wait-sem.c: Use in place of condition variables by initializing to 0. no lost wakeup problem!
- sempipe.c: need not protect reader and writer variables anymore!
- sem-mpmc.c: multiple producer, multiple consumer queue. Now need to protect reader and writer.

```
1  int sem_wait(sem_t *s) {
2      decrement the value of semaphore s by one
3      wait if value of semaphore s is negative
4  }
5
6  int sem_post(sem_t *s) {
7      increment the value of semaphore s by one
8      if there are one or more threads waiting, wake one
9  }
```

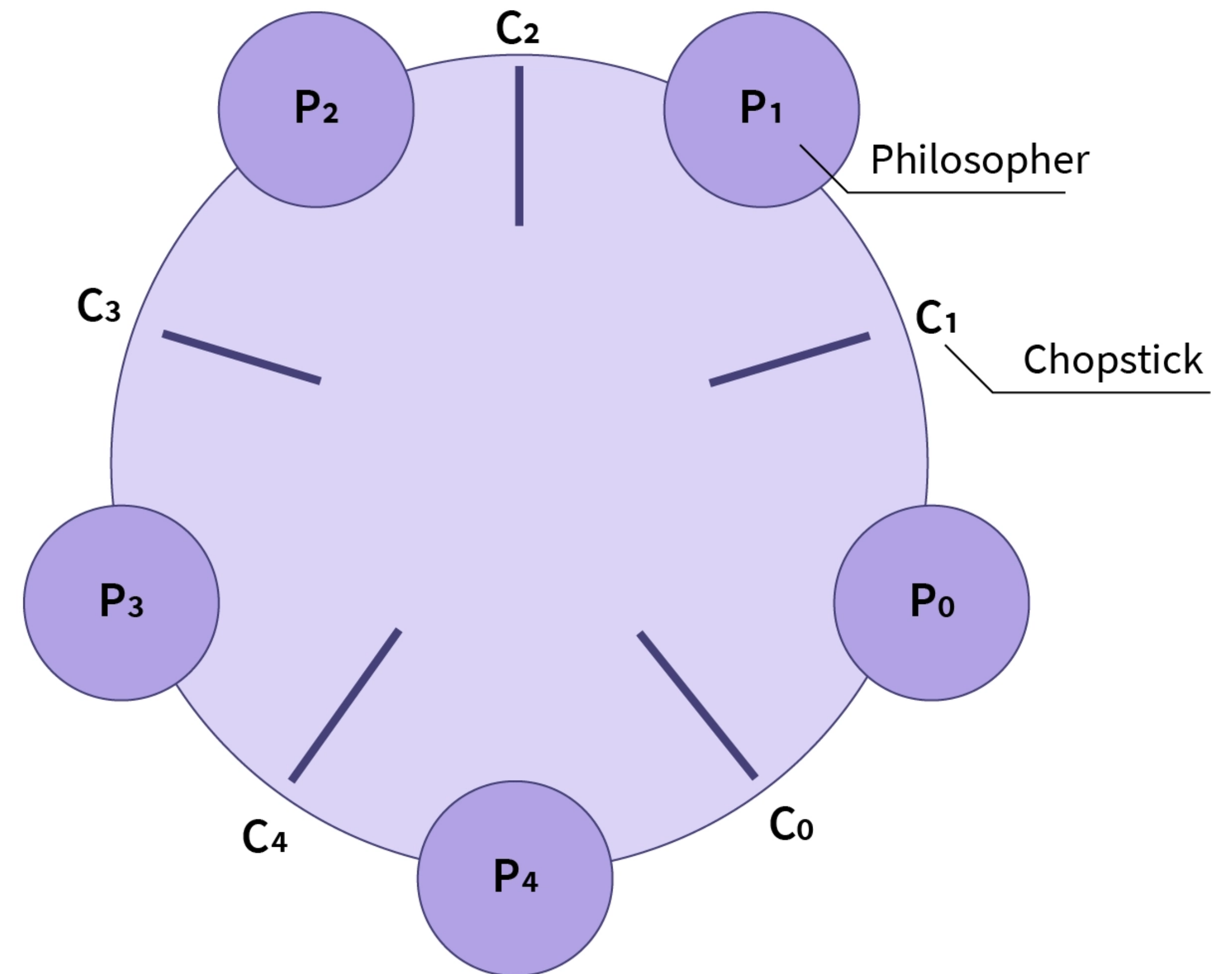
Reader-writer locks

- Multiple readers can acquire the lock at the same time. They are just reading.
- Only a single writer can acquire the lock. No reader should be holding the lock.
- `time ./rw-ctr`
- Locking primitives can be implemented using one another
 - `rw-using-sems`
 - `sems-using-lock-cv`

Dining philosophers

Deadlocks

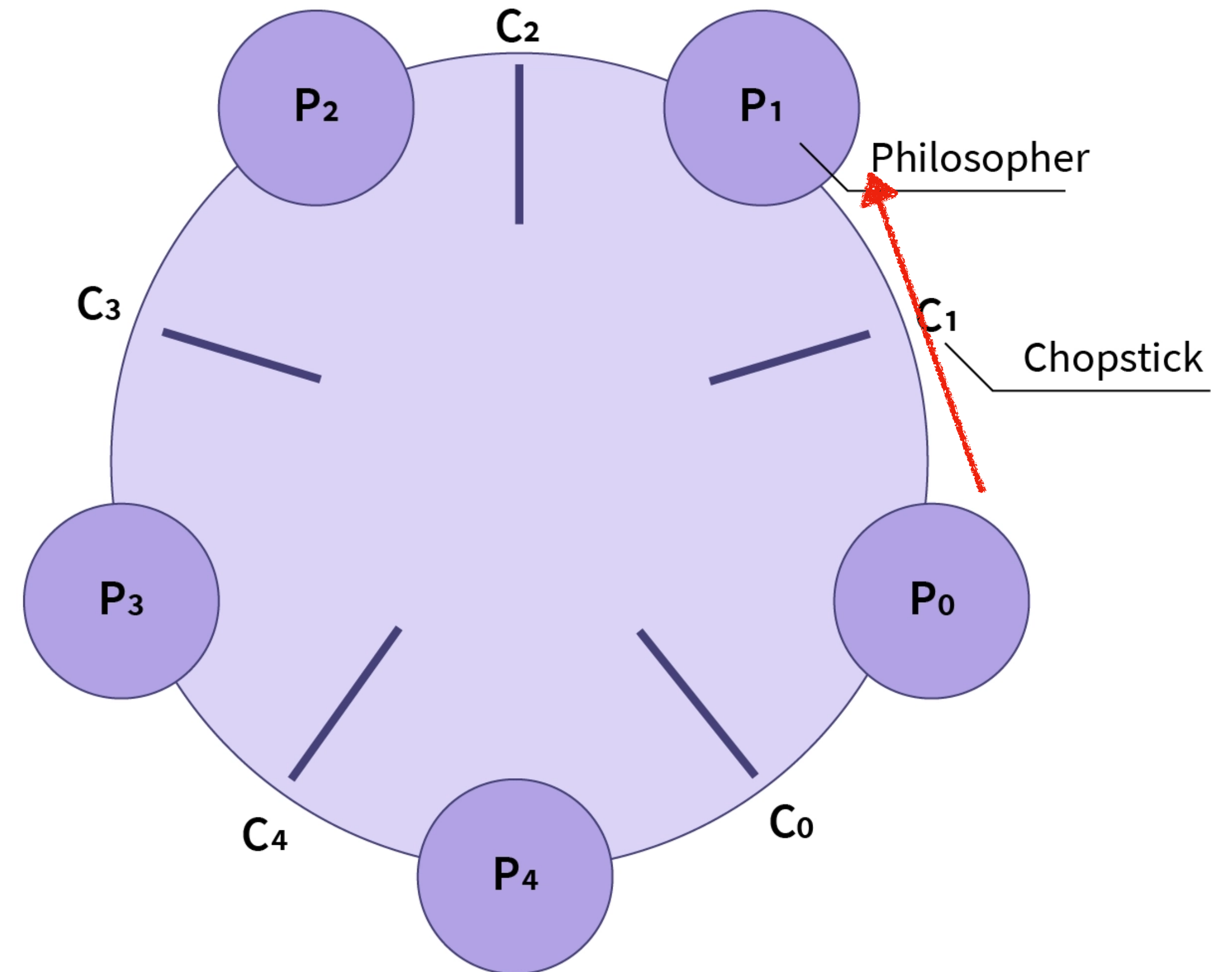
- Each philosopher randomly picks the left chopstick, then the right chopstick, eats, and releases both the chopsticks
- Example: dine-dead.c and dine.c



Dining philosophers

Deadlocks

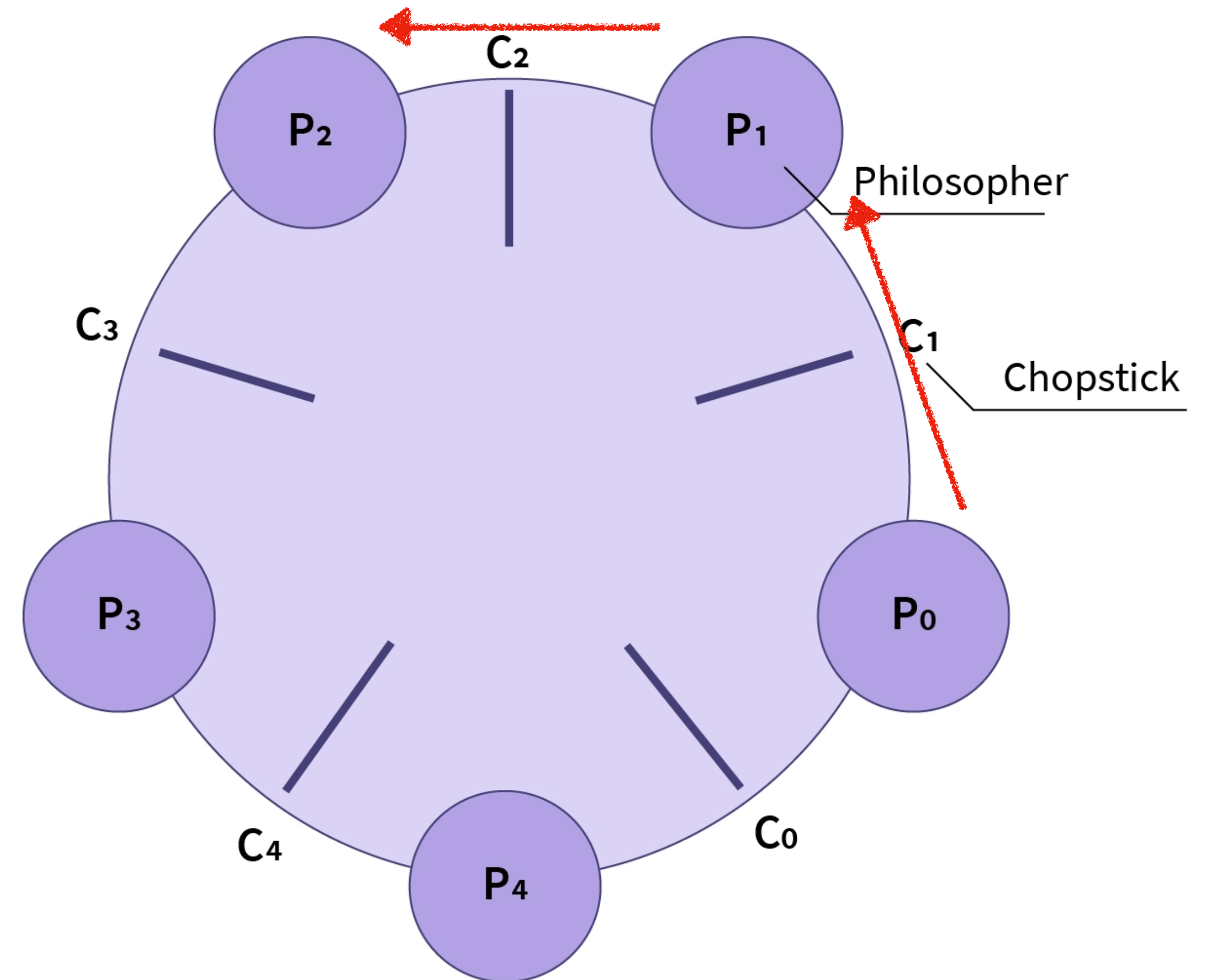
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Dining philosophers

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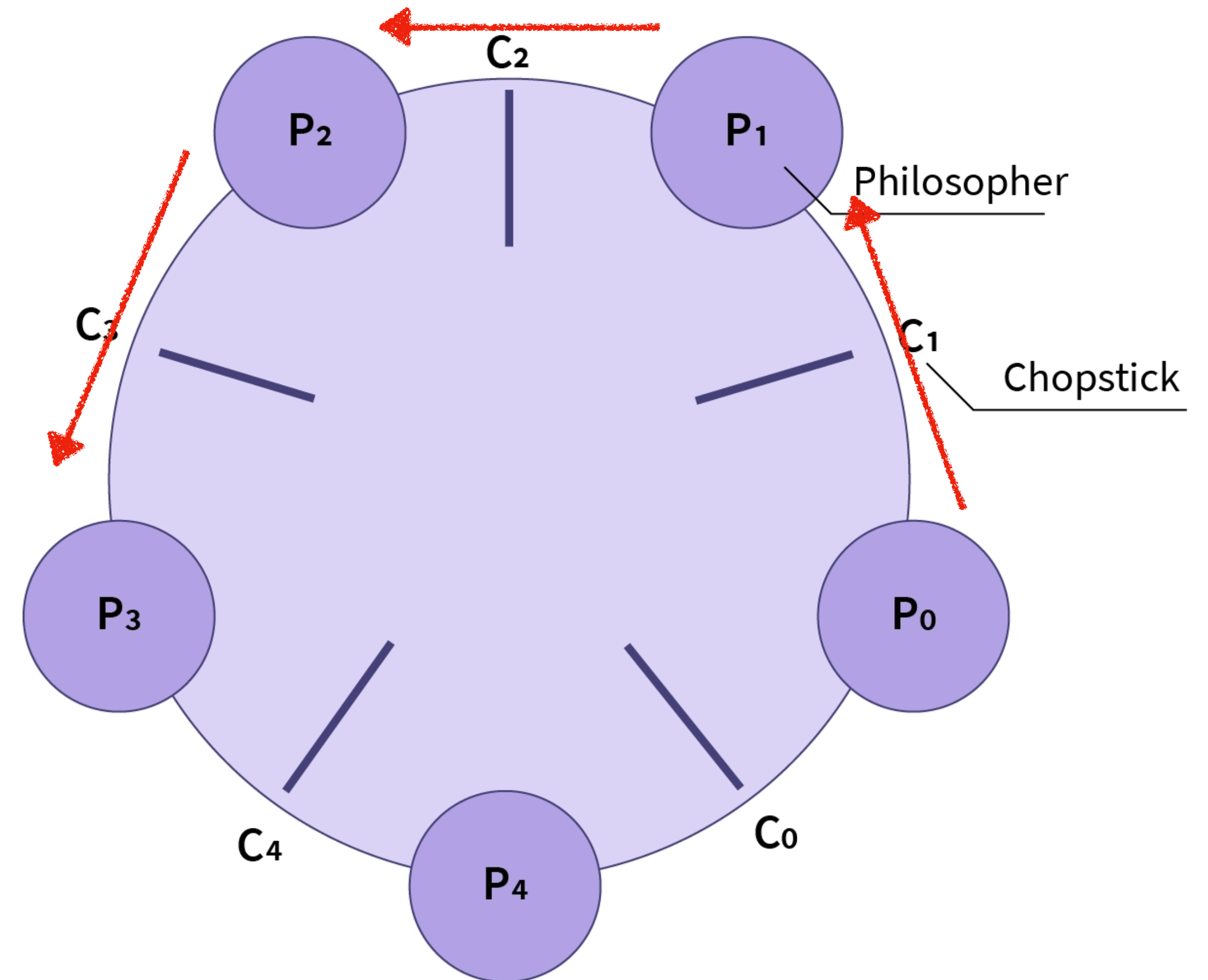
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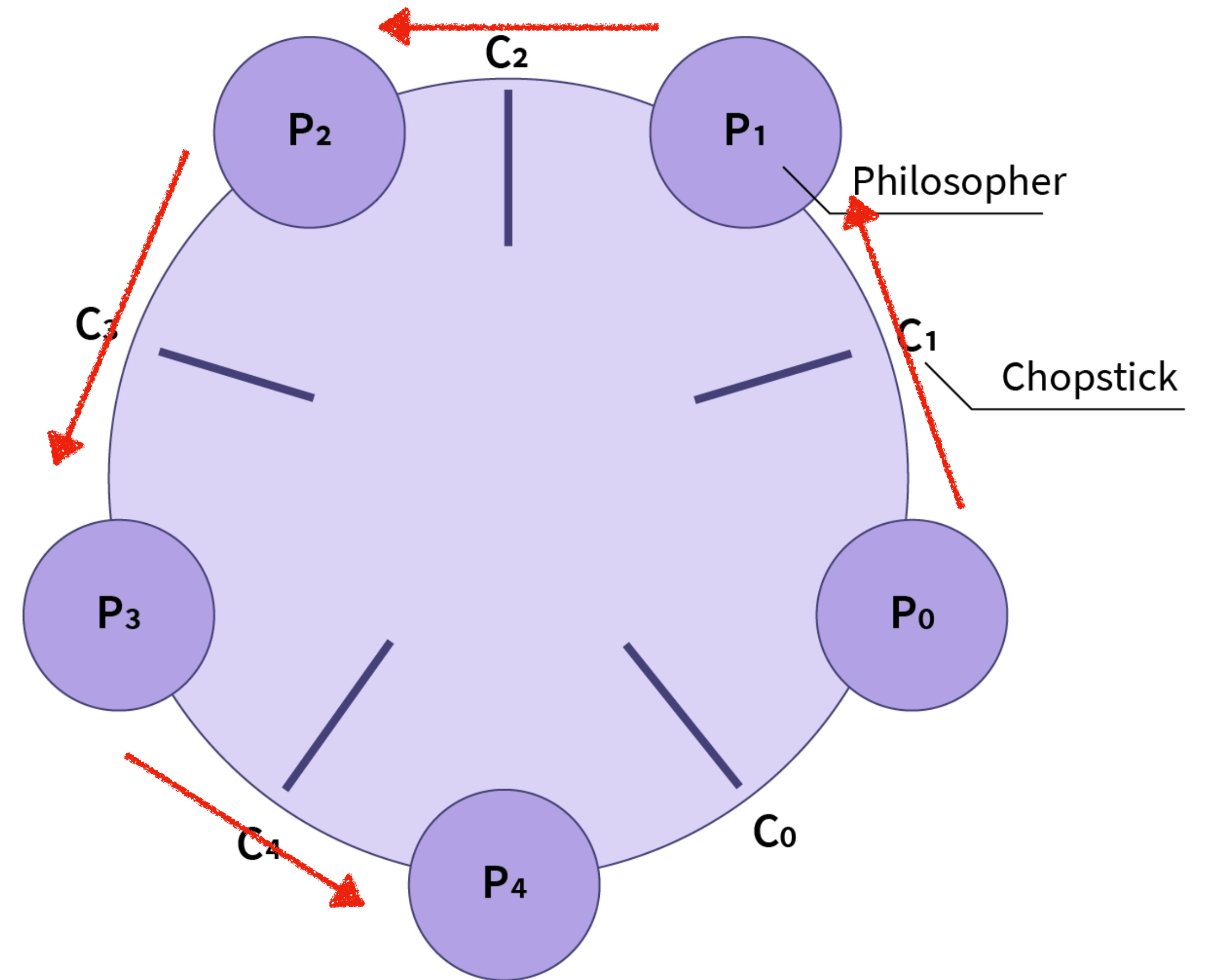
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Dining philosophers

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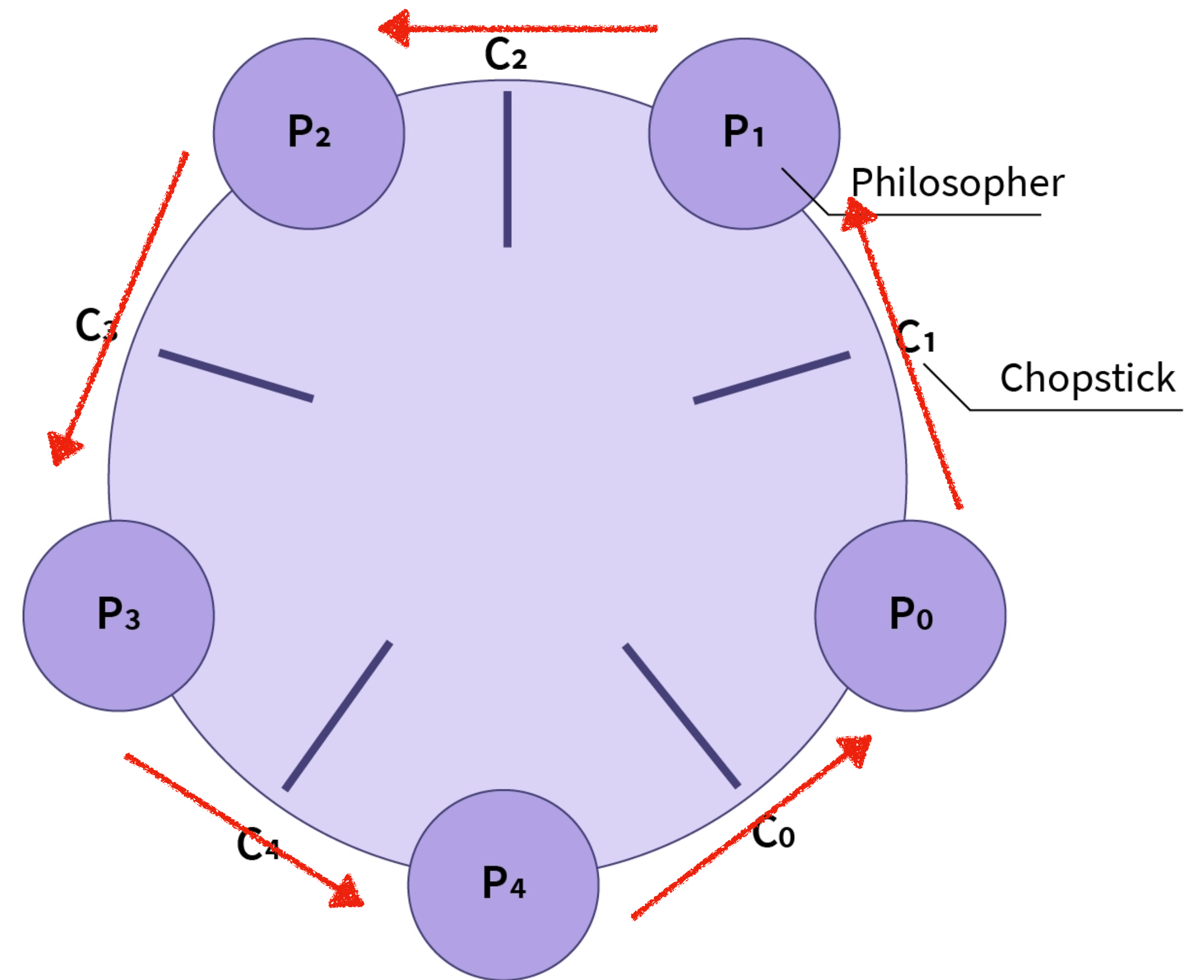
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Dining philosophers

Deadlocks

- Each philosopher randomly picks the left chopstick, then the right chopstick, eats, and releases both the chopsticks
- Example: dine-dead.c and dine.c



Deadlocks and locking discipline

- Acquire locks in the same order across all paths to avoid deadlocks.
- Example: `dead.c`, `dead-fix.c`
- Example lock order in xv6: `ptable.lock` is the last lock to be acquired

Summary

- Multi-processing hardware
 - xv6 setting up other processors
- Threads: shared address space, separate registers and stacks
 - Race conditions
- Design of locks
 - Sequential consistency and x86 memory model
 - Software barriers and hardware fences
 - Spin locks, conditional variables, hybrid locks, semaphores, read-write locks
- Difficulties with using locks: lost wakeup, deadlocks, locking discipline